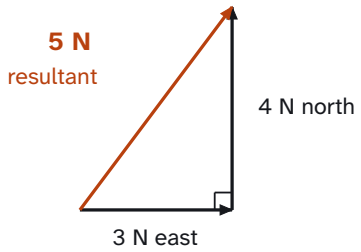
L3

The Level 3 study book

Understand the idea. Build the method. Earn the marks.

FIG. 01 / MECHANICS

Add components to find the resultant



Horizontal: 3 N

Vertical: 4 N

Magnitude: 5 N

Direction: 53.1° north of east

Takeaway: add vectors by direction; do not add 3 + 4 as force magnitudes.

Document	The Level 3 study book
Level / lessons	3 / 82
Edition	November 2026 / Bench Sheet
Pages	432

Start here

Begin with one lesson and a pencil. Follow the stages below, then practise without notes.

1. Read the Idea and describe it in your own words before looking at the rule.
2. Check what every symbol means and when the rule applies.
3. Cover the worked steps. Predict the next step, then compare.
4. Use the working grid to attempt Your turn. Check the answers in the lesson review section.
5. Attempt the subject tests without notes. Use the explanations to identify the exact step you missed.

Read the page at a glance

Rule

$$v = \frac{d}{t}$$

A framed rule is something to select and apply, not just memorise. Here, average speed is distance divided by elapsed time.

Your turn

Blue marks a task for you. Write a prediction or calculation before checking.

Watch out

An orange note marks a common error. Example: do not confuse distance with displacement.

Before you begin

- Have a calculator, pencil and the units used in your course.
- Review basic rearrangement and graph reading when a lesson lists them as pre-requisites.
- Use the formula sheet and symbol glossary for revision; the same letter can have different meanings in different lessons.

Use the linked contents and alphabetical topic index to find a lesson. Marking points are study guidance, not an official marking scheme.

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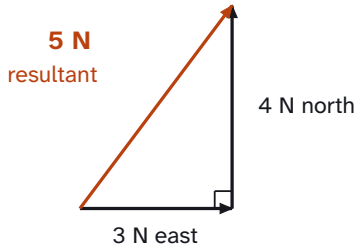
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Mechanics and Heat

FIG. 01 / MECHANICS

Add components to find the resultant



Horizontal: 3 N

Vertical: 4 N

Magnitude: 5 N

Direction: 53.1° north of east

Takeaway: add vectors by direction; do not add 3 + 4 as force magnitudes.

Motion, vectors, forces, work, energy, temperature and heat. New merged lectures are the main mechanics source; the older heat lecture supplies thermal topics.

Subject map

Motion, vectors, forces, work, energy, temperature and heat. New merged lectures are the main mechanics source; the older heat lecture supplies thermal topics.

Follow this reading route from foundations to applications. Each line links to its lesson; revisit the stated prerequisites whenever a step feels unfamiliar.

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09 Temperature, calorimetry and phase changes p.33

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Choose units and describe motion

MH / 01

- Distinguish distance from signed displacement.
- Calculate average speed and average velocity from a route.

You need: Add and subtract lengths; divide by elapsed time; convert minutes to seconds.

01 Idea

Position tells you where an object is. Displacement is final position minus initial position; distance counts the entire path. Average velocity uses displacement, while average speed uses distance. A negative velocity gives direction, not a negative speed.

Imagine laying a measuring tape along the whole route, then stretching a separate arrow directly from start to finish. The tape represents distance; the arrow represents displacement. Turning back adds tape length while shortening the arrow, so speed and velocity can describe the same journey differently.

- Distance adds each part of the route because every part was travelled.
- Displacement depends only on endpoints, so reversing direction changes its sign contribution.

02 Rule and symbols

Rule

$$\bar{v} = \Delta x / \Delta t$$

$$\text{average speed} = \text{distance} / \Delta t$$

Δx final position minus initial position / m

Δt elapsed time / s

$\bar{v}$ average velocity / m/s

03 Method

1. Choose east or right as positive and sketch the route.
2. Add every path length for distance; subtract start position from finish for displacement.
3. Divide distance and displacement separately by the same total time.
4. Attach a direction to velocity and a unit to every result.

Use this distinction whenever a route includes a turn, a return or a request for both speed and velocity.

04 Worked example

A student walks 30 m east and 10 m west in 20 s.

01 Take east as positive: the two legs are +30 m and -10 m.

02 Distance counts both legs: 30 + 10 = 40 m; displacement is 30 - 10 = +20 m.

03 Over 20 s, average speed is 40/20 = 2 m/s and average velocity is +20/20 = +1 m/s, east.

Watch out

Do not replace displacement with distance in a velocity question.

Exam habit

Write the two numerators before dividing: distance for speed, displacement for velocity. This earns method marks even if arithmetic slips.

05 Your turn

A cart travels 12 m right then 4 m left in 4 s. Find distance, displacement, average speed and average velocity.

If you get stuck, uncover one hint at a time:

1. Sketch the two legs and choose right as positive. Which quantity adds lengths and which adds signed changes?
2. Use 12 m and 4 m positively for distance; use a negative contribution for the leftward leg in displacement.
3. Divide each separate total by 4 s. Keep the direction on velocity and check that speed is nonnegative.

06 Check your understanding

Can average speed be nonzero while average velocity is zero?

Two walkers start and finish together in the same elapsed time but take routes of different lengths. Which averages must match?

1. Their average velocities
2. Their average speeds
3. Both averages

Answers and explanations: page 39.

Constant acceleration and graphs

MH / 02

- Select a constant-acceleration equation from the known quantities.
- Interpret slope and area on a velocity-time graph.

You need: Signed velocity; rectangle and triangle areas; rearranging a simple equation.

01 Idea

Acceleration measures the change of velocity per second. On a position-time graph, slope is velocity. On a velocity-time graph, slope is acceleration and signed area is displacement. The constant-acceleration equations require acceleration to remain constant.

Acceleration tells how much the velocity changes during each second. A velocity-time graph is like a running record of how quickly position accumulates: each thin time strip contributes velocity times time. The slope describes velocity change, whereas the accumulated signed area describes displacement.

- Constant acceleration makes velocity change evenly, so the velocity-time graph is a straight line.
- With positive forward velocity and negative acceleration, the car initially slows because the change opposes its motion.

02 Rule and symbols

Rule

$$v = u + at$$

$$s = ut + \tfrac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

u initial velocity / m/s

v final velocity / m/s

a constant acceleration / m/s^2

s displacement / m

t elapsed time / s

Use this when:

The equations in this lesson assume constant acceleration throughout the stated interval.

03 Method

1. List the known quantities with signs and identify the unknown.
2. Check that the acceleration is constant before using the equations.
3. Choose an equation without an extra unknown, then substitute with units.
4. Use graph slope or area as an independent interpretation of the result.

Look for a constant acceleration, a uniform slowing statement or a straight velocity-time segment.

04 Worked example

A car starts from rest and accelerates at 3 $\mathrm{m/s^2}$ for 4 s.

01 Starts from rest means u = 0; the example gives a = 3 m/s^2 and t = 4 s.

02 The velocity gain is at = 12 m/s, so v = 12 m/s.

03 The displacement is one half times 3 times 4^2 = 24 m.

04 On the velocity-time graph the triangle has base 4 s and height 12 m/s; its area is also 24 m.

Watch out

A straight horizontal velocity-time graph means zero acceleration, not zero velocity.

Exam habit

For a graph, state which axes you used before calculating slope or area; their units identify acceleration or displacement.

05 Your turn

A car slows from 20 to 8 m/s in 4 s with constant acceleration. Find acceleration and displacement.

If you get stuck, uncover one hint at a time:

1. List initial velocity, final velocity and elapsed time. Decide the sign of the velocity change before calculating.
2. Find acceleration from final minus initial velocity divided by 4 s.
3. For displacement use the average of 20 and 8 m/s multiplied by 4 s; this is the area of the trapezoid under the graph.

06 Check your understanding

A car has velocity +8 m/s and acceleration -2 m/s^2. Is it initially speeding up?

A car moves forward while its velocity-time line slopes downward but stays above zero. What happens?

1. It is stationary because acceleration is negative
2. It moves forward and slows down
3. It moves backward and speeds up

Answers and explanations: page 39.

Resolve vectors before adding

MH / 03

- Add perpendicular vectors using signed components.
- Give a resultant direction in the correct quadrant.

You need: Right-triangle trigonometry; Pythagoras; compass directions.

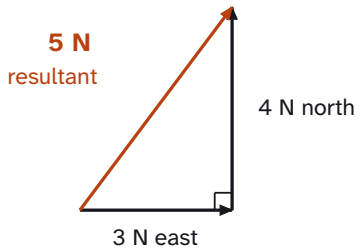
01 Idea

A vector has size and direction. Two vectors must be added through their signed components, not by simply adding magnitudes. The resultant points from the initial tail to the final head in a head-to-tail drawing.

Components are the separate instructions for moving horizontally and vertically. A component pair describes one arrow even when one instruction is negative. Magnitude measures the arrow length, while signs tell which quadrant it occupies; a calculator angle cannot replace that directional information.

FIG. 01 / MECHANICS

Add components to find the resultant



Horizontal: 3 N

Vertical: 4 N

Magnitude: 5 N

Direction: 53.1° north of east

Takeaway: add vectors by direction; do not add 3 + 4 as force magnitudes.

Figure 01. A 3 N east force and a 4 N north force give a 5 N resultant at 53.1° north of east. The force arrows use one common scale; the balancing force points in the opposite direction.

- Perpendicular components form a right triangle, so their squares combine to give magnitude.
- Squaring removes signs, so the quadrant must be recovered from the original component signs.

02 Rule and symbols

Rule

$$A_x = A\cos\theta$$
$$A_y = A\sin\theta$$
$$R = \sqrt{R_x^2 + R_y^2}$$

A_x horizontal vector component / same as vector

A_y vertical vector component / same as vector

R resultant magnitude / same as vector

θ angle from the stated axis / degrees or radians

03 Method

1. Draw axes and identify the reference axis for each angle.
2. Resolve each vector and assign signs to its components.
3. Add horizontal components and vertical components separately.
4. Calculate magnitude and use component signs to choose the direction.

Use components when vectors point in different directions, especially when an angle alone could be ambiguous.

04 Worked example

A 3 N force east and a 4 N force north produce a 5 N resultant, 53.1 degrees north of east.

01 East and north are perpendicular, so the example has Rx = 3 N and Ry = 4 N.

02 The magnitude is the square root of 3^2 + 4^2, giving 5 N.

03 The angle satisfies tan θ = 4/3, so θ = 53.1° north of east.

04 A balancing force reverses both components: 3 N west and 4 N south, with the same 5 N magnitude.

Watch out

The inverse tangent alone can return the wrong quadrant.

Exam habit

Report both an angle and its reference direction; an angle alone can leave the quadrant ambiguous.

05 Your turn

Find the magnitude and direction of a vector with components -3 m and +4 m.

If you get stuck, uncover one hint at a time:

1. Place the horizontal component to the left and the vertical component upward. Identify the quadrant first.
2. Use the right triangle for length and find the acute reference angle from the ratio 4/3.
3. Express that angle north of west, or convert it to a counterclockwise angle from positive x. State both magnitude and direction.

06 Check your understanding

Does a vector with negative horizontal and positive vertical components point northeast?

Two vectors have components (3,4) and (-3,4). What do they share?

1. Equal direction but different magnitude
2. Both magnitude and direction
3. Equal magnitude but different direction

Answers and explanations: page 39.

Projectiles: one time, two directions

MH / 04

- Separate a projectile into horizontal and vertical motions.
- Use the shared flight time to calculate horizontal range.

You need: Constant-acceleration motion; squaring and square roots; horizontal and vertical directions.

01 Idea

Ignoring air resistance, horizontal acceleration is zero and vertical acceleration is downward. Horizontal and vertical motion share the same time. At the top, only the vertical velocity becomes zero; horizontal velocity continues.

Imagine filming a dropped ball and a horizontally launched ball from the side. With equal starting height and no air drag, their vertical motions match. The launched ball also moves sideways, but that sideways motion does not delay gravity; both directions share one clock.

- Gravity changes vertical velocity, while zero horizontal force leaves horizontal velocity constant.
- Landing time is determined by the vertical fall; that same time determines the sideways range.

02 Rule and symbols

Rule

$$v_x = u\cos\theta$$
$$v_y = u\sin\theta - gt$$
$$y = u\sin\theta\, t - \tfrac{1}{2}gt^2$$

u_x initial horizontal velocity / m/s

u_y initial vertical velocity / m/s

g gravitational acceleration magnitude / m/s^2

t flight time / s

Use this when:

Ignore air resistance and take gravity as uniform; launch and landing heights need not match.

03 Method

1. Draw the launch point and select a positive vertical direction.
2. Write horizontal motion with zero acceleration and vertical motion with gravity.
3. Use the height information to find the positive physical flight time.
4. Use this same time in the horizontal displacement equation.

A horizontal launch from a table or cliff signals a vertical free-fall calculation followed by horizontal motion.

04 Worked example

A ball leaves a horizontal table at 3 m/s and takes 0.4 s to land.

01 A horizontal launch has initial vertical velocity zero; its horizontal velocity remains 3 m/s.

02 The stated time is 0.4 s, so horizontal range is 3 times 0.4 = 1.2 m.

03 The vertical drop is one half times 9.8 times 0.4^2 = 0.784 m.

04 The unequal range and drop are consistent because the two directions obey different motion equations.

Watch out

Do not set the total velocity to zero at the highest point.

Exam habit

If you choose upward positive, the downward displacement and acceleration must both carry negative signs.

05 Your turn

A ball is projected horizontally at 5 m/s from a height of 4.9 m. Ignore air resistance and use g = 9.8 $\mathrm{m/s^2}$. Find flight time and horizontal range.

If you get stuck, uncover one hint at a time:

1. Separate the height information from the sideways speed. A horizontal launch has zero initial vertical velocity.
2. Use the fall-distance relation with height 4.9 m and gravitational acceleration 9.8 m/s^2 to find the positive time.
3. Multiply that flight time by 5 m/s for range. Do not put the horizontal speed into the fall equation.

06 Check your understanding

At the highest point of an angled projectile, which velocity component is zero?

Two balls leave the same height at the same time, one dropped and one launched horizontally. Ignoring air drag, which lands first?

1. They land together
2. The horizontally launched ball lands later
3. The dropped ball lands later

Answers and explanations: page 39.

Newton laws and friction

MH / 05

- Construct a free-body diagram for one selected object.
- Use net force and friction to find acceleration.

You need: Signed addition; mass and weight; perpendicular directions.

01 Idea

A free-body diagram shows only forces acting on the selected object. Net force causes acceleration. Action-reaction forces act on different objects and therefore do not cancel on one free-body diagram. Static friction adjusts up to its maximum; kinetic friction applies during sliding.

A free-body diagram is an inventory of pushes and pulls on one object. The object accelerates according to the unbalanced part of that inventory. Friction opposes the sliding direction, and the normal force adjusts to the perpendicular force balance; neither is an additional forward driving force.

- The horizontal pull and kinetic friction oppose, so only their difference produces horizontal acceleration.
- With a horizontal pull and no vertical acceleration, the floor normal balances weight in this particular setup.

02 Rule and symbols

Rule

$$\sum F = ma$$

$$W = mg$$

$$f_k = \mu_k N$$

$$f_s \leq \mu_s N$$

m object mass / kg

N normal contact force / N

μ_k coefficient of kinetic friction / dimensionless

a acceleration / m/s^2

Use this when:

The example treats the block as sliding and the given 4 N as kinetic friction.

03 Method

1. Isolate one object and draw only the forces acting on it.
2. Choose axes and use the perpendicular force balance to obtain the normal force if needed.
3. Choose static or kinetic friction from the stated motion.
4. Add signed forces along the motion and divide the net force by mass.

Use a force diagram before calculating whenever a problem includes a pull, friction, contact or an incline.

04 Worked example

A 2 kg block on a horizontal surface is pulled with 10 N.

01 Choose the 10 N pull as positive; the 4 N sliding friction acts in the opposite direction.

02 The horizontal net force is 10 - 4 = 6 N.

03 For the 2 kg block, a = 6/2 = 3 m/s^2 in the direction of the pull.

04 Weight and normal force balance vertically under the horizontal-surface assumptions, so they do not enter the horizontal sum.

Watch out

The normal force is not always mg: angled pulls and inclines change it.

Exam habit

Show the signed force equation before substitution; do not write the applied force alone as ma.

05 Your turn

A 5 kg block slides to the right on a horizontal floor with coefficient of kinetic friction 0.2. It is pulled horizontally to the right with 20 N. Use g = 10 $\mathrm{m/s^2}$. Find normal force, friction and acceleration.

If you get stuck, uncover one hint at a time:

1. Draw weight, normal force, the rightward pull and leftward sliding friction. Separate vertical from horizontal equations.
2. Use vertical balance to obtain normal force, then multiply it by the kinetic coefficient 0.2.
3. Subtract friction from the 20 N pull and divide the net force by 5 kg. State the acceleration direction.

06 Check your understanding

A sliding block has a 7 N pull and 7 N friction. Must it be at rest?

A block already slides right. Its pull becomes smaller than the leftward kinetic friction. What happens initially?

1. It accelerates right because it is moving right
2. It keeps moving right but slows
3. It immediately moves left

Answers and explanations: page 39.

Circular motion and connected objects

MH / 06

- Identify the real forces that supply inward acceleration.
- Set up separate force equations for connected masses.

You need: Newton second law; force diagrams; uniform circular motion directions.

01 Idea

Changing direction is acceleration even at constant speed. Centripetal means inward; it names the required net force, not an extra physical force. In an ideal taut string over a massless frictionless pulley, connected objects share acceleration magnitude.

Changing direction is a change in velocity even when speed stays fixed. For circular motion, the real forces must supply that inward change. For connected masses, internal string forces can be eliminated by considering the whole system, while separate diagrams are needed to recover the tension.

- Tension acts on both connected masses, but it cancels from the whole-system driving-force calculation.
- The heavier side supplies the weight imbalance; both masses contribute inertia and therefore belong in the total mass.

02 Rule and symbols

Rule

$$a_r = v^2/r$$

$$\sum F_{\text{inward}} = mv^2/r$$

v speed along the circle / m/s

r circle radius / m

a_r inward acceleration / m/s^2

T string tension / N

Use this when:

Connected-object formulas assume a taut massless string and an ideal frictionless pulley when stated.

03 Method

1. For circular motion mark the centre and select inward as a positive radial direction.
2. Sum the actual radial force components and equate them to mv^2/r.
3. For connected objects draw a diagram for each mass and use compatible acceleration directions.
4. Use the ideal string constraint to share acceleration magnitude, then solve for acceleration and tension.

For circular paths look for the inward net force; for a pulley pair use one system equation and one mass equation.

04 Worked example

A 2 kg object moves at 3 m/s in a circle of radius 1.5 m.

01 In the circular example, v = 3 m/s and r = 1.5 m.

02 Squaring speed first gives ar = 9/1.5 = 6 m/s^2, directed toward the centre.

03 Multiplying by the 2 kg mass gives 12 N inward net force.

04 The problem would need a physical setup to decide whether tension, gravity or another force provides those 12 N.

Watch out

Do not add a separate centripetal force to tension or gravity.

Exam habit

Do not put tension and an extra centripetal force on the same diagram; centripetal describes the inward net force.

05 Your turn

An ideal Atwood machine has masses 2 kg and 3 kg. Use g = 10 $\mathrm{m/s^2}$. Find acceleration and tension.

If you get stuck, uncover one hint at a time:

1. Decide which mass descends. Draw one force diagram for each mass without adding any extra driving force.
2. Use the difference of the two weights as the system driving force and the sum of masses as system inertia.
3. After finding acceleration, apply the upward force equation to the rising 2 kg mass to obtain tension; it is not simply either weight.

06 Check your understanding

If the speed remains fixed and radius doubles, what happens to inward acceleration?

For an ideal Atwood machine, why is the total mass used in finding acceleration?

1. Only the heavier mass moves
2. Tension adds extra mass to the system
3. Both connected masses must accelerate

Answers and explanations: page 39.

Work and kinetic energy

MH / 07

- Determine the sign of work from force and displacement directions.
- Calculate net work from a change in kinetic energy.

You need: Cosine of common angles; squaring speed; positive and negative energy transfers.

01 Idea

Work transfers energy when a force has a component along displacement. A perpendicular force does no work. Net work equals the change in kinetic energy. For a varying force along a line, work is the signed area under the force-position graph.

Work measures how a force transfers energy along a displacement. Carrying a force at an angle is like splitting it into a useful along-path component and a sideways component. Net work concerns all forces together, so it is directly linked to the change in speed through kinetic energy.

- Only the component parallel to displacement contributes to work because the perpendicular component has no along-path displacement.
- Kinetic energy depends on speed squared, so equal speed increases at different starting speeds require different net work.

02 Rule and symbols

Rule

$$W = Fd\cos\theta$$

$$K = \tfrac{1}{2}mv^2$$

$$W_{\text{net}} = K_f - K_i$$

W work transferred by a force / J

F force magnitude / N

d displacement magnitude / m

K kinetic energy / J

03 Method

1. Identify the displacement and the angle between it and the force.
2. Use only the force component along displacement to calculate work.
3. Add work from all forces for net work, or use the kinetic-energy change.
4. Check that positive net work corresponds to an increase in speed.

Use the work-energy theorem when speeds and mass are given but the detailed force or time history is unnecessary.

04 Worked example

A 10 N force pulls through 3 m at 60 degrees to displacement.

01 The example states an angle of 60° between the 10 N force and 3 m displacement.

02 Only 10 cos 60° = 5 N acts along the displacement.

03 Work by this force is 5 times 3 = 15 J.

04 The remaining perpendicular component contributes zero work; 15 J is not automatically net work if other forces also do work.

Watch out

Use the force-displacement angle, not an unrelated angle in the drawing.

Exam habit

When using the energy theorem, square each speed before subtracting; $(vf - vi)^2$ is not $vf^2 - vi^2$.

05 Your turn

A 2 kg object speeds up from 3 to 5 m/s. Find the net work done.

If you get stuck, uncover one hint at a time:

1. This question gives mass and two speeds rather than a force and distance. Which energy theorem connects them directly?
2. Calculate final and initial kinetic energies separately, retaining each squared speed.
3. Subtract the initial energy from the final energy. A speed increase should give positive net work in joules.

06 Check your understanding

Can a nonzero force do zero work during a displacement?

The same object goes from speed 1 to 2 m/s, then from 2 to 3 m/s. Which stage requires more net work?

1. The second stage
2. The first stage
3. Both require equal net work

Answers and explanations: page 39.

Energy conservation and springs

MH / 08

- Track kinetic, gravitational and spring energy between two states.
- Identify when a loss term is required in an energy balance.

You need: Work and kinetic energy; choosing a height reference; spring compression or extension.

01 Idea

Energy can move between kinetic, gravitational and elastic stores. Mechanical energy stays constant only when the model excludes losses or other transfers. Choose a height reference; only height changes affect the result.

Energy bookkeeping compares two snapshots rather than following every instant of motion. A compressed spring initially stores energy; when it returns to its natural length, that store can become kinetic energy. The exchange is exact only after accounting for any friction, air drag or other transfer.

- At natural length the ideal spring deformation is zero, so its elastic energy term vanishes.
- On a level frictionless surface gravitational energy does not change, allowing spring energy to become block kinetic energy.

02 Rule and symbols

Rule

$$K_i + U_i + W_{\text{other}} = K_f + U_f$$

$$U_g = mgh$$

$$U_s = \tfrac{1}{2}kx^2$$

U_g gravitational potential energy / J

U_s elastic potential energy / J

k spring stiffness / N/m

x spring deformation from natural length / m

Use this when:

The falling-mass example neglects air resistance and other energy transfers; spring-energy formulas require the proportional linear range.

03 Method

1. Name initial and final states and choose a zero-height level.
2. Write the energy balance before removing terms that are zero.
3. Include friction or other transfers if the model contains them.
4. Solve for the unknown and check the physical sign and units.

Choose an energy balance when initial and final positions or spring deformations are given and time is not needed.

04 Worked example

A mass drops 5 m from rest without losses.

01 Set the final height to zero; the initial height is then 5 m.

02 The object starts from rest, so its initial energy is gravitational: mg times 5.

03 With no losses, mg times 5 equals one half mv^2; dividing by mass shows why mass cancels.

04 Using g = 10 m/s^2 gives v^2 = 100 and speed v = 10 m/s.

Watch out

Spring extension is measured from natural length, not total length.

Exam habit

For spring energy convert centimetres to metres before squaring the deformation.

05 Your turn

A spring of constant 200 N/m is compressed by 0.10 m and releases a 0.50 kg block on a frictionless horizontal surface. Find its speed when the spring returns to natural length.

If you get stuck, uncover one hint at a time:

1. Identify the initial energy store and the final energy store. Is there friction or a change in height?
2. Set the initial spring energy equal to the final block kinetic energy, using the compression in metres.
3. Insert spring constant 200 N/m, compression 0.10 m and mass 0.50 kg. Solve for the positive speed after isolating its square.

06 Check your understanding

Would the same loss-free 5 m fall give a heavier object a larger final speed?

The same compressed spring releases two blocks separately on a frictionless surface. Which block leaves faster?

1. Both have the same speed
2. The lighter block
3. The heavier block

Answers and explanations: page 39.

Temperature, calorimetry and phase changes

MH / 09

- Separate temperature change from phase change in a heating problem.
- Apply an energy balance with consistent mass and temperature units.

You need: Energy in joules; mass conversion; subtraction of temperatures.

01 Idea

Temperature describes thermal state; heat is energy transferred because of a temperature difference. During a phase change of a pure substance at fixed pressure, energy can transfer without a temperature change. In an insulated mixing model, heat lost equals heat gained.

Specific heat tells how much energy a material needs for a temperature rise. Latent heat describes a different job: changing phase. A heating problem becomes manageable when treated as a sequence of stages, each assigned the relation that matches what is physically changing.

- A liquid that warms without a phase change needs energy proportional to its mass and temperature rise.
- The Celsius and kelvin scales have equally sized intervals, so a temperature difference has the same numerical value in both.

02 Rule and symbols

Rule

$$Q = mc\Delta T$$

$$Q = mL$$

$$T_K = T_C + 273.15$$

Q energy transferred as heat / J

c specific heat capacity / J/(kg K)

L specific latent heat / J/kg

ΔT temperature change / K or °C difference

03 Method

1. Identify the initial and final phase and temperature.
2. Split the process into warming, phase-change and further-warming stages where needed.
3. Use mc△T for each warming stage and mL for each phase change.
4. Add the energy requirements or balance heat gained and lost in an insulated mixture.

Use specific heat when temperature changes within one phase; add latent heat only when the process crosses a phase transition.

04 Worked example

Heating 0.20 kg of water by 10 degrees C needs Q = 0.20 times 4200 times 10 = 8400 J.

01 The water remains liquid while warming, so the relevant relation is Q = mc△T.

02 A rise of 10°C is a temperature difference of 10 K.

03 For 0.20 kg and c = 4200 J/(kg K), Q = 0.20 times 4200 times 10 = 8400 J.

04 If the process also melts ice, that extra stage needs mL; it is not included in the 8400 J.

Watch out

A temperature difference of 10 degrees C equals 10 K; an absolute temperature does not.

Exam habit

On a heating curve label sloping segments mc△T and horizontal phase-change segments mL.

05 Your turn

How much energy heats 0.50 kg of water from 20 to 30 degrees C? Use c = 4200 J/(kg K).

If you get stuck, uncover one hint at a time:

1. Check whether the water changes phase. Identify the initial and final temperatures rather than using either as the temperature rise.
2. Subtract 20°C from 30°C and use the given liquid-water specific heat.
3. Multiply mass 0.50 kg, specific heat 4200 J/(kg K) and the temperature difference. Report energy in joules.

06 Check your understanding

Does liquid water warming from 20°C to 30°C have $\Delta T = 283.15$ K?

Two samples of the same liquid have the same temperature rise. One has twice the mass. How do their required heats compare?

1. Both need the same energy
2. The larger sample needs four times the energy
3. The larger sample needs twice the energy

Answers and explanations: page 39.

Heat transfer and thermal expansion

MH / 10

- Classify heat transfer as conduction, convection or radiation.
- Predict rate changes and calculate small linear expansion.

You need: Direct and inverse proportionality; temperature differences; metres and millimetres.

01 Idea

Conduction transfers energy through matter without bulk transport; convection uses fluid motion; radiation uses electromagnetic waves and can cross vacuum. Most solids expand on warming. A thermal insulator reduces transfer but does not create cold.

Thermal expansion is a small fractional change spread over an original length. A longer rod contains more length over which the same fractional change acts. Heat-transfer equations describe energy flow instead; first decide whether the question asks for a size change or a rate of energy transfer.

- For fixed material, expansion grows with both original length and temperature rise.
- The coefficient gives fractional expansion per degree, so its unit is inverse kelvin rather than metres.

02 Rule and symbols

Rule

$$\Delta L = \alpha L_0 \Delta T$$

$$P_{\text{cond}} = kA\Delta T/L$$

$$P_{\text{rad}} = \epsilon\sigma A(T^4 - T_s^4)$$

P heat-transfer rate / W

k thermal conductivity / W/(m K)

A transfer surface area / m^2

L wall thickness / m

α linear expansion coefficient / K^{-1}

Use this when:

The wall comparison uses steady one-dimensional conduction with unchanged material and temperature difference.

03 Method

1. Identify the physical mechanism and select the matching relation.
2. For a comparison, hold every unstated quantity fixed and change only the named factor.
3. For expansion use the original length and the temperature rise.
4. Use kelvin for absolute temperatures in radiation and convert the final length if requested.

Words such as rod length, expansion coefficient and temperature rise indicate a linear-expansion calculation.

04 Worked example

Doubling the thickness of an otherwise unchanged wall halves its conductive heat-transfer rate.

01 For conduction, the same wall material and temperature difference give P proportional to A/L.

02 Doubling thickness replaces L by 2L, so P becomes P/2.

03 Doubling area instead gives 2P.

04 If both area and thickness double together, the factors cancel and the rate is unchanged.

Watch out

Use kelvin in fourth-power radiation equations; Celsius produces incorrect results.

Exam habit

For a ratio question explicitly state what stays constant; otherwise the proportionality is incomplete.

05 Your turn

A rod is 2 m long and has expansion coefficient 0.000012 per K. It warms by 50 K. Find the increase in length in metres and millimetres.

If you get stuck, uncover one hint at a time:

1. Identify original length and temperature rise; the question asks for the increase rather than the final length.
2. Multiply coefficient by original length 2 m and temperature rise 50 K.
3. The product is in metres. Multiply by 1000 to express the same increase in millimetres; do not add the original length unless asked.

06 Check your understanding

A wall is made three times thicker with all other quantities unchanged. What is its new conduction rate?

Two equal-length rods undergo the same temperature rise. The first has twice the expansion coefficient. What follows?

1. The first length increase is twice as large
2. Their length increases must match
3. The first final length is twice as large

Answers and explanations: page 39.

Lesson review and answers

Choose units and describe motion

Return to lesson: page 9.

Your turn: Yes. A return trip has zero displacement but a positive total distance.

Concept check: 1. Correct. Matching endpoints give equal displacement and matching times give equal average velocity.

Written method: Distance = 16 m; displacement = +8 m; speed = 16/4 = 4 m/s; velocity = 8/4 = +2 m/s. Award one mark for each quantity.

Constant acceleration and graphs

Return to lesson: page 12.

Your turn: No. Acceleration opposes its current velocity, so its speed initially decreases.

Concept check: 2. Correct. Positive velocity means forward motion; a decreasing positive velocity means lower speed.

Written method: a = (8-20)/4 = -3 $\mathrm{m/s^2}$. s = [(20+8)/2] times 4 = 56 m. Two marks for each calculation, including the sign and unit.

Resolve vectors before adding

Return to lesson: page 15.

Your turn: No. It points northwest, in quadrant II.

Concept check: 3. Correct. Squared components give equal lengths, while horizontal signs place them in different quadrants.

Written method: Magnitude = 5 m. The vector is in quadrant II: 126.9 degrees counterclockwise from positive x, or 53.1 degrees north of west. Two marks for magnitude and two for a correct direction.

Projectiles: one time, two directions

Return to lesson: page 18.

Your turn: Only the vertical component; horizontal velocity remains constant in this model.

Concept check: 1. Correct. They share initial vertical velocity and vertical acceleration, so their fall times match.

Written method: 4.9 = 0.5 times 9.8 times t^2, so t = 1 s. Range = 5 times 1 = 5 m. Two marks per result with the equation shown.

Newton laws and friction

Return to lesson: page 21.

Your turn: No. Zero net force gives zero acceleration; it may continue moving at constant velocity.

Concept check: 2. Correct. Velocity initially remains rightward, but the leftward net force reduces that velocity.

Written method: N = 50 N, friction = 0.2 times 50 = 10 N, net force = 10 N, a = 10/5 = 2 $\mathrm{m/s^2}$. One mark for each step.

Circular motion and connected objects

Return to lesson: page 24.

Your turn: It halves because v^2/r is inversely proportional to radius at fixed speed.

Concept check: 3. Correct. Both masses share an acceleration magnitude, so both contribute to the system inertia.

Written method: Net driving force on the two-mass system is (3-2) times 10 = 10 N. Total mass = 5 kg, so a = 2 $\mathrm{m/s^2}$. For the rising 2 kg mass, T-20 = 2 times 2, hence T = 24 N. Two marks per result.

Work and kinetic energy

Return to lesson: page 27.

Your turn: Yes. A force perpendicular to the displacement does zero work.

Concept check: 1. Correct. The squared-speed increase is larger during the second stage.

Written method: Net work = 0.5 times 2 times (25-9) = 16 J. Give marks for the energy theorem, both squared speeds, substitution and unit.

Energy conservation and springs

Return to lesson: page 30.

Your turn: No. Mass cancels from mgh = one half mv^2.

Concept check: 2. Correct. The same spring energy gives a larger speed when shared by a smaller mass.

Written method: Spring energy = 0.5 times 200 times 0.10 squared = 1 J. Set 1 = 0.5 times 0.50 times v squared to get v = 2 m/s. Two marks for each stage.

Temperature, calorimetry and phase changes

Return to lesson: page 33.

Your turn: No. ΔT = 10 K. The Celsius-to-kelvin offset cancels when subtracting two temperatures.

Concept check: 3. Correct. With unchanged specific heat and temperature rise, heat is proportional to mass.

Written method: Temperature rise = 10 K. Q = mc times temperature rise = 0.50 times 4200 times 10 = 21000 J. Four marks: model, temperature difference, substitution, result/unit.

Heat transfer and thermal expansion

Return to lesson: page 36.

Your turn: One third of the original rate.

Concept check: 1. Correct. Length increase is proportional to the coefficient with the other factors fixed.

Written method: Increase = 0.000012 times 2 times 50 = 0.0012 m = 1.2 mm. Award marks for formula, substitution, metres and conversion.

Mechanics and Heat / Test A

25 marks / Suggested time: 25 minutes. MCQs: 1 mark each. Written problems: 4 marks each. Attempt without notes.

Part I / Choose one answer

1. A runner completes one lap and returns to the start. Average velocity is:

   A. Always positive

   B. Impossible to define

   C. Zero

   D. Equal to average speed

2. The area under a velocity-time graph gives:

   A. Displacement

   B. Force

   C. Speed at the final instant

   D. Acceleration

3. Two perpendicular forces of 6 N and 8 N have resultant magnitude:

   A. 10 N

   B. 48 N

   C. 2 N

   D. 14 N

4. At the top of a projectile path, acceleration is:

   A. Zero

   B. Horizontal

   C. Upward

   D. Downward with magnitude g

5. An object moving at constant velocity has:

   A. A forward net force

   B. Increasing acceleration

   C. Zero net force

D. No forces acting at all

Part II / Show your reasoning

1. A cart travels 12 m right then 4 m left in 4 s. Find distance, displacement, average speed and average velocity.

2. A car slows from 20 to 8 m/s in 4 s with constant acceleration. Find acceleration and displacement.

3. Find the magnitude and direction of a vector with components -3 m and +4 m.

4. A ball is projected horizontally at 5 m/s from a height of 4.9 m. Ignore air resistance and use g = 9.8 $\mathrm{m/s^2}$. Find flight time and horizontal range.

5. A 5 kg block slides to the right on a horizontal floor with coefficient of kinetic friction 0.2. It is pulled horizontally to the right with 20 N. Use g = 10 $\mathrm{m/s^2}$. Find normal force, friction and acceleration.

Mechanics and Heat / Test B

25 marks / Suggested time: 25 minutes. MCQs: 1 mark each. Written problems: 4 marks each. Attempt without notes.

Part I / Choose one answer

1. Doubling speed at fixed radius changes centripetal acceleration by a factor of:

   A. Two

   B. Four

   C. Eight

   D. One half

2. A force perpendicular to displacement does work equal to:

   A. Fd

   B. Zero

   C. -Fd

   D. Half of Fd

3. If the extension of a Hooke-law spring doubles while k stays constant, its stored energy becomes:

   A. Four times as large

   B. Twice as large

   C. Unchanged

   D. Half as large

4. During melting at fixed pressure, added heat mainly:

   A. Raises temperature continuously

   B. Changes phase

   C. Decreases mass

   D. Disappears

5. Which heat-transfer mechanism can operate through a vacuum?

   A. Fluid mixing

   B. Convection

C. Conduction through air

D. Radiation

Part II / Show your reasoning

1. An ideal Atwood machine has masses 2 kg and 3 kg. Use g = 10 $\mathrm{m/s^2}$. Find acceleration and tension.

2. A 2 kg object speeds up from 3 to 5 m/s. Find the net work done.

3. A spring of constant 200 N/m is compressed by 0.10 m and releases a 0.50 kg block on a frictionless horizontal surface. Find its speed when the spring returns to natural length.

4. How much energy heats 0.50 kg of water from 20 to 30 degrees C? Use c = 4200 J/(kg K).

5. A rod is 2 m long and has expansion coefficient 0.000012 per K. It warms by 50 K. Find the increase in length in metres and millimetres.

Mechanics and Heat / Test A / Answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1

MCQ: C. The net displacement is zero even though the distance travelled is not.

Written solution: Distance = 16 m; displacement = +8 m; speed = 16/4 = 4 m/s; velocity = 8/4 = +2 m/s. Award one mark for each quantity.

Return to lesson: page 9.

Question 2

MCQ: A. Area has units (m/s) times s = m and includes the sign of velocity.

Written solution: a = (8-20)/4 = -3 $\mathrm{m/s^2}$. s = [(20+8)/2] times 4 = 56 m. Two marks for each calculation, including the sign and unit.

Return to lesson: page 12.

Question 3

MCQ: A. The Pythagorean calculation gives the square root of 36+64 = 10 N.

Written solution: Magnitude = 5 m. The vector is in quadrant II: 126.9 degrees counterclockwise from positive x, or 53.1 degrees north of west. Two marks for magnitude and two for a correct direction.

Return to lesson: page 15.

Question 4

MCQ: D. Gravity continues to act at the top.

Written solution: 4.9 = 0.5 times 9.8 times t^2, so t = 1 s. Range = 5 times 1 = 5 m. Two marks per result with the equation shown.

Return to lesson: page 18.

Question 5

MCQ: C. Constant velocity means zero acceleration, so forces balance.

Written solution: N = 50 N, friction = 0.2 times 50 = 10 N, net force = 10 N, a = 10/5 = 2 $\mathrm{m/s^2}$. One mark for each step.

Return to lesson: page 21.

Mechanics and Heat / Test B / Answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1

MCQ: B. The speed is squared.

Written solution: Net driving force on the two-mass system is (3-2) times 10 = 10 N. Total mass = 5 kg, so a = 2 $\mathrm{m/s^2}$. For the rising 2 kg mass, T-20 = 2 times 2, hence T = 24 N. Two marks per result.

Return to lesson: page 24.

Question 2

MCQ: B. The cosine of 90 degrees is zero.

Written solution: Net work = 0.5 times 2 times (25-9) = 16 J. Give marks for the energy theorem, both squared speeds, substitution and unit.

Return to lesson: page 27.

Question 3

MCQ: A. Spring energy is proportional to extension squared.

Written solution: Spring energy = 0.5 times 200 times 0.10 squared = 1 J. Set 1 = 0.5 times 0.50 times v squared to get v = 2 m/s. Two marks for each stage.

Return to lesson: page 30.

Question 4

MCQ: B. Energy changes the phase while temperature remains at the melting point.

Written solution: Temperature rise = 10 K. Q = mc times temperature rise = 0.50 times 4200 times 10 = 21000 J. Four marks: model, temperature difference, substitution, result/unit.

Return to lesson: page 33.

Question 5

MCQ: D. Electromagnetic radiation needs no material medium.

Written solution: Increase = 0.000012 times 2 times 50 = 0.0012 m = 1.2 mm. Award marks for formula, substitution, metres and conversion.

Return to lesson: page 36.

Formula sheet

Choose units and describe motion

p.9

Rule

$$\bar{v} = \Delta x / \Delta t$$

$$\text{average speed} = \text{distance} / \Delta t$$

Constant acceleration and graphs

p.12

Rule

$$v = u + at$$

$$s = ut + \tfrac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

Resolve vectors before adding

p.15

Rule

$$A_x = A\cos\theta$$

$$A_y = A\sin\theta$$

$$R = \sqrt{R_x^2 + R_y^2}$$

Projectiles: one time, two directions

p.18

Rule

$$v_x = u\cos\theta$$

$$v_y = u\sin\theta - gt$$

$$y = u\sin\theta\, t - \tfrac{1}{2}gt^2$$

Newton laws and friction

p.21

Rule

$$\sum F = ma$$

$$W = mg$$

$$f_k = \mu_k N$$

$$f_s \le \mu_s N$$

Circular motion and connected objects

p.24

Rule

$$a_r = v^2/r$$

$$\sum F_{\text{inward}} = mv^2/r$$

Work and kinetic energy

p.27

Rule

$$W = Fd\cos\theta$$

$$K = \tfrac{1}{2}mv^2$$

$$W_{\text{net}} = K_f - K_i$$

Energy conservation and springs

p.30

Rule

$$K_i + U_i + W_{\text{other}} = K_f + U_f$$

$$U_g = mgh$$

$$U_s = \tfrac{1}{2}kx^2$$

Temperature, calorimetry and phase changes

p.33

Rule

$$Q = mc\Delta T$$

$$Q = mL$$

$$T_K = T_C + 273.15$$

Heat transfer and thermal expansion

p.36

Rule

$$\Delta L = \alpha L_0 \Delta T$$

$$P_{\text{cond}} = kA\Delta T/L$$

$$P_{\text{rad}} = \epsilon\sigma A(T^4 - T_s^4)$$

Symbol glossary

Symbols are listed with their lesson context. A symbol may represent different quantities in different rules.

Choose units and describe motion

Δx final position minus initial position / m

Δt elapsed time / s

$\bar{v}$ average velocity / m/s

Constant acceleration and graphs

u initial velocity / m/s

v final velocity / m/s

a constant acceleration / m/s^2

s displacement / m

t elapsed time / s

Resolve vectors before adding

A_x horizontal vector component / same as vector

A_y vertical vector component / same as vector

R resultant magnitude / same as vector

θ angle from the stated axis / degrees or radians

Projectiles: one time, two directions

u_x initial horizontal velocity / m/s

u_y initial vertical velocity / m/s

g gravitational acceleration magnitude / m/s^2

t flight time / s

Newton laws and friction

m object mass / kg

N normal contact force / N

μ_k coefficient of kinetic friction / dimensionless

a acceleration / m/s^2

Circular motion and connected objects

v speed along the circle / m/s

r circle radius / m

a_r inward acceleration / m/s^2

T string tension / N

Work and kinetic energy

W work transferred by a force / J

F force magnitude / N

d displacement magnitude / m

K kinetic energy / J

Energy conservation and springs

U_g gravitational potential energy / J

U_s elastic potential energy / J

k spring stiffness / N/m

x spring deformation from natural length / m

Temperature, calorimetry and phase changes

Q energy transferred as heat / J

c specific heat capacity / J/(kg K)

L specific latent heat / J/kg

ΔT temperature change / K or °C difference

Heat transfer and thermal expansion

P heat-transfer rate / W

k thermal conductivity / W/(m K)

A transfer surface area / m^2

L wall thickness / m

α linear expansion coefficient / K^{-1}

Source and coverage notes

Motion, vectors, forces, work, energy, temperature and heat. New merged lectures are the main mechanics source; the older heat lecture supplies thermal topics.

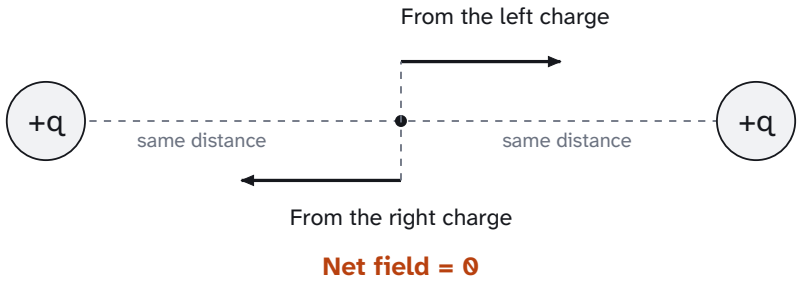
Open original course folder

The source audit inventories 159 non-programming files; 95 yielded substantial extracted text. Scanned diagrams and handwritten or image-only material were not fully transcribed. Check lecturer-specific derivations and conventions. The practice tests are original study material, not predictions of an examination.

Electricity and Magnetism

FIG. 02 / ELECTRICITY

Equal source fields cancel at the midpoint



Takeaway: equal field magnitudes cancel only when their directions oppose.

Concept-first coverage of the course plan. Chapters 23-31 have lecture support; inductance, AC and EM waves are course-plan extensions without separate lecture files in this folder. Advanced field integrals are recognition topics, not worked derivations.

Subject map

Concept-first coverage of the course plan. Chapters 23-31 have lecture support; inductance, AC and EM waves are course-plan extensions without separate lecture files in this folder. Advanced field integrals are recognition topics, not worked derivations.

Follow this reading route from foundations to applications. Each line links to its lesson; revisit the stated prerequisites whenever a step feels unfamiliar.

01 Charges, conductors and Coulomb force p.62

02 Electric field and superposition p.65

03 Flux and Gauss law p.69

04 Symmetry and conductors p.72

05 Potential is not field p.75

06 Capacitors and dielectrics p.78

07 Current, resistance and electrical power p.82

08 DC networks, Kirchhoff rules and RC behaviour p.85

09 Magnetic force: direction before calculation p.90

10 Sources of magnetic fields p.93

11 Induction and Lenz law p.96

12 Inductance, AC and electromagnetic waves p.99

Charges, conductors and Coulomb force

EM / 01

- Predict attraction or repulsion from charge signs.
- Use simple Coulomb-force ratios without unnecessary arithmetic.

You need: Positive and negative charge; squares; micro means one millionth.

01 Idea

Charge is conserved and occurs in integer multiples of the elementary charge. Ordinary charging transfers electrons. Like charges repel and unlike charges attract. Conductors let charge move readily; insulators restrict it. A neutral object can still be attracted by polarization.

Coulomb-force ratios let you compare two situations without knowing every charge value. Imagine moving the same two charges farther apart: the effect spreads according to the square of separation. Charge signs tell attraction or repulsion, while charge sizes and separation set the magnitude.

- Doubling separation squares the distance factor, making the force one quarter as large.
- Changing only one charge magnitude scales the force directly without changing the separation factor.

02 Rule and symbols

Rule

$$F = k|q_1 q_2|/r^2$$

$$k = 8.99 \times 10^9 \ \mathrm{N\,m^2/C^2}$$

F electrostatic force magnitude / N

q_1, q_2 source charges / C

r separation of charges / m

Use this when:

Coulomb calculations here use point charges, or a stated model equivalent to point charges.

03 Method

1. Mark the signs and positions of both charges.
2. Predict attraction or repulsion before calculating magnitude.
3. For a ratio, compare the charge product and the square of the separation.
4. For a numerical force, convert charges and lengths to SI units first.

Use a ratio when a force is already known and only a charge or separation changes.

04 Worked example

Two charges at distance r exert force F.

01 Keep both charges fixed and replace the separation r by 2r.

02 The denominator becomes $(2r)^2 = 4r^2$, so the force becomes F/4.

03 If instead one charge doubles at the original separation, its product with the other charge doubles, so force becomes 2F.

04 Changing separation or charge magnitude alone does not change attraction into repulsion.

Watch out

A negative charge does not mean a negative force magnitude.

Exam habit

A magnitude answer is positive; state attraction or repulsion separately to communicate direction.

05 Your turn

Two charges exert 12 N at separation r. What is the force at separation 2r, and at r if one charge is tripled?

If you get stuck, uncover one hint at a time:

1. Treat the two requested situations separately, each starting from the original 12 N force.
2. For doubled separation, square the separation-change factor before applying it to the force.
3. For one charge tripled at the original separation, apply only the charge-change factor. State both results with newtons.

06 Check your understanding

Tripling separation with fixed charges changes the force by what factor?

Two unlike charges are moved farther apart without changing their signs. What changes?

1. The attraction becomes weaker
2. Attraction becomes repulsion
3. The force keeps the same magnitude

Answers and explanations: page 102.

Electric field and superposition

EM / 02

- Combine electric fields by their directions at an observation point.
- Explain why a negative test charge feels force opposite the field.

You need: Vector addition; positive and negative charges; multiplication.

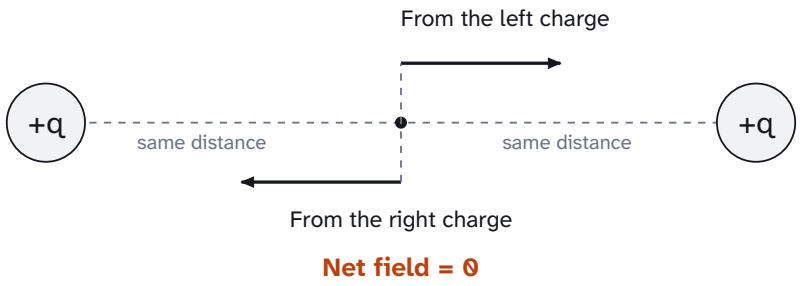
01 Idea

The electric field gives force per unit positive test charge. Field arrows point away from positive sources and toward negative sources. The field exists whether or not a test charge is present. Add fields as vectors, so equal magnitudes can cancel.

A field is a direction-and-strength instruction already present at a location. Its arrow says how a positive test charge would be pushed. An actual negative charge reads the same instruction with the direction reversed, so finding a field and finding a charge force are separate steps.

FIG. 02 / ELECTRICITY

Equal source fields cancel at the midpoint



Takeaway: equal field magnitudes cancel only when their directions oppose.

Figure 02. Equal positive charges give zero net field at the midpoint, so the force on a test charge there is zero. The two field arrows are offset vertically only for clarity; both represent fields at the midpoint.

- Multiplying by a negative charge reverses direction but does not make the force magnitude negative.
- The field is defined by the source arrangement and does not require the test charge to be present.

02 Rule and symbols

Rule

$$E = F/q_{\text{test}}$$
$$E = k|Q|/r^2$$
$$\vec{F} = q\vec{E}$$

E electric field magnitude / N/C

q charge experiencing the force / C

F force magnitude / N

03 Method

1. Choose the observation point and draw the field arrow from each source there.
2. Point away from a positive source and toward a negative source.
3. Add signed collinear fields or use vector components when needed.
4. Only after finding the net field, use the test-charge sign to obtain force direction.

Separate field direction from force direction whenever the question supplies a negative charge.

04 Worked example

At the midpoint between equal positive charges, the two electric fields oppose and cancel.

01 At the midpoint between equal positive charges, both source distances are the same, so field magnitudes match.

02 The left positive charge points the field right; the right positive charge points it left. The net field is zero.

03 Replace the right source by an equal negative charge: its field now points toward it, also to the right.

04 The two fields therefore reinforce, pointing from the positive source toward the negative source.

Watch out

Zero net field at a point does not mean each source field is zero.

Exam habit

Draw field arrows before multiplying by a test charge; otherwise a negative test

charge can lead to two direction reversals.

05 Your turn

A field of 200 N/C points east. Find the force on a charge of -0.003 C.

If you get stuck, uncover one hint at a time:

1. The field direction is east. Decide the force direction using the sign of the supplied charge before calculating.
2. Multiply the field magnitude 200 N/C by the charge magnitude 0.003 C.
3. State the positive force magnitude in newtons and its compass direction. The minus sign belongs to the directional reasoning, not a magnitude.

06 Check your understanding

If the net field is zero at a point, is a charge placed there forced in either direction by that field?

Two small test charges, one positive and one negative with equal magnitudes, are placed separately at the same point in a nonzero electric field. Their forces have:

1. Different magnitudes solely because their signs differ
2. Equal magnitudes and opposite directions
3. Equal magnitudes and the same direction

Answers and explanations: page 102.

Flux and Gauss law

EM / 03

- Use the surface normal to determine electric flux.
- Distinguish enclosed charge from outside charge in Gauss law.

You need: Surface area; normal means perpendicular; cosine at 0° and 90°.

01 Idea

Electric flux measures the field passing through an oriented surface. The angle is measured from the surface normal, not the surface itself. Net flux through a closed surface depends on enclosed charge. An outside charge can affect the field at the surface but contributes zero net closed-surface flux.

Flux counts the field crossing a surface rather than the field lying along it. Imagine rain falling through a hoop: turning the hoop changes how much rain crosses its opening even when the rain itself stays the same. A surface normal records that orientation.

- A field parallel to the normal crosses the full effective area.
- A field tangent to the surface has no crossing component, even if its magnitude is large.

02 Rule and symbols

Rule

$$\Phi = EA\cos\theta$$

$$\Phi_{\text{closed}} = Q_{\text{enclosed}}/\epsilon_0$$

Φ electric flux / N m^2/C

E electric field magnitude / N/C

A surface area / m^2

Q_{enclosed} net charge inside a closed surface / C

03 Method

1. Decide whether the question concerns one flat surface or a complete closed surface.
2. For a flat surface, draw the normal and identify its angle to the field.
3. For a closed surface, add only the enclosed charges with signs.
4. Interpret positive closed flux as net outward flux and negative as net inward flux.

Use flat-surface flux when area, a uniform field and orientation are given; use enclosed charge for net closed-surface flux.

04 Worked example

A cube enclosing +q has net outward flux $\frac{q}{\varepsilon_0}$.

01 The cube is a closed surface enclosing net charge +q.

02 Gauss law fixes its total outward flux as q divided by ϵ0.

03 Moving q off-centre leaves enclosed charge unchanged, so total flux is unchanged.

04 The nearby faces can now have stronger local fields; Gauss law does not say each face receives an equal share after the move.

Watch out

Zero net closed-surface flux does not imply zero field everywhere.

Exam habit

The angle in EA cos θ is to the normal; a field parallel to the surface gives zero flux.

05 Your turn

A uniform field of 50 N/C passes through a flat area of 0.20 square metres. The chosen surface normal points in the same direction as the field. Find flux. What if the normal is perpendicular to the field?

If you get stuck, uncover one hint at a time:

1. Draw the chosen normal as an arrow. In the first case it points with the field, so identify their angle.
2. Multiply field 50 N/C by area 0.20 m^2 and the cosine of that normal angle.
3. For the second case change only the cosine factor. Keep flux units distinct from field units.

06 Check your understanding

Can an external charge produce a nonzero field on a box while contributing zero net flux through it?

A flat surface rotates from normal parallel to the field to field tangent to the surface. What happens to flux magnitude?

1. It stays constant because field strength is unchanged
2. It increases because the surface turns
3. It falls from its maximum to zero

Answers and explanations: page 102.

Symmetry and conductors

EM / 04

- State the electrostatic properties of conducting material.
- Use the inverse-square field rule outside a spherical conductor.

You need: Electric field; conductor versus insulator; inverse-square ratios.

01 Idea

Gauss law is always valid, but it easily gives field magnitude only when symmetry makes the surface calculation simple. In electrostatic equilibrium the field inside conducting material is zero and excess charge lies on its surface. A conductor is an equipotential. For the outside-sphere inverse-square rule, assume an isolated conducting sphere with no external fields.

On an isolated conducting sphere, charge can rearrange until the conductor is in electrostatic equilibrium. Outside this symmetric arrangement, its field follows the same distance pattern as one central point charge. The relevant distance therefore starts at the centre, not at the surface.

- Spherical symmetry makes all outside points at the same centre distance equivalent.
- The inverse-square outside field weakens by a factor of four when centre distance doubles.

02 Rule and symbols

Rule

$$E_{\text{outside sphere}} = k|Q|/r^2$$
$$(r > R)$$

Q net sphere charge / C

R sphere radius / m

r distance from sphere centre / m

E electric field magnitude / N/C

Use this when:

The sphere is isolated with spherical symmetry and is in electrostatic equilibrium.

03 Method

1. Check that the conductor is in electrostatic equilibrium.
2. Distinguish a point in the conducting material from a point outside the sphere.
3. For an outside point measure distance from the centre, not from the surface.
4. Use the point-charge field or a simple ratio, then describe the radial direction.

Use the outside-sphere comparison only when the sphere is isolated as stated and both points lie outside it.

04 Worked example

A charged conducting sphere behaves outside like a point charge at its centre.

01 Outside a spherically symmetric charged conducting sphere, the field has the same form as a charge at its centre.

02 Compare two outside distances r and 2r.

03 The square of the distance grows by a factor of four, so the field magnitude becomes E/4.

04 This comparison uses centre distances; doubling the gap from the surface would generally give a different ratio.

Watch out

Do not extend the zero-field-in-metal statement to every cavity containing a charge.

Exam habit

Write r > R beside the outside-sphere formula to show that its domain has been checked.

05 Your turn

An isolated charged conducting sphere, with no external fields, has field magnitude 80 N/C at r = 0.20 m. Find the field at r = 0.40 m. Both distances are from the centre and outside the sphere.

If you get stuck, uncover one hint at a time:

1. Check that both stated distances start at the centre and exceed the sphere radius as specified.
2. Compare 0.40 m with 0.20 m and square that distance ratio.
3. Divide the original 80 N/C by the squared distance factor. No unknown total charge is needed.

06 Check your understanding

A field is 80 N/C at 0.20 m from the centre. What is it at 0.40 m if both points are outside?

Why can doubling the gap above a sphere surface differ from doubling the distance used in its field formula?

1. The formula measures distance from the centre
2. The formula always uses distance from the nearest charge on the surface
3. Outside fields never depend on distance

Answers and explanations: page 102.

Potential is not field

EM / 05

- Add scalar potentials with their charge signs.
- Relate potential difference to energy change and field work.

You need: Scalar versus vector; signed energy changes; potential energy.

01 Idea

Electric potential is energy per unit charge, a scalar. Electric field is a vector. Add potentials with charge signs, but add fields with directions. Work by the electric field equals the decrease in electric potential energy.

Potential is a location-based energy-per-charge value, rather like height in a gravitational comparison. A change in potential tells how the potential energy of a particular charge changes. The electric field work has the opposite sign because energy given to kinetic motion comes from a decrease in potential energy.

- A positive charge moving to higher potential gains electric potential energy.
- Positive potential-energy gain means the electric field does negative work over that move.

02 Rule and symbols

Rule

$$V = kQ/r \text{ (zero at infinity)}$$

$$\Delta U = q\Delta V$$

$$W_{\text{field}} = -\Delta U$$

V electric potential / V = J/C

ΔV final minus initial potential / V

ΔU change in potential energy / J

q moving charge / C

03 Method

1. Separate a field question, requiring direction, from a potential question, requiring signed scalar addition.
2. Choose the potential reference supplied by the problem.
3. Calculate energy change from charge times final-minus-initial potential.
4. Reverse the sign of that energy change to obtain work by the electric field.

When a question gives a charge and a potential difference, use the energy change directly rather than inventing a field or travel distance.

04 Worked example

Equal positive charges can give zero field at their midpoint but positive potential there.

01 At the midpoint of two equal positive charges, equal field arrows point in opposite directions and cancel.

02 Each positive charge contributes a positive potential at that point when zero potential is chosen at infinity.

03 The two potentials add rather than cancel because potential has no directional arrow.

04 Thus zero net field and positive potential can occur at the same point without contradiction.

Watch out

A charge moving to lower potential does not always lose potential energy; a negative charge does the opposite.

Exam habit

Write ΔV = Vfinal - Vinitial before calculating; swapping the endpoints reverses both energy and work signs.

05 Your turn

A +2 C charge moves through a potential difference of +3 V. Find its potential-energy change and the work done by the field.

If you get stuck, uncover one hint at a time:

1. Treat the stated +3 V as final potential minus initial potential. Identify the moving charge sign.
2. Multiply charge +2 C by the potential difference to obtain the signed potential-energy change.
3. Reverse that energy-change sign for work done by the field. Report each quantity in joules and label them separately.

06 Check your understanding

A +2 C charge moves through $\Delta V = +3$ V. What are ΔU and work by the field?

A negative charge moves through a positive potential difference. What is the sign of its potential-energy change?

1. Always zero
2. Negative
3. Positive

Answers and explanations: page 102.

Capacitors and dielectrics

EM / 06

- Distinguish series and parallel capacitor connections.
- Calculate stored charge and identify the effect of a dielectric.

You need: Potential difference; charge; reciprocals and addition.

01 Idea

A capacitor stores separated charge and electric-field energy. Capacitance describes charge stored per volt. Parallel capacitors share voltage and their capacitances add. Series capacitors carry equal charge magnitude and have a smaller equivalent capacitance.

Capacitance describes how much separated charge corresponds to each volt across two plates. A capacitor with a larger capacitance can store more charge at the same voltage. It is the connection to a battery or isolation of the capacitor that decides which quantity stays fixed during a change.

- At fixed voltage, increasing capacitance increases stored charge.
- A microfarad multiplied by a volt gives a microcoulomb, so compatible prefixes can simplify a calculation.

02 Rule and symbols

Rule

$$Q = CV$$

$$U = \tfrac{1}{2}CV^2$$

$$C_{\parallel} = C_1 + C_2$$

$$1/C_s = 1/C_1 + 1/C_2$$

C capacitance / F

Q stored charge magnitude on either plate / C

V potential difference between plates / V

U stored electric energy / J

03 Method

1. Identify shared nodes: parallel capacitors share voltage, while ideal series capacitors carry equal charge magnitudes.
2. Add capacitances for parallel or add their reciprocals for series.
3. Use Q = CV with compatible prefixes such as μF times V giving μC.
4. For dielectric questions first check whether voltage or charge is held fixed.

Use the charge-capacitance-voltage relation when any two of those three quantities are supplied.

04 Worked example

Two 6 μF capacitors give 12 μF in parallel and 3 μF in series.

01 For two 6 μF capacitors in parallel, capacitances add: 6 + 6 = 12 μF.

02 For series, 1/Cs = 1/6 + 1/6 = 1/3 in reciprocal μF, so Cs = 3 μF.

03 The series result is smaller than either individual capacitor, providing a quick plausibility check.

04 Inserting a suitable dielectric increases capacitance; the resulting charge or voltage change depends on the connection to a battery.

Watch out

Do not use resistor series/parallel rules for capacitors.

Exam habit

For series, invert the reciprocal sum at the end; stopping at 1/C is a common lost result mark.

05 Your turn

A 4 μF capacitor is connected to 12 V. Find the charge in microcoulombs.

If you get stuck, uncover one hint at a time:

1. The capacitor is connected across 12 V. Identify whether charge follows by multiplying or dividing capacitance and voltage.
2. Use capacitance 4 microfarads and the supplied voltage.
3. A result from microfarads times volts is in microcoulombs. If using farads instead, convert the final coulombs back consistently.

06 Check your understanding

What charge magnitude is stored by 4 μF at 12 V?

A capacitor stays connected to an ideal battery while its capacitance increases. What happens?

1. Charge stays fixed and voltage increases
2. Both voltage and charge must stay fixed
3. Voltage stays fixed and stored charge increases

Answers and explanations: page 102.

What changes a capacitor’s capacitance?

These simple comparisons cover the geometry and dielectric ideas in the lecture without a derivation.

- Use the first formula for parallel plates with a small gap compared with plate size, so edge effects can be ignored.
- Here A is overlapping plate area, d is separation, and ε describes the material filling the gap.
- Doubling area doubles capacitance. Doubling separation halves it.

- A dielectric filling the gap increases capacitance. Battery attached: voltage stays fixed. Disconnected and isolated: charge stays fixed.
- $C = \dfrac{\varepsilon A}{d}$
- $Q = CV$

An isolated charged capacitor has capacitance $2\,\mu\mathrm{F}$ and voltage $12\,\mathrm{V}$. A dielectric triples its capacitance. Find the new charge and voltage.

1. Initial charge: $Q = CV = 2 \times 12 = 24\,\mu\mathrm{C}$.
2. Isolation keeps the charge at $24\,\mu\mathrm{C}$.
3. New capacitance: $6\,\mu\mathrm{F}$; new voltage: $V = 24/6 = 4\,\mathrm{V}$.

Charge stays $24\,\mu\mathrm{C}$; voltage falls to $4\,\mathrm{V}$.

Current, resistance and electrical power

EM / 07

- Find current, voltage and power for an ohmic resistor.
- Predict how wire length and area change resistance.

You need: Charge flow; multiplication and division; circuit voltage.

01 Idea

Current is the rate of charge flow. Conventional current follows positive-charge flow, opposite electron drift in a metal. Resistance depends on material and geometry. Ohm law describes an ohmic element under fixed physical conditions.

Voltage is the energy transferred per unit charge, while current counts charge passing each second. Multiplying them gives energy per second, which is power. Resistance connects the current through a resistor with the voltage across that same resistor; those locations matter in a circuit.

- For a given resistor current, the voltage drop follows its resistance.
- Power follows both the amount of charge passing each second and the energy transferred by each unit charge.

02 Rule and symbols

Rule

$$I = \Delta Q/\Delta t$$

$$V = IR$$

$$R = \rho L/A$$

$$P = VI$$

I conventional current / A

R resistance / Ω

V potential difference / V

P electrical power / W

ρ resistivity / Ω m

Use this when:

Ohm law is used for an ohmic resistor under unchanged physical conditions.

03 Method

1. Label voltage across the resistor and current through it.
2. Use V = IR to find the missing electrical quantity.
3. Calculate power using VI or an equivalent resistor relation.
4. For wire comparisons hold material and temperature fixed before changing length or area.

When current and resistance are supplied, find the resistor voltage first, then use voltage times current for power.

04 Worked example

A 6 Ω resistor across 12 V carries 2 A and dissipates 24 W.

01 The 6 Ω resistor has 12 V across it, so I = 12/6 = 2 A.

02 Its electrical power is VI = 12 times 2 = 24 W.

03 Power of 24 W means 24 joules of energy are transferred each second.

04 A longer wire with unchanged material, area and temperature has larger resistance because R is proportional to length.

Watch out

More resistance lowers power at fixed voltage but raises power at fixed current.

Exam habit

Do not confuse power in watts with energy in joules; multiply power by time when energy is requested.

05 Your turn

A 3 Ω resistor carries 2 A. Find voltage and power.

If you get stuck, uncover one hint at a time:

1. Mark the given 2 A as current through the 3-ohm resistor. Which relation gives its voltage drop?
2. Multiply current by resistance to obtain voltage.
3. Use that voltage with the same current to obtain power. Check that the units are volts for the first result and watts for the second.

06 Check your understanding

What happens to resistance if a wire length doubles but material, area and temperature stay fixed?

The current through the same fixed resistor doubles. What happens to its power?

1. It becomes four times as large
2. It doubles
3. It stays fixed because resistance stays fixed

Answers and explanations: page 102.

DC networks, Kirchhoff rules and RC behaviour

EM / 08

- Reduce simple resistor networks and recover branch currents.
- Describe the initial and final behaviour of an RC charging circuit.

You need: Ohm law; series and parallel connections; charge and energy conservation.

01 Idea

Series resistors share current; parallel resistors share voltage. Kirchhoff junction rules express charge conservation, and loop rules express energy conservation. An initially uncharged capacitor charges gradually through a resistor and eventually blocks steady DC in the ideal model.

In a series path, charge has no branch in which to split, so the same current passes through each resistor. The source voltage is shared as energy drops along the path. A larger resistance takes a larger voltage share because the current is common.

- Series resistances add because their voltage drops add at the same current.
- Checking that the individual drops sum to the source voltage tests the energy bookkeeping.

02 Rule and symbols

Rule

$$R_s = R_1 + R_2$$

$$1/R_p = 1/R_1 + 1/R_2$$

$$\tau = RC$$

R_{eq} equivalent resistance / Ω

I current / A

V voltage / V

τ RC time constant / s

03 Method

1. Identify series elements by shared current and parallel elements by shared voltage.
2. Reduce the network and calculate total source current.
3. Return to the original circuit to calculate branch currents or voltage drops.
4. Check junction current sums and loop voltage sums; for RC questions distinguish transient from long-time behaviour.

Use series reduction when every charge passing one element must pass the next without a junction.

04 Worked example

Two 6 Ω resistors in parallel give 3 Ω.

01 Both 6 Ω resistors are connected across the same two nodes, so they are parallel.

02 Their equal-resistor equivalent is 6/2 = 3 Ω.

03 The 12 V source therefore supplies 12/3 = 4 A total.

04 Each branch still has 12 V and carries 12/6 = 2 A; the branch currents sum to the 4 A source current.

Watch out

Adjacent components in a drawing are not necessarily in series.

Exam habit

After finding an equivalent resistance, redraw or relabel the original circuit before assigning individual voltage drops.

05 Your turn

A 4 Ω and 8 Ω resistor are in series across 24 V. Find total current and each voltage drop.

If you get stuck, uncover one hint at a time:

1. Confirm there is one series path. Add the two resistances before finding source current.
2. Divide the 24 V source voltage by the total resistance.
3. Use that same current with each resistance separately. Verify that the two voltage drops add to 24 V.

06 Check your understanding

After a long time, does an ideal capacitor in a DC series RC circuit carry a steady current?

A 4-ohm resistor and an 8-ohm resistor carry the same series current. Which has the larger voltage drop?

1. The drops must be equal because current is equal
2. The 8-ohm resistor
3. The 4-ohm resistor

Answers and explanations: page 102.

One time constant: the 63% and 37% rules

Recognize how quickly a capacitor charges or discharges without deriving an exponential.

- Use an ideal series resistor-capacitor circuit with constant R and C. The time constant τ is measured in seconds.
- Charging from zero: after one time constant, capacitor voltage and charge reach about 63% of their final values.
- Discharging: after one time constant, about 37% of the starting voltage and charge remain.
- A larger resistance or capacitance makes the change slower.
- $\tau = RC$
- $1 - e^{-1} \approx 0.632$
- $e^{-1} \approx 0.368$

A capacitor charges from zero toward $10\,\mathrm{V}$. Its time constant is $2\,\mathrm{s}$. Estimate its voltage after $2\,\mathrm{s}$.

1. The elapsed time is one time constant.
2. Multiply final voltage by 0.632: $10 \times 0.632 = 6.32\,\mathrm{V}$.

About $6.3\,\mathrm{V}$. It is not fully charged after one time constant.

Kirchhoff signs and a real battery

Choosing consistent signs is the main skill. A negative current answer tells you the actual direction.

- At a junction, charge does not keep building up: add entering and leaving currents separately.
- Choose a loop direction. Across a resistor, travelling with the assumed current gives $-IR$; travelling against it gives $+IR$.
- Across a battery, going from its negative to positive terminal gives $+\mathcal{E}$; the reverse gives $-\mathcal{E}$.
- For a battery supplying current, internal resistance r causes the terminal voltage to be less than its emf $\mathcal{E}$.
- current in $=$ current out
- $\sum \Delta V = 0$

- $V_{\text{terminal}} = \mathcal{E} - Ir$ (battery delivering current)

A battery has emf $12\,\text{V}$ and internal resistance $1\,\Omega$. It supplies a $5\,\Omega$ resistor. Find current and terminal voltage.

1. Use one loop: $12 - I(1) - I(5) = 0$.
2. Total resistance is $6\,\Omega$, so $I = 12/6 = 2\,\text{A}$.
3. Terminal voltage is $12 - (2)(1) = 10\,\text{V}$. Check the external resistor: $IR = (2)(5) = 10\,\text{V}$.

Current $2\,\text{A}$; terminal voltage $10\,\text{V}$.

Magnetic force: direction before calculation

EM / 09

- Determine when magnetic force vanishes or is largest.
- Explain circular motion without a change in kinetic energy.

You need: Perpendicular directions; circular acceleration; charge sign.

01 Idea

A magnetic field exerts force on a moving charge when its motion has a perpendicular component. The force is perpendicular to velocity and therefore does no work on a point charge. It can change direction without changing speed.

Magnetic force acts sideways relative to moving charge. For a straight current-carrying wire, the current direction plays the corresponding role. Geometry matters: a wire across the field has maximum magnetic force, whereas a wire along the field has no perpendicular component and no magnetic force.

- A perpendicular arrangement uses the full field-current-length product because the sine factor is one.
- Reversing current reverses force direction but does not change magnitude when other factors stay fixed.

02 Rule and symbols

Rule

$$F = |q|vB\sin\theta$$

$$F_{\text{wire}} = BIL\sin\theta$$

q particle charge / C

v particle speed / m/s

B magnetic field magnitude / T

F magnetic force magnitude / N

θ angle between velocity and field / degrees or radians

03 Method

1. Decide whether the object is a moving charge or a current-carrying wire.
2. Identify the angle to the magnetic field and the perpendicular motion or wire length.
3. Use the right-hand rule for positive charge or conventional current; reverse for negative charge.
4. Calculate magnitude and check that the force is perpendicular to motion.

Use the wire-force relation when current, straight wire length and magnetic field are given; do not substitute particle speed.

04 Worked example

A positive charge moving perpendicular to a uniform magnetic field follows a circular path if no other force acts.

01 For the positive charge in the example, velocity is perpendicular to the uniform field, so the magnetic force magnitude is qvB.

02 At each point the force is perpendicular to velocity and turns the particle inward.

03 Because the force does no work, speed and kinetic energy remain constant.

04 With no other forces, the continual turning gives a circular path in the plane perpendicular to the field.

Watch out

Do not apply the right-hand direction unchanged to an electron.

Exam habit

A direction answer needs the charge sign; a right-hand-rule result for a positive charge must be reversed for an electron.

05 Your turn

A straight 0.50 m wire carries 2 A perpendicular to a 0.30 T field. Find force magnitude.

If you get stuck, uncover one hint at a time:

1. This is a current-carrying wire, not a single moving particle. Choose the relation involving field, current and length.
2. The wire is perpendicular to the field, so the angle factor is one.
3. Multiply 0.30 T, 2 A and 0.50 m. The question asks for magnitude, so no charge sign is needed.

06 Check your understanding

Does a charge moving exactly along a magnetic field feel magnetic force?

A straight wire initially lies perpendicular to a uniform field. It rotates until parallel while current stays fixed. Its magnetic force magnitude:

1. Stays unchanged
2. Doubles
3. Falls to zero

Answers and explanations: page 102.

Sources of magnetic fields

EM / 10

- Relate a current to the direction of the magnetic field it produces.
- Use simple distance and current ratios for a long wire.

You need: Current; inverse proportionality; circular field-line diagrams.

01 Idea

Currents produce magnetic fields. Around a long straight wire the field forms circles. Inside an ideal long solenoid the field is approximately uniform. For two parallel wires, currents in the same direction attract and opposite currents repel.

A straight current-carrying wire has magnetic field lines that circle the wire. Moving farther away weakens the field, but its distance pattern differs from the field of a point charge. Recognizing the source geometry prevents using the familiar but incorrect inverse-square rule here.

- For an ideal long wire, magnetic field magnitude is inversely proportional to perpendicular distance.
- At unchanged current, a threefold increase in distance therefore gives a threefold decrease rather than a ninefold decrease.

02 Rule and symbols

Rule

$$B_{\text{wire}} = \mu_0 I/(2\pi r)$$

$$B_{\text{solenoid}} = \mu_0 n I$$

B magnetic field magnitude / T

I wire current / A

r perpendicular distance from wire / m

n solenoid turns per unit length / m^{-1}

Use this when:

The wire is sufficiently long and the observation point sufficiently far from its ends for the long-wire approximation.

03 Method

1. Identify whether the source is a straight wire, a solenoid or a pair of wires.
2. For a straight wire, use the right-hand grip rule to draw field circles.
3. Apply the matching field relation; for ratios hold unstated factors fixed.
4. For paired wires determine attraction or repulsion from relative current directions.

The phrase long straight wire signals the current-over-distance rule, not the point-charge inverse-square rule.

04 Worked example

Doubling distance from a long wire halves its magnetic field; doubling current doubles the field.

01 For the long wire, B is proportional to I/r.

02 Moving from r to 2r at the same current gives half the original field.

03 Doubling current at fixed r instead doubles the field.

04 If current and distance both double, the two factors cancel and field magnitude stays the same.

Watch out

The wire field varies as inverse distance, not inverse distance squared.

Exam habit

Use 1/r for the long-wire field; the $1/r^2$ rule belongs to a different source geometry.

05 Your turn

The field is 30 microtesla at 2 cm from a long straight wire. Find the field at 6 cm at the same current.

If you get stuck, uncover one hint at a time:

1. Keep the current fixed as stated. Compare the new perpendicular distance with the original distance.
2. Use an inverse-distance ratio for a long wire; do not square the ratio.
3. Apply that factor to 30 microtesla. Centimetres can remain in the ratio because both distances use the same unit.

06 Check your understanding

A wire gives 30 μT at 2 cm. What field does it give at 6 cm with the same current?

Current and perpendicular distance from a long wire both double. What happens to magnetic field magnitude?

1. It stays unchanged
2. It doubles
3. It halves

Answers and explanations: page 102.

Induction and Lenz law

EM / 11

- Identify which change produces an induced emf.
- Use Lenz law to choose the induced current direction.

You need: Magnetic field direction; loop area; clockwise versus counterclockwise viewed from a stated side.

01 Idea

An induced emf requires a change in magnetic flux, which can arise from changing field, area or orientation. Lenz law sets the direction: the induced effect opposes the change of flux. A static magnet near a stationary loop does not produce a sustained induced current.

Induction responds to a changing magnetic flux through a loop, rather than merely to the presence of a magnet. A faster change gives a larger average emf. A coil repeats the same flux-change effect through many turns, so their induced emfs add when the turns are linked alike.

- More turns multiply the emf when each turn experiences the same flux change.
- Spreading the same flux change over a longer time reduces the average emf.

02 Rule and symbols

Rule

$$|\mathcal{E}| = N|\Delta\Phi_B|/\Delta t$$

$$\Phi_B = BA\cos\theta$$

Φ_B magnetic flux through one turn / Wb

N number of coil turns / dimensionless

$\mathcal{E}$ induced emf / V

Δt time interval of flux change / s

03 Method

1. State the original flux direction and whether its magnitude increases or decreases.
2. Choose an induced field that opposes that change.
3. Use the grip rule to convert the required field to current direction, naming the viewing side.
4. For magnitude, multiply the flux change per turn by the number of turns and divide by elapsed time.

Look for a change in field, loop area or loop orientation over a stated time; identify whether flux is per turn.

04 Worked example

If outward flux increases, the induced field points inward.

01 In the example, outward flux through the loop is increasing.

02 The induced effect must oppose the increase, so its own field points inward.

03 Viewed from the side toward which the original flux points outward, the inward field requires clockwise current.

04 If the outward flux were decreasing instead, the induced field and current direction would both reverse.

Watch out

Opposes the change is not the same as always opposes the original field.

Exam habit

Say oppose the change in flux, not oppose the flux: a decreasing field produces a reinforcing induced field.

05 Your turn

A 20-turn coil has flux per turn changing by 0.03 Wb in 0.10 s. Find average emf magnitude.

If you get stuck, uncover one hint at a time:

1. The problem gives flux change per turn. Decide where the 20 turns enter the total effect.
2. Multiply turn count by flux change per turn, then divide by the 0.10 s interval.
3. Use the magnitude of the change because magnitude is requested. A current direction would require separate Lenz-law information.

06 Check your understanding

Does a stationary loop near an unchanging stationary magnet have a sustained induced emf?

The same flux change per turn occurs in half the time. With turn count unchanged, average emf magnitude:

1. Stays unchanged
2. Doubles
3. Halves

Answers and explanations: page 102.

Inductance, AC and electromagnetic waves

EM / 12

- Convert sinusoidal peak voltage to RMS and calculate resistive heating.
- Relate electromagnetic frequency to vacuum wavelength.

You need: Square roots; frequency and wavelength; electrical power.

01 Idea

Course-plan extension: an inductor resists changes of current and stores magnetic energy. AC reverses direction periodically. RMS values describe equivalent heating for resistive loads. In vacuum an EM wave has mutually perpendicular electric field, magnetic field and travel direction.

An alternating voltage has a changing instantaneous value, so its peak is not its steady heating equivalent. RMS voltage compresses that varying waveform into the DC voltage that would produce the same average heating in a resistor. The familiar peak conversion depends on the waveform being sinusoidal.

- Resistive heating depends on voltage squared, so simply averaging positive and negative voltage would miss the heating.
- A sinusoid has RMS voltage below its peak because it spends only part of each cycle near the peak.

02 Rule and symbols

Rule

$$U_L = \tfrac{1}{2} L I^2$$

$$V_{\mathrm{rms}} = V_{\mathrm{peak}} / \sqrt{2} \text{ (sinusoidal)}$$

$$c = f\lambda \text{ (vacuum)}$$

V_{peak} maximum sinusoidal voltage magnitude / V

V_{rms} root-mean-square voltage / V

L inductance / H

f wave frequency / Hz

λ wavelength / m

Use this when:

The peak-to-RMS conversion assumes a sinusoid; the wavelength calculation assumes propagation in vacuum.

03 Method

1. Identify whether the question concerns an inductor, sinusoidal AC or an electromagnetic wave.
2. For sinusoidal AC divide peak voltage by the square root of two.
3. Use RMS values for average power in a pure resistor.
4. For vacuum waves calculate wavelength as c divided by frequency and convert metres to the requested unit.

Use RMS values for average resistive power; use the peak-to-RMS conversion only for a specified sinusoid.

04 Worked example

A sinusoidal supply with peak voltage 141.4 V has RMS voltage 100 V.

01 For the stated sinusoidal supply, 141.4 divided by $\sqrt{2}$ is approximately 100 V RMS.

02 This RMS value matches the heating effect of 100 V DC in the same pure resistor.

03 For the separate light example, wavelength is 3.00×10^{8} divided by 6×10^{14} = 5×10^{-7} m.

04 Since one nanometre is 10^{-9} m, the wavelength is 500 nm.

Watch out

An ideal transformer does not continuously transform steady DC after switching transients.

Exam habit

Keep this extension conceptual: use RMS for resistor power and $c = f\lambda$ for waves; do not mix peak and RMS values.

05 Your turn

A sinusoidal voltage has peak value 20 V. Find its RMS value, then the average power in a 10 Ω resistor.

If you get stuck, uncover one hint at a time:

1. Separate the request into RMS voltage and average resistor power. The supplied 20 V is a peak value.
2. Divide the peak by the square root of two to obtain the sinusoidal RMS value.
3. Square that RMS voltage and divide by 10 ohms. Keep a few extra digits until the final rounding.

06 Check your understanding

A 20 V peak sinusoid is applied to 10 Ω. What is the average power?

A symmetric sinusoidal voltage has zero average voltage over a cycle. Does a resistor receive zero average power?

1. Yes; positive and negative heating cancel
2. Only the positive half-cycle heats it
3. No; both voltage polarities produce heating

Answers and explanations: page 102.

Lesson review and answers

Charges, conductors and Coulomb force

Return to lesson: page 62.

Your turn: It becomes one ninth as large.

Concept check: 1. Correct. Opposite signs still attract, but larger separation reduces magnitude.

Written method: At 2r: 12/4 = 3 N. Tripling one charge at r: 36 N. Two marks for each ratio and result.

Electric field and superposition

Return to lesson: page 65.

Your turn: No. The electric force qE is zero there in the stated source-field model.

Concept check: 2. Correct. The field is the same and charge magnitudes match; opposite signs reverse force direction.

Written method: Magnitude = 0.003 times 200 = 0.60 N. Direction is west because the charge is negative. Two marks for magnitude and two for direction with reasoning.

Flux and Gauss law

Return to lesson: page 69.

Your turn: Yes. Field enters and leaves; the total closed-surface flux from that external charge cancels.

Concept check: 3. Correct. The crossing component decreases to zero as the normal becomes perpendicular to the field.

Written method: Parallel normal: flux = 50 times 0.20 = 10 N m squared/C. Perpendicular normal: cosine 90 degrees = 0, so flux = 0. Two marks per case.

Symmetry and conductors

Return to lesson: page 72.

Your turn: 20 N/C: the centre distance doubles and the field is divided by four.

Concept check: 1. Correct. Centre distance equals sphere radius plus the surface gap, so the two doubling operations differ.

Written method: Inverse-square scaling gives 80 times (0.20/0.40) squared = 20

N/C. Credit the outside-sphere condition and inverse-square reasoning.

Potential is not field

Return to lesson: page 75.

Your turn: $\triangle$U = +6 J and work by the field = -6 J.

Concept check: 2. Correct. Charge times potential difference is negative for a negative charge and positive difference.

Written method: Energy change = 2 times 3 = +6 J. Work by field = -6 J. Two marks each; the signs are essential.

Capacitors and dielectrics

Return to lesson: page 78.

Your turn: 48 μC, from Q = CV.

Concept check: 3. Correct. The battery fixes voltage and supplies the extra charge associated with larger capacitance.

Written method: Q = CV = 4 times 12 = 48 microcoulombs. In SI this is 0.000048 C. Credit formula, substitution, prefix and result.

Current, resistance and electrical power

Return to lesson: page 82.

Your turn: Resistance doubles.

Concept check: 1. Correct. Both current and its corresponding voltage double, so their product quadruples.

Written method: V = IR = 6 V. P = VI = 12 W (or I^2R). Two marks per result.

DC networks, Kirchhoff rules and RC behaviour

Return to lesson: page 85.

Your turn: No. Once charged, it blocks steady DC in the ideal model.

Concept check: 2. Correct. At common current, voltage drop is proportional to resistance.

Written method: Total resistance = 12 Ω; current = 2 A; drops are 8 V and 16 V. One mark each; the drops must sum to 24 V.

Magnetic force: direction before calculation

Return to lesson: page 90.

Your turn: No. The angle is 0° or 180°, so its sine is zero.

Concept check: 3. Correct. At parallel alignment the perpendicular component and sine factor vanish.

Written method: F = BIL = 0.30 times 2 times 0.50 = 0.30 N. Four marks for formula, perpendicular condition, substitution and unit.

Sources of magnetic fields

Return to lesson: page 93.

Your turn: 10 μT because distance triples.

Concept check: 1. Correct. Doubling current and distance multiplies the numerator and denominator by the same factor.

Written method: B is inversely proportional to r, so B = 30 times 2/6 = 10 microtesla. Credit model, ratio, arithmetic and unit.

Induction and Lenz law

Return to lesson: page 96.

Your turn: No. Its magnetic flux is not changing.

Concept check: 2. Correct. Emf depends on the rate of flux change, so halving time doubles that rate.

Written method: Magnitude = 20 times 0.03/0.10 = 6 V. The direction needs the direction of the flux change; it cannot be inferred from these magnitudes alone.

Inductance, AC and electromagnetic waves

Return to lesson: page 99.

Your turn: 20 W: $V_{rms}^2/R = (20^2/2)/10$.

Concept check: 3. Correct. The heating depends on voltage squared, which is nonnegative for either polarity.

Written method: RMS = 20/square root of 2 = 14.14 V. Average power = RMS squared/10 = 20 W. Two marks per stage.

Electricity and Magnetism / Test A

30 marks / Suggested time: 30 minutes. MCQs: 1 mark each. Written problems: 4 marks each. Attempt without notes.

Part I / Choose one answer

1. A neutral object loses electrons. It becomes:

   A. Negative

   B. Positive

   C. Uncharged forever

   D. Magnetic only

2. A negative test charge in a field directed right feels force:

   A. Left

   B. Zero

   C. Right

   D. Up

3. A charge outside a closed empty box gives net flux through the box of:

   A. $\frac{q}{\varepsilon_0}$

   B. Always positive

   C. Always negative

   D. Zero

4. Inside the material of a conductor in electrostatic equilibrium, the electric field is:

   A. Equal to its surface value

   B. Infinite

   C. Zero

   D. Always outward

5. For equal and opposite point charges, take potential as zero at infinity. At their midpoint the potential is:

   A. Zero

   B. Undefined

C. Positive

D. Negative

6. Adding a dielectric to an isolated charged capacitor increases C. Its voltage:

A. Decreases

B. Increases

C. Must stay fixed

D. Becomes infinite

Part II / Show your reasoning

1. Two charges exert 12 N at separation r. What is the force at separation 2r, and at r if one charge is tripled?

2. A field of 200 N/C points east. Find the force on a charge of -0.003 C.

3. A uniform field of 50 N/C passes through a flat area of 0.20 square metres. The chosen surface normal points in the same direction as the field. Find flux. What if the normal is perpendicular to the field?

4. An isolated charged conducting sphere, with no external fields, has field magnitude 80 N/C at r = 0.20 m. Find the field at r = 0.40 m. Both distances are from the centre and outside the sphere.

5. A +2 C charge moves through a potential difference of +3 V. Find its potential-energy change and the work done by the field.

6. A 4 $\mu\mathrm{F}$ capacitor is connected to 12 V. Find the charge in microcoulombs.

Electricity and Magnetism / Test B

30 marks / Suggested time: 30 minutes. MCQs: 1 mark each. Written problems: 4 marks each. Attempt without notes.

Part I / Choose one answer

1. At fixed voltage, doubling resistance makes power:

   A. Unchanged

   B. Double

   C. Half

   D. Four times

2. After a long time in an ideal DC charging circuit, capacitor current is:

   A. Maximum

   B. Negative forever

   C. Zero

   D. Equal to the battery voltage divided by the resistor resistance

3. A charge moving parallel to a magnetic field experiences magnetic force:

   A. qvB

   B. Twice qvB

   C. Opposite its velocity

   D. Zero

4. Two parallel wires carrying currents in the same direction:

   A. Repel

   B. Attract

   C. Have no interaction

   D. Must carry zero field

5. Which situation produces an induced emf?

   A. No change of area, field or angle

   B. Changing magnetic flux

   C. A stationary unchanging field only

D. Constant flux

6. In vacuum, electric and magnetic fields of a plane EM wave are:

A. Parallel to each other

B. Longitudinal only

C. Zero

D. Perpendicular to each other and to travel

Part II / Show your reasoning

1. A 3 Ω resistor carries 2 A. Find voltage and power.

2. A 4 Ω and 8 Ω resistor are in series across 24 V. Find total current and each voltage drop.

3. A straight 0.50 m wire carries 2 A perpendicular to a 0.30 T field. Find force magnitude.

4. The field is 30 microtesla at 2 cm from a long straight wire. Find the field at 6 cm at the same current.

5. A 20-turn coil has flux per turn changing by 0.03 Wb in 0.10 s. Find average emf magnitude.

6. A sinusoidal voltage has peak value 20 V. Find its RMS value, then the average power in a 10 Ω resistor.

Electricity and Magnetism / Test A / Answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1

MCQ: B. Removing negative charge leaves a positive net charge.

Written solution: At 2r: 12/4 = 3 N. Tripling one charge at r: 36 N. Two marks for each ratio and result.

Return to lesson: page 62.

Question 2

MCQ: A. The negative charge reverses the field direction in $F = qE$.

Written solution: Magnitude = 0.003 times 200 = 0.60 N. Direction is west because the charge is negative. Two marks for magnitude and two for direction with reasoning.

Return to lesson: page 65.

Question 3

MCQ: D. No charge is enclosed, so total entering and leaving flux cancels.

Written solution: Parallel normal: flux = 50 times 0.20 = 10 N m squared/C. Perpendicular normal: cosine 90 degrees = 0, so flux = 0. Two marks per case.

Return to lesson: page 69.

Question 4

MCQ: C. Free charges redistribute until the internal field is cancelled.

Written solution: Inverse-square scaling gives 80 times (0.20/0.40) squared = 20 N/C. Credit the outside-sphere condition and inverse-square reasoning.

Return to lesson: page 72.

Question 5

MCQ: A. The signed scalar potentials cancel when distances and charge magnitudes match.

Written solution: Energy change = 2 times 3 = +6 J. Work by field = -6 J. Two marks each; the signs are essential.

Return to lesson: page 75.

Question 6

MCQ: A. Charge remains fixed and $V = \frac{Q}{C}$.

Written solution: Q = CV = 4 times 12 = 48 microcoulombs. In SI this is 0.000048 C. Credit formula, substitution, prefix and result.

Return to lesson: page 78.

Electricity and Magnetism / Test B / Answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1

MCQ: C. $P = \frac{V^2}{R}$, so doubling the denominator halves it.

Written solution: V = IR = 6 V. P = VI = 12 W (or I^2R). Two marks per result.

Return to lesson: page 82.

Question 2

MCQ: C. After a long time, the capacitor voltage equals the battery voltage. The resistor voltage and circuit current are zero. During charging, charge moves in the wires onto and off the plates; it does not cross the ideal dielectric.

Written solution: Total resistance = 12 Ω; current = 2 A; drops are 8 V and 16 V. One mark each; the drops must sum to 24 V.

Return to lesson: page 85.

Question 3

MCQ: D. The sine of the angle is zero.

Written solution: F = BIL = 0.30 times 2 times 0.50 = 0.30 N. Four marks for formula, perpendicular condition, substitution and unit.

Return to lesson: page 90.

Question 4

MCQ: B. Each current experiences the magnetic field of the other wire.

Written solution: B is inversely proportional to r, so B = 30 times 2/6 = 10 microtesla. Credit model, ratio, arithmetic and unit.

Return to lesson: page 93.

Question 5

MCQ: B. Faraday law depends on the rate of flux change.

Written solution: Magnitude = 20 times 0.03/0.10 = 6 V. The direction needs the direction of the flux change; it cannot be inferred from these magnitudes alone.

Return to lesson: page 96.

Question 6

MCQ: D. A plane electromagnetic wave in vacuum is transverse.

Written solution: RMS = 20/square root of 2 = 14.14 V. Average power = RMS squared/10 = 20 W. Two marks per stage.

Return to lesson: page 99.

Formula sheet

Charges, conductors and Coulomb force

p.62

Rule

$$F = k|q_1 q_2|/r^2$$

$$k = 8.99 \times 10^9 \ \mathrm{N\,m^2/C^2}$$

Electric field and superposition

p.65

Rule

$$E = F/q_{\text{test}}$$

$$E = k|Q|/r^2$$

$$\vec{F} = q\vec{E}$$

Flux and Gauss law

p.69

Rule

$$\Phi = EA\cos\theta$$

$$\Phi_{\text{closed}} = Q_{\text{enclosed}}/\epsilon_0$$

Symmetry and conductors

p.72

Rule

$$E_{\text{outside sphere}} = k|Q|/r^2$$

$$(r > R)$$

Potential is not field

p.75

Rule

$$V = kQ/r \text{ (zero at infinity)}$$

$$\Delta U = q\Delta V$$

$$W_{\text{field}} = -\Delta U$$

Capacitors and dielectrics

p.78

Rule

$$Q = CV$$

$$U = \tfrac{1}{2}CV^2$$

$$C_{\parallel} = C_1 + C_2$$

$$1/C_s = 1/C_1 + 1/C_2$$

Current, resistance and electrical power

p.82

Rule

$$I = \Delta Q / \Delta t$$

$$V = IR$$

$$R = \rho L / A$$

$$P = VI$$

DC networks, Kirchhoff rules and RC behaviour

p.85

Rule

$$R_s = R_1 + R_2$$

$$1/R_p = 1/R_1 + 1/R_2$$

$$\tau = RC$$

Magnetic force: direction before calculation

p.90

Rule

$$F = |q| v B \sin\theta$$

$$F_{\text{wire}} = BIL \sin\theta$$

Sources of magnetic fields

p.93

Rule

$$B_{\text{wire}} = \mu_0 I / (2\pi r)$$

$$B_{\text{solenoid}} = \mu_0 n I$$

Induction and Lenz law

p.96

Rule

$$|\mathcal{E}| = N |\Delta \Phi_B| / \Delta t$$

$$\Phi_B = BA \cos\theta$$

Inductance, AC and electromagnetic waves

p.99

Rule

$$U_L = \tfrac{1}{2} L I^2$$

$$V_{\text{rms}} = V_{\text{peak}} / \sqrt{2} \text{ (sinusoidal)}$$

$$c = f\lambda \text{ (vacuum)}$$

Symbol glossary

Symbols are listed with their lesson context. A symbol may represent different quantities in different rules.

Charges, conductors and Coulomb force

F electrostatic force magnitude / N

q_1, q_2 source charges / C

r separation of charges / m

Electric field and superposition

E electric field magnitude / N/C

q charge experiencing the force / C

F force magnitude / N

Flux and Gauss law

Φ electric flux / N m^2/C

E electric field magnitude / N/C

A surface area / m^2

Q_{enclosed} net charge inside a closed surface / C

Symmetry and conductors

Q net sphere charge / C

R sphere radius / m

r distance from sphere centre / m

E electric field magnitude / N/C

Potential is not field

V electric potential / V = J/C

ΔV final minus initial potential / V

ΔU change in potential energy / J

q moving charge / C

Capacitors and dielectrics

C capacitance / F

Q stored charge magnitude on either plate / C

V potential difference between plates / V

U stored electric energy / J

Current, resistance and electrical power

I conventional current / A

R resistance / Ω

V potential difference / V

P electrical power / W

ρ resistivity / Ω m

DC networks, Kirchhoff rules and RC behaviour

R_{eq} equivalent resistance / Ω

I current / A

V voltage / V

τ RC time constant / s

Magnetic force: direction before calculation

q particle charge / C

v particle speed / m/s

B magnetic field magnitude / T

F magnetic force magnitude / N

θ angle between velocity and field / degrees or radians

Sources of magnetic fields

B magnetic field magnitude / T

I wire current / A

r perpendicular distance from wire / m

n solenoid turns per unit length / m^{-1}

Induction and Lenz law

Φ_B magnetic flux through one turn / Wb

N number of coil turns / dimensionless

$\mathcal{E}$ induced emf / V

Δt time interval of flux change / s

Inductance, AC and electromagnetic waves

V_{peak} maximum sinusoidal voltage magnitude / V

V_{rms} root-mean-square voltage / V

L inductance / H

f wave frequency / Hz

λ wavelength / m

Source and coverage notes

Concept-first coverage of the course plan. Chapters 23-31 have lecture support; inductance, AC and EM waves are course-plan extensions without separate lecture files in this folder. Advanced field integrals are recognition topics, not worked derivations.

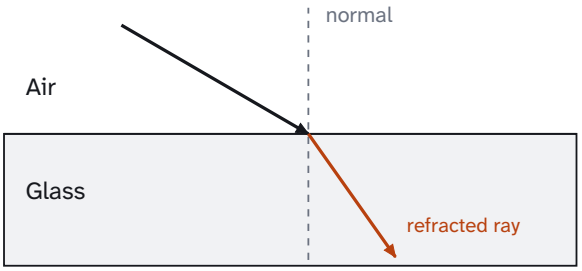
Open original course folder

The source audit inventories 159 non-programming files; 95 yielded substantial extracted text. Scanned diagrams and handwritten or image-only material were not fully transcribed. Check lecturer-specific derivations and conventions. The practice tests are original study material, not predictions of an examination.

Optics and Waves

FIG. 03 / OPTICS

A ray bends toward the normal in glass



At the boundary

Frequency: unchanged

Speed: decreases

Wavelength: decreases

Takeaway: measure angles from the normal, not from the glass surface.

The introduction and four current sections: wave motion; electromagnetic theory, photons and light; geometrical optics; physical optics. Derivations are replaced by plain-language interpretation and short exam calculations.

Subject map

The introduction and four current sections: wave motion; electromagnetic theory, photons and light; geometrical optics; physical optics. Derivations are replaced by plain-language interpretation and short exam calculations.

Follow this reading route from foundations to applications. Each line links to its lesson; revisit the stated prerequisites whenever a step feels unfamiliar.

01 Wave language and simple harmonic motion p.125

02 Read a travelling wave without deriving it p.128

03 Wavefronts, spreading and group velocity p.132

04 Superposition, standing waves and beats p.135

05 Light, photons and the electromagnetic picture p.138

06 Reflection, refraction and total internal reflection p.141

07 Mirrors, lenses and image signs p.145

08 Optical systems, prisms and instruments p.148

09 Interference and diffraction p.151

10 Polarization and Malus law p.155

11 Scattering, apparent depth and lens power p.158

12 Recognize advanced optics without long derivations p.161

Wave language and simple harmonic motion

OW / 01

- Convert an oscillation count into frequency and period.
- Relate displacement, speed and acceleration at key SHM positions.

You need: One oscillation is one complete cycle; reciprocal quantities; displacement from equilibrium.

01 Idea

A wave carries a disturbance and energy. Mechanical waves need a medium; electromagnetic waves can travel through vacuum. Transverse displacement is perpendicular to travel; longitudinal displacement is parallel. Simple harmonic motion has a restoring acceleration proportional and opposite to displacement.

One complete oscillation returns the motion to the same position and direction. Period asks how long that return takes; frequency asks how many returns occur each second. They describe the same repeating motion from opposite viewpoints.

- More cycles within the same time means larger frequency.
- A larger frequency leaves less time for each cycle, so period decreases.

02 Rule and symbols

Rule

$$T = 1/f$$

$$f = N/t$$

$$a = -\omega^2 x$$

N number of complete cycles / dimensionless

t total elapsed time / s

f cycles per second / Hz

T time for one cycle / s

ω angular frequency / rad/s

03 Method

1. Count complete cycles rather than individual crossings of equilibrium.
2. Divide cycle count by total time for frequency.
3. Find period as total time per cycle or the reciprocal of frequency.
4. For SHM compare the point with equilibrium and use restoring acceleration toward equilibrium.

A counted number of complete oscillations over a measured time signals a frequency and period calculation.

04 Worked example

If 20 oscillations take 5 s, frequency is 4 Hz and period is 0.25 s.

01 Twenty complete oscillations in 5 s means 20/5 = 4 cycles per second, so f = 4 Hz.

02 One cycle takes 5/20 = 0.25 s, giving T = 0.25 s.

03 At an extreme the object must turn around, so its instantaneous speed is zero.

04 Its displacement magnitude is largest there; restoring acceleration therefore has maximum magnitude and points toward equilibrium.

Watch out

The direction of particle motion is not necessarily the direction of wave travel.

Exam habit

Write f = N/t and T = t/N side by side to prevent interchanging hertz and seconds.

05 Your turn

A source makes 50 complete oscillations in 10 s. Find frequency and period.

If you get stuck, uncover one hint at a time:

1. Identify the complete cycle count and total time. Which ratio counts cycles per second?
2. Divide 50 cycles by 10 s for frequency.
3. Use total time divided by cycle count for period, or take the reciprocal of frequency. Check hertz and seconds respectively.

06 Check your understanding

At SHM equilibrium is acceleration magnitude maximum?

A source doubles its period. How does its frequency change?

1. It halves
2. It doubles
3. It stays unchanged

Answers and explanations: page 164.

Read a travelling wave without deriving it

OW / 02

- Read amplitude, wavelength and frequency from a travelling-wave expression.
- Use v = fλ and distinguish pattern motion from particle motion.

You need: Sine-wave shape; π; coefficients of x and t; SI units.

01 Idea

Amplitude is maximum displacement, wavelength is crest-to-crest separation, and frequency counts cycles per second. Wave speed is the speed of the pattern, not the oscillating particle. In a sine wave, the coefficients of position and time reveal wavelength and period.

The source decides how often a wave repeats, while the medium sets its pattern speed under the stated conditions. Wavelength is the distance the pattern advances during one cycle. At fixed speed, more cycles each second mean each cycle occupies less distance.

- Frequency times wavelength gives distance travelled each second.
- In the same nondispersive medium, a frequency increase changes wavelength while speed stays fixed.

02 Rule and symbols

Rule

$$v = f\lambda$$

$$k = 2\pi/\lambda$$

$$\omega = 2\pi f$$

$$y = A\sin(kx - \omega t + \phi)$$

A amplitude or maximum displacement / m

k wave number / rad/m

ω angular frequency / rad/s

λ wavelength / m

f frequency / Hz

Use this when:

The example takes x and y in metres and t in seconds.

03 Method

1. Match the wave to y = A sin(kx - ωt + ϕ).
2. Read A directly, then divide 2π by k to obtain wavelength.
3. Divide ω by 2π to obtain frequency.
4. Multiply frequency by wavelength for pattern speed and check the units.

Use the fixed-speed relationship when frequency changes but the same nondispersive medium is retained.

04 Worked example

For $y = 0.02\sin(4\pi x - 20\pi t)$, amplitude is 0.02 m, wavelength is 0.50 m, frequency is 10 Hz and speed is 5 m/s.

01 Compare y = 0.02 sin(4πx - 20πt) with the standard form: A = 0.02 m, k = 4π rad/m and ω = 20π rad/s.

02 Wavelength is 2π/(4π) = 0.50 m.

03 Frequency is 20π/(2π) = 10 Hz.

04 Pattern speed is 10 times 0.50 = 5 m/s; this is not the maximum transverse speed of an individual particle.

Watch out

The coefficient of time is angular frequency, not ordinary frequency.

Exam habit

Identify coefficients before calculating; treating k as wavelength or ω as ordinary frequency loses the factor 2π.

05 Your turn

A wave has frequency 8 Hz and wavelength 0.75 m. Find speed. If frequency becomes 12 Hz in the same nondispersive medium, find wavelength.

If you get stuck, uncover one hint at a time:

1. Use the original frequency and wavelength to establish speed first.
2. Retain that speed because the medium remains nondispersive and unchanged.
3. Divide the established speed by 12 Hz for the new wavelength; do not reuse the old wavelength with the new frequency.

06 Check your understanding

Is the coefficient 20π in the example the frequency in hertz?

A wave enters a medium where its speed is lower while frequency stays fixed. Its wavelength:

1. Must remain fixed
2. Becomes shorter
3. Becomes longer

Answers and explanations: page 164.

Phase means position within a cycle

A phase difference tells whether two oscillations are at the same or different stages of a cycle.

- One full cycle is 2π radians. Half a cycle is π radians.
- At a fixed time, points one wavelength apart are in phase. Points half a wavelength apart are opposite in phase.
- At one place, a full period brings the oscillation back to the same phase.
- These formulas give the size of the phase difference. Its sign depends on the order of comparison and the travelling-wave convention.
- $\Delta\phi = 2\pi\dfrac{\Delta x}{\lambda}$ (phase-difference magnitude at one time)
- $\Delta\phi = 2\pi\dfrac{\Delta t}{T}$ (phase-change magnitude at one place)

Two points on a sinusoidal wave are separated by $\lambda/4$. Find their phase-difference magnitude at the same time.

1. The separation is one quarter of a full spatial cycle.
2. Therefore $\Delta\phi = 2\pi \times (1/4) = \pi/2$.

$\pi/2$ radians, or $90°$. They are neither in phase nor exactly opposite in phase.

Wavefronts, spreading and group velocity

OW / 03

- Predict intensity and amplitude changes for spherical spreading.
- Distinguish wavefront, ray, phase velocity and group velocity.

You need: Surface area grows with radius squared; square roots; amplitude versus intensity.

01 Idea

A ray indicates propagation direction and is normal to a wavefront. Plane wavefronts are flat, spherical wavefronts spread from a point source, and cylindrical wavefronts spread around a line source. Phase velocity tracks a crest; group velocity tracks the envelope of a narrow wave packet.

Picture fixed source power being shared over an expanding spherical shell. The larger shell carries the same total power when nothing is absorbed, but each square metre receives a smaller share. Geometric spreading can therefore reduce intensity without dissipating the total power.

- Sphere area grows with radius squared.
- With total power fixed, intensity decreases by the same area factor.

02 Rule and symbols

Rule

$$I \propto A^2$$

$$I_{\text{spherical}} \propto 1/r^2$$

I energy-transfer rate per unit area / W/m^2

A wave amplitude / depends on disturbance

r distance from point source / m

Use this when:

The spreading example assumes an isotropic point source, no absorption and unchanged medium properties.

03 Method

1. Identify whether the source produces plane, spherical or cylindrical spreading.
2. For an isotropic point source use intensity proportional to $1/r^2$.
3. Convert an intensity ratio to an amplitude ratio by taking its square root.
4. In description questions, identify a crest for phase velocity and the packet envelope for group velocity.

An isotropic point source with negligible absorption permits inverse-square spreading; other geometries need their own model.

04 Worked example

For an isotropic point source without absorption, doubling distance reduces intensity to one quarter and amplitude to one half.

01 The same source power crosses a sphere around the source at each distance.

02 Doubling radius makes the sphere area four times as large.

03 The available power per square metre is therefore one quarter as large.

04 Because intensity is proportional to amplitude squared, the amplitude ratio is $\sqrt{(1/4)} = 1/2$.

Watch out

The ideal plane-wave model does not have spherical inverse-square spreading.

Exam habit

Do not apply the inverse-square intensity rule automatically to a plane wave or line source.

05 Your turn

At distance r an isotropic source has intensity 18 W per square metre. Find intensity at 3r, ignoring absorption.

If you get stuck, uncover one hint at a time:

1. The new distance is three times the original. Ask how a sphere area changes when radius triples.
2. Square the distance factor for the inverse-square intensity change.
3. Apply that factor to 18 W/m^2 and retain intensity units; this is not a total-power calculation.

06 Check your understanding

An intensity of 18 W/m^2 is measured at r. What is it at 3r?

Without absorption, which quantity stays the same on complete spheres farther from an ideal point source?

1. Intensity per square metre
2. Wave amplitude at each point
3. Total power crossing each sphere

Answers and explanations: page 164.

Superposition, standing waves and beats

OW / 04

- Calculate beat frequency from two nearby frequencies.
- Connect fixed-string harmonics with allowed wavelengths.

You need: Adding displacements; wavelength; frequency; fixed endpoints.

01 Idea

Overlapping disturbances add algebraically. In-phase equal waves reinforce; opposite-phase equal waves cancel. A standing wave forms when matching waves travel in opposite directions. Nodes stay at zero displacement. Beats are slow changes in loudness from nearby frequencies.

A string fixed at both ends must have nodes at its endpoints. Only patterns fitting whole numbers of loops between them can persist as normal modes. One loop spans half a wavelength, so a string length is not automatically a complete wavelength.

- Fixed endpoints restrict which wavelengths fit.
- At fixed wave speed, fitting more loops shortens wavelength and increases frequency.

02 Rule and symbols

Rule

$$f_{\text{beat}} = |f_1 - f_2|$$

$\lambda_n = 2L/n$ (string fixed at both ends)

f_{beat} number of loudness cycles per second / Hz

L vibrating string length / m

n harmonic number, 1, 2, 3, ... / dimensionless

λ_n wavelength of harmonic n / m

03 Method

1. For superposition add instantaneous signed displacements at the same point.
2. For beats take the positive difference of the two nearby frequencies.
3. For a string fixed at both ends draw nodes at both ends.
4. Fit n half-wavelengths into the length, then use wave speed to find each frequency.

For a string fixed at both ends, draw its node-and-loop pattern before selecting the harmonic wavelength.

04 Worked example

Two tones at 256 and 260 Hz produce 4 beats per second.

01 The tones have frequencies 256 Hz and 260 Hz.

02 Their frequency difference is 4 Hz, so there are four loudness beats each second.

03 For the separate string example, the fundamental has one loop between two end nodes.

04 That loop occupies half a wavelength, so $\lambda 1 = 2L$ and $f1 = v/(2L)$.

Watch out

A node is not a point where two travelling waves are individually absent.

Exam habit

A beat frequency is a difference, not the average of the two tone frequencies.

05 Your turn

A 1 m string fixed at both ends supports waves at 40 m/s. Find its fundamental frequency and second harmonic.

If you get stuck, uncover one hint at a time:

1. Draw the fixed endpoints and the single-loop fundamental.
2. The 1 m string contains half a wavelength. Find the full wavelength before using the 40 m/s speed.
3. Divide speed by wavelength for the fundamental frequency, then use the harmonic relationship for the second harmonic.

06 Check your understanding

For a fixed string, how does the second-harmonic frequency compare with the fundamental?

Adjacent nodes on an ideal standing wave are separated by:

1. Half a wavelength
2. One wavelength
3. One quarter wavelength

Answers and explanations: page 164.

Light, photons and the electromagnetic picture

OW / 05

- Compare photon energy using frequency or vacuum wavelength.
- Distinguish beam brightness from energy per photon.

You need: Direct and inverse proportionality; wave frequency; energy transfer.

01 Idea

Light behaves as an electromagnetic wave in propagation and exchanges energy in photons. Photon energy increases with frequency. In the photoelectric effect, emission requires enough energy per photon; increasing brightness alone cannot overcome a frequency below threshold.

Photon energy belongs to an individual packet, whereas intensity describes energy arriving per area each second. Increasing frequency makes each packet more energetic. Making a fixed-colour beam brighter can instead increase the number of packets arriving, without changing the energy in each.

- Photon energy increases directly with frequency.
- Vacuum wavelength decreases as frequency increases because vacuum propagation speed is fixed.

02 Rule and symbols

Rule

$$E_\gamma = hf = hc/\lambda_0$$

$\lambda_0 =$ vacuum wavelength

$$c = 3.00 \times 10^8 \text{ m/s}$$

E_γ energy of one photon / J

h Planck constant / J s

f light frequency / Hz

λ vacuum wavelength / m

03 Method

1. Decide whether the question asks about one photon or the whole beam.
2. Use E_γ proportional to frequency, or inversely proportional to vacuum wavelength.
3. For brightness comparisons separate photon energy from the number arriving each second.
4. For photoelectric emission compare energy per photon with the threshold requirement.

Use frequency ratios to compare photon energies and vacuum wavelengths without inserting unnecessary constants.

04 Worked example

Blue light has shorter wavelength than red light and therefore higher photon energy.

01 Blue light has a shorter vacuum wavelength than red light.

02 Since $E_\gamma = hc/\lambda$, the smaller wavelength corresponds to a larger energy per photon.

03 Increasing red-beam brightness at the same colour increases the energy arriving per second, usually through more photons.

04 Each red photon still has the same lower energy; brightness alone does not change its frequency.

Watch out

Higher intensity does not mean higher energy per photon. In E = hc/lambda, use the vacuum wavelength. At an ordinary stationary boundary, frequency and photon energy stay the same even though wavelength in the medium changes.

Exam habit

Use the phrase energy per photon when discussing threshold effects; total beam energy does not answer the same question.

05 Your turn

Light A has twice the frequency of light B. Compare photon energy and vacuum wavelength.

If you get stuck, uncover one hint at a time:

1. The supplied frequency ratio allows comparison without calculating Planck constant.
2. Photon energy follows the same ratio as frequency.
3. Vacuum wavelength follows the reciprocal frequency ratio. State which light has the larger energy and which has the shorter wavelength.

06 Check your understanding

If light A has twice light B frequency, how do photon energy and vacuum wavelength compare?

Light crosses a stationary ordinary air-glass boundary. The transmitted photon energy:

1. Increases solely because the ray bends
2. Stays the same because frequency stays the same
3. Decreases because wavelength in glass is shorter

Answers and explanations: page 164.

Reflection, refraction and total internal reflection

OW / 06

- Track speed, frequency and wavelength across a refracting boundary.
- Recognize the two conditions for total internal reflection.

You need: Normal to a surface; sine; reciprocal refractive index.

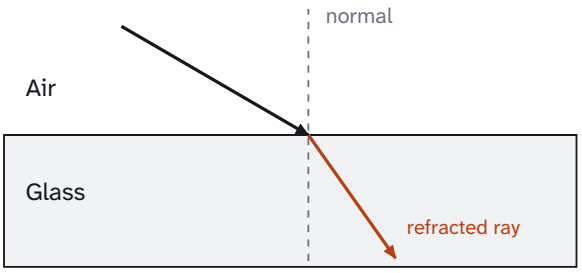
01 Idea

Angles of incidence, reflection and refraction are measured from the normal. At a stationary interface frequency stays the same, while speed and wavelength may change. When light enters a higher refractive index at a nonzero angle, the ray bends toward the normal. Total internal reflection needs travel from higher to lower index and an incidence angle greater than the critical angle.

A ray entering a lower-index medium bends away from the normal. As incidence angle increases, the transmitted direction approaches the boundary. At the critical angle the limiting transmitted ray runs along that boundary; beyond it the ordinary transmitted ray cannot propagate away into the second medium.

FIG. 03 / OPTICS

A ray bends toward the normal in glass



At the boundary

Frequency: unchanged

Speed: decreases

Wavelength: decreases

Takeaway: measure angles from the normal, not from the glass surface.

Figure 03. At oblique incidence from air into glass, the ray bends toward the normal. At a stationary boundary its frequency is unchanged, while speed and wavelength decrease.

- Only higher-to-lower index travel can reach the critical-angle condition.
- Total internal reflection requires incidence above, rather than merely equal to, the critical angle.

02 Rule and symbols

Rule

$$\theta_i = \theta_r$$

$$n = c/v$$

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

$$\sin\theta_c = n_2/n_1$$

n refractive index / dimensionless

c vacuum light speed / m/s

v light speed in medium / m/s

θ_c critical incidence angle / degrees or radians

Use this when:

The boundary is stationary and the media are treated as uniform; air index is approximated as 1.

03 Method

1. Draw the surface normal and measure all ray angles from it.
2. Use v = c/n to compare speeds; frequency stays fixed at a stationary boundary.
3. Use wavelength = speed/frequency to compare wavelengths.
4. For total internal reflection check higher-to-lower index travel, then compare incidence with the critical angle.

The words critical angle signal a check of travel direction followed by the lower-to-higher index sine ratio.

04 Worked example

From air into glass with index 1.5, light slows to $2 \times 10^8\ \mathrm{m/s}$.

01 Air is approximated by n = 1 and glass has n = 1.5.

02 In glass, speed is $3.00 \times 10^8 / 1.5 = 2.00 \times 10^8$ m/s.

03 The frequency is set by the source and remains the same across the boundary.

04 Because speed drops while frequency stays fixed, wavelength in glass is two thirds of its air value.

Watch out

The critical angle is an incidence angle in the higher-index medium.

Exam habit

At the critical angle the refracted ray runs along the boundary; total internal reflection occurs above that angle.

05 Your turn

Find the critical angle for glass of index 1.5 into air of index 1.0.

If you get stuck, uncover one hint at a time:

1. Confirm that light travels from glass index 1.5 into air index 1.0.
2. Divide the lower index by the higher index to obtain the critical-angle sine.
3. Take inverse sine and report an angle from the normal. Check calculator degree mode if using degrees.

06 Check your understanding

Can light travelling from air into glass undergo total internal reflection at that interface?

At exactly the critical angle, the refracted ray travels:

1. Along the normal

2. Back along the incident ray
3. Along the boundary

Answers and explanations: page 164.

Mirrors, lenses and image signs

OW / 07

- Use the stated image-distance sign convention consistently.
- Infer image reality, orientation and relative size from di and m.

You need: Reciprocals; positive and negative numbers; a simple ray sketch.

01 Idea

A real image is formed where rays actually meet and can be projected on a screen. A virtual image is where rays appear to originate. Use one sign convention consistently: real object distance positive, real image distance positive; converging focal length positive and diverging focal length negative.

A lens equation locates where rays meet or appear to originate. Magnification then compares image height with object height, including orientation. A negative magnification records an inverted image; it does not mean the physical image has a negative length.

- A converging lens with an object at twice focal length gives a real image at twice focal length on the other side.
- Magnitude of magnification gives relative size, while its sign gives orientation.

02 Rule and symbols

Rule

$$1/f = 1/d_o + 1/d_i$$

$$m = -d_i/d_o$$

f signed focal length / cm or m consistently

d_o object distance / same length unit as f

d_i signed image distance / same length unit as f

m transverse magnification / dimensionless

Use this when:

Use the lesson convention: real object and real image distances positive; converging focal length positive.

03 Method

1. Write the sign convention and assign f and do before substitution.
2. Rearrange the lens equation to isolate 1/di and then invert.
3. Calculate m = -di/do.
4. Use di sign for real or virtual, m sign for orientation, and magnitude of m for size.

Use one consistent sign convention when focal length and object distance are given, then interpret the numerical result.

04 Worked example

For a converging lens with focal length 10 cm and object distance 30 cm, image distance is 15 cm and magnification is -0.5: real, inverted, reduced.

01 A converging lens has f = +10 cm and the real object has do = +30 cm.

02 Then 1/di = 1/10 - 1/30 = 1/15 per cm, so di = +15 cm.

03 Magnification is -15/30 = -0.5.

04 Positive image distance means real; negative magnification means inverted; magnitude 0.5 means half the object height.

Watch out

Do not call every image behind a lens virtual; interpret actual ray convergence and your convention.

Exam habit

After subtraction, invert 1/di to get di; include all three requested image descriptions.

05 Your turn

A converging lens has focal length 12 cm and an object at 24 cm. Find image distance, magnification and image type.

If you get stuck, uncover one hint at a time:

1. Assign positive focal length to the converging lens and positive distance to the real object.
2. Subtract the reciprocal of 24 cm from the reciprocal of 12 cm, then invert for image distance.
3. Calculate signed magnification and interpret distance sign, orientation and size separately.

06 Check your understanding

What does m = +2 tell you about an image?

Why can a real image be projected onto a screen?

1. Actual rays meet at its position
2. Its magnification must be positive
3. All images lie on the lens surface

Answers and explanations: page 164.

Optical systems, prisms and instruments

OW / 08

- Explain the different roles of objective and eyepiece.
- Calculate telescope angular magnification for normal adjustment.

You need: Focal length; image formation; ratios of lengths.

01 Idea

A prism refracts light at two surfaces; dispersion separates colours because refractive index depends on wavelength. In a multi-lens system, the first image becomes the object for the next lens. A microscope and a telescope both use an objective and eyepiece, but they serve different object distances.

A telescope objective forms an intermediate image of a distant scene, and an eyepiece enlarges the angle under which it is viewed. Angular magnification compares viewing angles, so it is different from the height ratio for one ordinary lens image.

- For normal adjustment, a shorter eyepiece focal length gives larger angular magnification with the same objective.
- A focal-length ratio is dimensionless when both lengths use the same unit.

02 Rule and symbols

Rule

$$m_{\text{total}} = m_1 m_2$$

$$|M_{\text{telescope}}| = f_o / f_e \text{ (normal adjustment)}$$

f_o objective focal length / cm or m consistently

f_e eyepiece focal length / same length unit as fo

M angular magnification / dimensionless

m_1, m_2 transverse magnifications of successive stages / dimensionless

Use this when:

The telescope is in normal adjustment, giving the final image at infinity.

03 Method

1. Identify the instrument and whether it views a distant or nearby object.
2. For a normal-adjustment astronomical telescope divide objective focal length by eyepiece focal length for magnification magnitude.
3. State image orientation separately.
4. For a general multi-lens system locate the first image before using it as the next object.

Use the focal-length ratio for an astronomical telescope in normal adjustment, not for an arbitrary two-lens system.

04 Worked example

An astronomical telescope with objective focal length 80 cm and eyepiece 4 cm has angular magnification magnitude 20.

01 The telescope objective focal length is 80 cm and the eyepiece focal length is 4 cm.

02 Their ratio is 80/4 = 20, so the angular magnification magnitude is 20.

03 An astronomical telescope with this usual two-converging-lens arrangement gives an inverted image.

04 The ratio needs no length conversion because both focal lengths use the same unit.

Watch out

Linear image magnification and angular magnification are different quantities.

Exam habit

Use the telescope focal-length ratio only for the stated telescope model, not as a universal magnification formula.

05 Your turn

A telescope objective has focal length 100 cm and eyepiece 5 cm. Find angular magnification magnitude. What happens if the eyepiece is replaced by 10 cm?

If you get stuck, uncover one hint at a time:

1. Identify the objective focal length and eyepiece focal length.
2. Divide 100 cm by 5 cm for initial magnification magnitude.
3. Replace only the denominator with 10 cm and compare ratios. Keep image orientation separate from the requested magnitude.

06 Check your understanding

Keeping the objective fixed, does doubling the eyepiece focal length increase or decrease telescope magnification?

Which eyepiece gives greater magnification with the same objective in normal adjustment?

1. Both give equal magnification
2. The shorter-focal-length eyepiece
3. The longer-focal-length eyepiece

Answers and explanations: page 164.

Interference and diffraction

OW / 09

- Predict how slit geometry changes an interference or diffraction pattern.
- Distinguish double-slit bright fringes from single-slit dark fringes.

You need: Wavelength; coherent sources; nanometres and millimetres; small angles.

01 Idea

Interference occurs when overlapping waves combine. A stable visible pattern needs a fixed phase relationship (coherence). Diffraction is spreading associated with an opening or obstacle. Narrower slits spread light more. Young double-slit bright fringes are evenly spaced when the angles are small.

Coherent slits create bright and dark directions through path differences between overlapping waves. A screen turns those angles into distances. Moving the screen farther away enlarges the visible spacing even though the slit geometry and bright angular directions remain unchanged.

- Longer wavelength increases the angular change needed between neighbouring bright fringes.
- A more distant screen maps the same angular separation to a larger physical distance.

02 Rule and symbols

Rule

$$\Delta y = \lambda L / d$$

$d \sin\theta = m\lambda$ (double-slit bright)

$a \sin\theta = m\lambda$ (single-slit dark, $m \neq 0$)

Δy double-slit fringe spacing / m

L slit-to-screen distance / m

d separation of double slits / m

a single-slit width / m

λ wavelength / m

Use this when:

Even double-slit spacing uses the small-angle, distant-screen approximation and coherent illumination.

03 Method

1. Identify whether the setup is double-slit interference or single-slit diffraction.
2. Label d as slit separation and a as slit width; they describe different distances.
3. For double-slit spacing use λL/d with all lengths in consistent units.
4. For a single slit remember that a smaller width moves the first minima outward and broadens the centre.

For two coherent narrow slits and a distant screen, use wavelength, slit separation and screen distance to find spacing.

04 Worked example

Doubling screen distance doubles double-slit fringe spacing.

01 Double-slit spacing is proportional to screen distance when wavelength and slit separation stay fixed.

02 Replacing L by 2L therefore doubles fringe spacing.

03 In the separate single-slit comparison, the first dark direction obeys sin θ = λ/a.

04 Halving a makes that angle larger, so the bright central maximum broadens.

Watch out

The double-slit bright condition is not the single-slit bright condition.

Exam habit

Label whether you are locating a bright double-slit fringe or a dark single-slit fringe before using the order number.

05 Your turn

In a double-slit experiment, wavelength is 500 nm, screen distance 2 m and slit separation 0.50 mm. Find fringe spacing.

If you get stuck, uncover one hint at a time:

1. Distinguish slit separation from slit width; this is a double-slit spacing question.
2. Convert 500 nm and 0.50 mm to metres before combining them with 2 m.
3. Multiply wavelength by screen distance and divide by slit separation. Convert the final spacing to millimetres if helpful.

06 Check your understanding

If double-slit separation doubles with screen distance and wavelength fixed, what happens to fringe spacing?

Only the double-slit screen distance doubles. The bright-fringe angular directions:

1. Double along with screen spacing
2. All become zero
3. Stay unchanged while screen spacing grows

Answers and explanations: page 164.

Recognize gratings, Michelson paths and thin films

All three appear in the physical-optics section. Start by identifying which arrangement the question describes.

- A diffraction grating has many equally spaced slits. It makes sharp bright directions that separate wavelengths.
- A Michelson interferometer splits light into two arms. Each beam travels to a mirror and back, so mirror movement changes a round-trip path.
- A thin film produces interference between reflections from its two surfaces. Reflection can add a phase reversal, so do not copy the ordinary double-slit bright rule.
- For fringe counting in a simple Michelson setup, each one-wavelength change in path difference shifts the pattern by one fringe.
- $\Delta\text{path} = 2\Delta x$ (one Michelson mirror moves)

In a Michelson interferometer using $500\,\text{nm}$ light in air, one mirror moves $250\,\text{nm}$ along its arm. What path change and fringe shift result?

1. Round-trip path change: $2 \times 250 = 500\,\text{nm}$.
2. This equals one wavelength.

The path changes by $500\,\text{nm}$, so one fringe passes a fixed observation point.

Polarization and Malus law

OW / 10

- Apply the initial half-intensity rule to unpolarized light.
- Apply Malus law using the angle between polarization and analyzer axes.

You need: Cosine of common angles; intensity versus field amplitude; sequential fractions.

01 Idea

Polarization describes the electric-field direction of a transverse light wave. An ideal polarizer transmits half the intensity of unpolarized light. An analyzer then transmits a fraction set by the angle between its axis and the incident polarization. Linear, circular and elliptical polarization describe different field-tip motions.

The first ideal polarizer turns unpolarized light into linearly polarized light while transmitting half its intensity. The analyzer receives that output. It then passes a fraction determined by the angle to the already established polarization, so the first and second stages use different rules.

- The half-intensity rule applies to initially unpolarized input.
- For linearly polarized input, the analyzer transmits a squared-cosine fraction.

02 Rule and symbols

Rule

$$I = I_0 \cos^2 \theta$$

$$I_{\text{after first}} = I_{\text{unpolarized}}/2$$

I_0 intensity incident on the analyzer / W/m^2 or stated units

I intensity transmitted by analyzer / same as I0

θ angle between incident polarization and analyzer axis / degrees or radians

Use this when:

The calculation assumes ideal polarizers and initially unpolarized illumination.

03 Method

1. Check whether the first incident beam is unpolarized or already linearly polarized.
2. For unpolarized light through an ideal first polarizer divide intensity by two.
3. Use that transmitted intensity as I0 for the analyzer.
4. Multiply by $\cos^2\theta$, where θ is between the two relevant axes.

For successive polarizers, identify input polarization and process each element in order.

04 Worked example

Unpolarized intensity 100 units passes a polarizer then an analyzer at 60 degrees.

01 The initial 100 intensity units are unpolarized, so the first ideal polarizer transmits 50 units.

02 The analyzer axis is at 60° to the first polarizer axis.

03 Cos 60° is 0.5 and its square is 0.25.

04 The analyzer therefore transmits 50 times 0.25 = 12.5 units, one eighth of the original intensity.

Watch out

Do not halve already polarized intensity again before applying Malus law.

Exam habit

Square the cosine, not the angle; and do not apply an extra half factor to an already polarized analyzer input.

05 Your turn

Unpolarized intensity 80 W per square metre passes through a polarizer and an analyzer at 45 degrees. Find the final intensity.

If you get stuck, uncover one hint at a time:

1. Start with the unpolarized 80 W/m^2 and apply the first-polarizer rule.
2. Use that transmitted intensity as the analyzer input.
3. Multiply by the squared cosine of 45 degrees. Keep the two stages visible rather than applying an extra half factor.

06 Check your understanding

Two ideal polarizers have perpendicular axes. What does the second transmit from the first?

A second ideal polarizer is aligned with the first. What fraction of the first output passes it?

1. All of it
2. Half of it
3. None of it

Answers and explanations: page 164.

Scattering, apparent depth and lens power

OW / 11

- Calculate apparent depth near normal viewing and signed lens power.
- Explain why small-particle scattering favours shorter wavelengths.

You need: Refraction; inverse relationships; centimetres-to-metres conversion.

01 Idea

Small particles compared with wavelength produce Rayleigh scattering, which favours shorter wavelengths strongly. Droplets comparable to or larger than visible wavelengths can scatter a broad range of colours, helping explain white clouds. Refraction makes a submerged object appear shallower when viewed near the normal from air.

Lens power describes how strongly a lens converges or diverges rays. A shorter focal-length magnitude means stronger bending and greater power magnitude. The negative sign of a diverging lens communicates the direction of its optical effect, rather than the physical size of the lens.

- Reciprocals make smaller focal-length magnitudes correspond to larger power magnitudes.
- Dioptres use metres in the focal-length reciprocal, so centimetres must be converted first.

02 Rule and symbols

Rule

$$I_{\text{Rayleigh}} \propto 1/\lambda^4$$

$$d_{\text{apparent}} \approx d_{\text{real}}/n$$

$$P_{\text{lens}} = 1/f \text{ (f in metres)}$$

d_{real} actual depth / m

d_{apparent} apparent depth / m

n liquid refractive index relative to air approximation / dimensionless

P lens optical power / D = m^{-1}

f signed focal length / m

Use this when:

The apparent-depth approximation assumes viewing from air near the normal; the Rayleigh rule applies to particles much smaller than wavelength.

03 Method

1. Select the relevant model: small-particle scattering, near-normal depth or lens power.
2. For depth viewed from air divide real depth by the liquid index.
3. For power convert focal length to metres before taking its reciprocal.
4. Retain the focal-length sign so converging power is positive and diverging power negative.

A supplied focal length and a request for optical power call for a signed reciprocal in metre units.

04 Worked example

A pool 2 m deep in water of index 1.33 appears about 1.50 m deep from air near the normal.

01 For the pool, near-normal apparent depth is 2/1.33 ≈ 1.50 m, so the bottom appears shallower.

02 The index does not change the real 2 m depth; it changes the directions of the emerging rays.

03 For the separate converging lens, f = +0.25 m.

04 Its power is 1/0.25 = +4 m^{-1}, written +4 dioptres.

Watch out

A focal length of 25 cm is not 25 m; using it directly gives the wrong lens power.

Exam habit

A reciprocal of a focal length in centimetres does not directly give dioptres; convert to metres first.

05 Your turn

A diverging lens has focal length -50 cm. Find its optical power.

If you get stuck, uncover one hint at a time:

1. Retain the negative focal-length sign because the lens diverges.
2. Convert -50 cm into metres.
3. Take the signed reciprocal and report dioptres. A positive result would conflict with the stated lens type.

06 Check your understanding

A diverging lens has focal length -50 cm. What is its power?

Which diverging lens has greater power magnitude: focal length -0.25 m or -0.50 m?

1. They match because both diverge
2. The -0.25 m lens
3. The -0.50 m lens

Answers and explanations: page 164.

Recognize advanced optics without long derivations

OW / 12

- Recognize what a quarter-wave plate changes about polarized light.
- Use Brewster angle and plane-wave relations at recognition level.

You need: Linear and circular polarization; phase as position within an oscillation cycle; refractive index.

01 Idea

The wave equation describes a disturbance propagating in space and time. Complex notation is a bookkeeping method for amplitude and phase; the measured field is real. Maxwell laws connect electric and magnetic fields. Birefringent materials have different refractive behaviour for different polarizations. At Brewster incidence the reflected component can be linearly polarized.

Brewster incidence is a special transparent-interface geometry. In ordinary transparent nonmagnetic media, the reflected light at that angle lacks the component polarized in the incidence plane. This concerns polarization of the reflected component, not complete reflection of all incoming energy.

- The refractive-index ratio sets the tangent of the Brewster angle from the normal.
- Brewster incidence and total internal reflection describe different conditions and effects.

02 Rule and symbols

Rule

$$\tan\theta_B = n_2/n_1$$

$E = cB$ (plane wave in vacuum)

θ_B Brewster incidence angle from the normal / degrees or radians

n_1, n_2 incident and transmitted medium indices / dimensionless

E plane-wave electric-field magnitude / V/m

B plane-wave magnetic-field magnitude / T

Use this when:

The wave plate is operating at its intended wavelength; Brewster law here uses ordinary transparent nonmagnetic media.

03 Method

1. Identify whether the question concerns phase delay, interface polarization or vacuum wave fields.
2. For a quarter-wave plate, compare the two orthogonal component amplitudes and their phase difference.
3. For Brewster incidence use the index ratio with inverse tangent.
4. For a plane vacuum wave connect field magnitudes using E = cB without a long derivation.

Use Brewster angle when an interface question links refractive indices and linearly polarized reflection.

04 Worked example

A quarter-wave plate can turn suitable linearly polarized light into circular polarization by introducing a quarter-cycle phase difference between equal orthogonal components.

01 Resolve suitable incident linear polarization along the plate principal axes.

02 Equal component amplitudes are required for the stated circular-polarization result.

03 A quarter-wave plate adds a quarter-cycle, or $\pi/2$, relative phase delay.

04 Equal amplitudes with this delay give circular polarization; unequal amplitudes generally give elliptical polarization.

Watch out

These recognition notes do not replace a lecturer-assigned derivation; return to the source if such derivations are explicitly assessed.

Exam habit

For a recognition answer name the condition and consequence; deriving Maxwell equations is unnecessary for these exercises.

05 Your turn

For light from air into glass of index 1.5, estimate Brewster angle.

If you get stuck, uncover one hint at a time:

1. Choose the Brewster tangent relation, not the critical-angle sine relation.
2. Place glass index 1.5 over air index approximately 1.0.
3. Take inverse tangent of that ratio and measure the result from the normal. Check degree mode if reporting degrees.

06 Check your understanding

Is a quarter-cycle phase delay alone enough to guarantee circular polarization?

For unpolarized light at Brewster incidence in ordinary transparent nonmagnetic media:

1. All incident light must be reflected
2. Total internal reflection must occur
3. The reflected light is linearly polarized

Answers and explanations: page 164.

Lesson review and answers

Wave language and simple harmonic motion

Return to lesson: page 125.

Your turn: No. Displacement and restoring acceleration are zero; speed is maximum.

Concept check: 1. Correct. Twice the time per cycle gives half as many cycles each second.

Written method: Frequency = 50/10 = 5 Hz. Period = 10/50 = 0.20 s. Two marks each.

Read a travelling wave without deriving it

Return to lesson: page 128.

Your turn: No. It is angular frequency; divide it by 2π to get 10 Hz.

Concept check: 2. Correct. At fixed frequency, lower speed means less distance travelled per cycle.

Written method: Speed = 8 times 0.75 = 6 m/s. New wavelength = 6/12 = 0.50 m. Two marks each.

Wavefronts, spreading and group velocity

Return to lesson: page 132.

Your turn: 2 W/m^2: divide by 3^2 = 9.

Concept check: 3. Correct. The unchanged source power is shared over each complete sphere.

Written method: Inverse-square spreading gives 18/9 = 2 W per square metre. The distance triples, so the area over which power spreads increases ninefold.

Superposition, standing waves and beats

Return to lesson: page 135.

Your turn: It is twice the fundamental frequency in the ideal uniform-string model.

Concept check: 1. Correct. One loop between neighbouring nodes occupies half a wavelength.

Written method: Fundamental wavelength = 2 m; frequency = 40/2 = 20 Hz. Second harmonic = 40 Hz. Two marks per frequency with reasoning.

Light, photons and the electromagnetic picture

Return to lesson: page 138.

Your turn: A has twice the photon energy and half the vacuum wavelength.

Concept check: 2. Correct. Unchanged frequency gives unchanged photon energy.

Written method: A has twice the photon energy and half the vacuum wavelength. Two marks for each relationship.

Reflection, refraction and total internal reflection

Return to lesson: page 141.

Your turn: No. Total internal reflection requires travel from higher to lower refractive index.

Concept check: 3. Correct. The transmitted angle from the normal is 90 degrees.

Written method: Sine of critical angle = $\frac{1}{1}$.5 = 0.6667, so critical angle = 41.8 degrees. Total internal reflection occurs above this angle. Credit condition, equation, value and interpretation.

Mirrors, lenses and image signs

Return to lesson: page 145.

Your turn: It is upright and twice the object height; the sign convention and lens setup determine its image position.

Concept check: 1. Correct. A screen intercepts the rays where they form the image.

Written method: 1/di = 1/12 - 1/24 = 1/24, so di = 24 cm. m = -1. Image is real, inverted and equal size. One mark each for distance, magnification, real and inverted/equal size.

Optical systems, prisms and instruments

Return to lesson: page 148.

Your turn: It halves the magnification magnitude.

Concept check: 2. Correct. The eyepiece focal length is the denominator of the ratio.

Written method: Initial magnitude = 100/5 = 20. New magnitude = 100/10 = 10. Doubling eyepiece focal length halves angular magnification.

Interference and diffraction

Return to lesson: page 151.

Your turn: It halves.

Concept check: 3. Correct. Wavelength and slit separation set directions; screen distance converts direction to position.

Written method: Convert wavelength to 0.0000005 m and separation to 0.0005 m. Spacing = (0.0000005 times 2)/0.0005 = 0.002 m = 2 mm. Credit both conversions, calculation and result.

Polarization and Malus law

Return to lesson: page 155.

Your turn: Zero intensity in the ideal model, because $\cos^2 90° = 0$.

Concept check: 1. Correct. The input is already aligned, so the squared cosine of zero is one.

Written method: First polarizer gives 40. Analyzer gives 40 times cosine squared 45 degrees = 20 W per square metre. Two marks for each stage.

Scattering, apparent depth and lens power

Return to lesson: page 158.

Your turn: -2 D, since -50 cm = -0.50 m.

Concept check: 2. Correct. The smaller focal-length magnitude has the larger reciprocal magnitude.

Written method: Convert to -0.50 m. Power=1/(-0.50)=-2 dioptres. Credit conversion, formula, sign and unit.

Recognize advanced optics without long derivations

Return to lesson: page 161.

Your turn: No. The two orthogonal components must also have equal amplitudes.

Concept check: 3. Correct. The reflected component lacks one linear polarization component.

Written method: Tangent of Brewster angle=1.5/1.0. The angle is arctan(1.5)=56.3 degrees. Credit correct ratio, inverse tangent, result and interpretation as incidence angle.

Optics and Waves / Test A

30 marks / Suggested time: 30 minutes. MCQs: 1 mark each. Written problems: 4 marks each. Attempt without notes.

Part I / Choose one answer

1. Which wave can propagate through vacuum?

   A. A water wave

   B. Light

   C. A stretched-string wave

   D. Sound in air

2. With wave speed fixed, doubling frequency makes wavelength:

   A. Half as large

   B. Unchanged

   C. Four times as large

   D. Twice as large

3. Group velocity describes the motion of:

   A. The wave-packet envelope

   B. The source itself

   C. One particle only

   D. A stationary node

4. Two tones at 440 and 443 Hz produce a beat frequency of:

   A. 1320 Hz

   B. 883 Hz

   C. 3 Hz

   D. 441.5 Hz

5. At the same medium, which has greater photon energy?

   A. Lower-frequency light

   B. Both always equal

   C. Higher-frequency light

D. Longer-wavelength light

6. Light enters glass from air. Which quantity remains unchanged?

A. Ray direction in every case

B. Speed

C. Wavelength

D. Frequency

Part II / Show your reasoning

1. A source makes 50 complete oscillations in 10 s. Find frequency and period.

2. A wave has frequency 8 Hz and wavelength 0.75 m. Find speed. If frequency becomes 12 Hz in the same nondispersive medium, find wavelength.

3. At distance r an isotropic source has intensity 18 W per square metre. Find intensity at 3r, ignoring absorption.

4. A 1 m string fixed at both ends supports waves at 40 m/s. Find its fundamental frequency and second harmonic.

5. Light A has twice the frequency of light B. Compare photon energy and vacuum wavelength.

6. Find the critical angle for glass of index 1.5 into air of index 1.0.

Optics and Waves / Test B

30 marks / Suggested time: 30 minutes. MCQs: 1 mark each. Written problems: 4 marks each. Attempt without notes.

Part I / Choose one answer

1. An image with magnification -2 is:

   A. Upright and smaller

   B. Inverted and twice as tall

   C. Virtual in every case

   D. Upright and twice as tall

2. In a simple astronomical telescope, the objective usually has:

   A. A longer focal length than the eyepiece

   B. Zero focal length

   C. The same role as a polarizer

   D. No effect on the image

3. If double-slit separation increases while wavelength and screen distance stay fixed, fringe spacing:

   A. Becomes negative

   B. Increases

   C. Stays fixed

   D. Decreases

4. Two ideal crossed polarizers transmit from an initially unpolarized beam:

   A. One quarter

   B. Zero

   C. All its intensity

   D. Half

5. The unit of optical lens power is:

   A. Hertz

   B. Watt

C. Dioptre

D. Newton

6. A quarter-wave plate acts primarily by changing the relative:

A. Charge of light

B. Speed of the source

C. Mass of photons

D. Phase of orthogonal components

Part II / Show your reasoning

1. A converging lens has focal length 12 cm and an object at 24 cm. Find image distance, magnification and image type.

2. A telescope objective has focal length 100 cm and eyepiece 5 cm. Find angular magnification magnitude. What happens if the eyepiece is replaced by 10 cm?

3. In a double-slit experiment, wavelength is 500 nm, screen distance 2 m and slit separation 0.50 mm. Find fringe spacing.

4. Unpolarized intensity 80 W per square metre passes through a polarizer and an analyzer at 45 degrees. Find the final intensity.

5. A diverging lens has focal length -50 cm. Find its optical power.

6. For light from air into glass of index 1.5, estimate Brewster angle.

Optics and Waves / Test A / Answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1

MCQ: B. Light is electromagnetic and does not require a material medium.

Written solution: Frequency = 50/10 = 5 Hz. Period = 10/50 = 0.20 s. Two marks each.

Return to lesson: page 125.

Question 2

MCQ: A. Wavelength equals speed divided by frequency.

Written solution: Speed = 8 times 0.75 = 6 m/s. New wavelength = 6/12 = 0.50 m. Two marks each.

Return to lesson: page 128.

Question 3

MCQ: A. The group is the envelope formed by combining nearby frequencies.

Written solution: Inverse-square spreading gives 18/9 = 2 W per square metre. The distance triples, so the area over which power spreads increases ninefold.

Return to lesson: page 132.

Question 4

MCQ: C. Beat frequency is the absolute difference.

Written solution: Fundamental wavelength = 2 m; frequency = 40/2 = 20 Hz. Second harmonic = 40 Hz. Two marks per frequency with reasoning.

Return to lesson: page 135.

Question 5

MCQ: C. Photon energy is proportional to frequency.

Written solution: A has twice the photon energy and half the vacuum wavelength.

Two marks for each relationship.

Return to lesson: page 138.

Question 6

MCQ: D. The boundary does not change the frequency of the incident oscillation.

Written solution: Sine of critical angle = $\frac{1}{1}$.5 = 0.6667, so critical angle = 41.8 degrees. Total internal reflection occurs above this angle. Credit condition, equation, value and interpretation.

Return to lesson: page 141.

Optics and Waves / Test B / Answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1

MCQ: B. The minus sign indicates inversion and magnitude 2 gives the size ratio.

Written solution: 1/di = 1/12 - 1/24 = 1/24, so di = 24 cm. m = -1. Image is real, inverted and equal size. One mark each for distance, magnification, real and inverted/equal size.

Return to lesson: page 145.

Question 2

MCQ: A. The long-focus objective and short-focus eyepiece give a large angular-magnification ratio.

Written solution: Initial magnitude = 100/5 = 20. New magnitude = 100/10 = 10. Doubling eyepiece focal length halves angular magnification.

Return to lesson: page 148.

Question 3

MCQ: D. Fringe spacing is inversely proportional to slit separation.

Written solution: Convert wavelength to 0.0000005 m and separation to 0.0005 m. Spacing = (0.0000005 times 2)/0.0005 = 0.002 m = 2 mm. Credit both conversions, calculation and result.

Return to lesson: page 151.

Question 4

MCQ: B. Their axes differ by 90 degrees and $\cos^2 90^\circ$ is zero.

Written solution: First polarizer gives 40. Analyzer gives 40 times cosine squared 45 degrees = 20 W per square metre. Two marks for each stage.

Return to lesson: page 155.

Question 5

MCQ: C. Lens power is inverse focal length in metres, expressed in dioptres.

Written solution: Convert to -0.50 m. Power=1/(-0.50)=-2 dioptres. Credit conversion, formula, sign and unit.

Return to lesson: page 158.

Question 6

MCQ: D. A controlled phase delay changes the polarization state.

Written solution: Tangent of Brewster angle=1.5/1.0. The angle is arctan(1.5)=56.3 degrees. Credit correct ratio, inverse tangent, result and interpretation as incidence angle.

Return to lesson: page 161.

Formula sheet

Wave language and simple harmonic motion

p.125

Rule

$$T = 1/f$$

$$f = N/t$$

$$a = -\omega^2 x$$

Read a travelling wave without deriving it

p.128

Rule

$$v = f\lambda$$

$$k = 2\pi/\lambda$$

$$\omega = 2\pi f$$

$$y = A\sin(kx - \omega t + \phi)$$

Wavefronts, spreading and group velocity

p.132

Rule

$$I \propto A^2$$

$$I_{\text{spherical}} \propto 1/r^2$$

Superposition, standing waves and beats

p.135

Rule

$$f_{\text{beat}} = |f_1 - f_2|$$

$\lambda_n = 2L/n$ (string fixed at both ends)

Light, photons and the electromagnetic picture

p.138

Rule

$$E_\gamma = hf = hc/\lambda_0$$

$\lambda_0 =$ vacuum wavelength

$$c = 3.00 \times 10^8 \text{ m/s}$$

Reflection, refraction and total internal reflection

p.141

Rule

$$\theta_i = \theta_r$$

$$n = c/v$$

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

$$\sin\theta_c = n_2/n_1$$

Mirrors, lenses and image signs

p.145

Rule

$$1/f = 1/d_o + 1/d_i$$

$$m = -d_i/d_o$$

Optical systems, prisms and instruments

p.148

Rule

$$m_{\text{total}} = m_1 m_2$$

$|M_{\text{telescope}}| = f_o/f_e$ (normal adjustment)

Interference and diffraction

p.151

Rule

$$\Delta y = \lambda L/d$$

$d \sin\theta = m\lambda$ (double-slit bright)

$a \sin\theta = m\lambda$ (single-slit dark, $m \neq 0$)

Polarization and Malus law

p.155

Rule

$$I = I_0 \cos^2\theta$$

$$I_{\text{after first}} = I_{\text{unpolarized}}/2$$

Scattering, apparent depth and lens power

p.158

Rule

$$I_{\text{Rayleigh}} \propto 1/\lambda^4$$

$$d_{\text{apparent}} \approx d_{\text{real}}/n$$

$P_{\text{lens}} = 1/f$ (f in metres)

Recognize advanced optics without long derivations

p.161

Rule

$$\tan\theta_B = n_2/n_1$$

$E = cB$ (plane wave in vacuum)

Symbol glossary

Symbols are listed with their lesson context. A symbol may represent different quantities in different rules.

Wave language and simple harmonic motion

N number of complete cycles / dimensionless

t total elapsed time / s

f cycles per second / Hz

T time for one cycle / s

ω angular frequency / rad/s

Read a travelling wave without deriving it

A amplitude or maximum displacement / m

k wave number / rad/m

ω angular frequency / rad/s

λ wavelength / m

f frequency / Hz

Wavefronts, spreading and group velocity

I energy-transfer rate per unit area / W/m^2

A wave amplitude / depends on disturbance

r distance from point source / m

Superposition, standing waves and beats

f_{beat} number of loudness cycles per second / Hz

L vibrating string length / m

n harmonic number, 1, 2, 3, … / dimensionless

λ_n wavelength of harmonic n / m

Light, photons and the electromagnetic picture

E_γ energy of one photon / J

h Planck constant / J s

f light frequency / Hz

λ vacuum wavelength / m

Reflection, refraction and total internal reflection

n refractive index / dimensionless

c vacuum light speed / m/s

v light speed in medium / m/s

θ_c critical incidence angle / degrees or radians

Mirrors, lenses and image signs

f signed focal length / cm or m consistently

d_o object distance / same length unit as f

d_i signed image distance / same length unit as f

m transverse magnification / dimensionless

Optical systems, prisms and instruments

f_o objective focal length / cm or m consistently

f_e eyepiece focal length / same length unit as fo

M angular magnification / dimensionless

m_1, m_2 transverse magnifications of successive stages / dimensionless

Interference and diffraction

Δy double-slit fringe spacing / m

L slit-to-screen distance / m

d separation of double slits / m

a single-slit width / m

λ wavelength / m

Polarization and Malus law

I_0 intensity incident on the analyzer / W/m^2 or stated units

I intensity transmitted by analyzer / same as I0

θ angle between incident polarization and analyzer axis / degrees or radians

Scattering, apparent depth and lens power

d_{real} actual depth / m

d_{apparent} apparent depth / m

n liquid refractive index relative to air approximation / dimensionless

P lens optical power / D = m^{-1}

f signed focal length / m

Recognize advanced optics without long derivations

θ_B Brewster incidence angle from the normal / degrees or radians

n_1, n_2 incident and transmitted medium indices / dimensionless

E plane-wave electric-field magnitude / V/m

B plane-wave magnetic-field magnitude / T

Source and coverage notes

The introduction and four current sections: wave motion; electromagnetic theory, photons and light; geometrical optics; physical optics. Derivations are replaced by plain-language interpretation and short exam calculations.

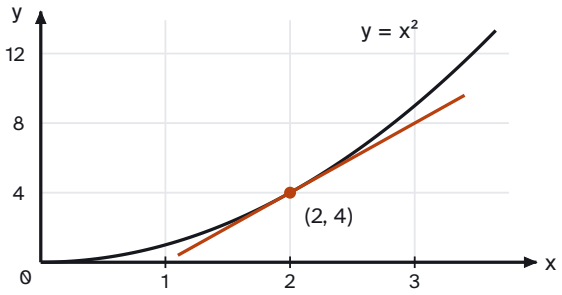
Open original course folder

The source audit inventories 159 non-programming files; 95 yielded substantial extracted text. Scanned diagrams and handwritten or image-only material were not fully transcribed. Check lecturer-specific derivations and conventions. The practice tests are original study material, not predictions of an examination.

Calculus I

FIG. 04 / CALCULUS

A tangent measures the local rate of change



At x = 2

Slope = 2x = 4

Tangent equation:
y - 4 = 4(x - 2)

Takeaway: use coordinate changes for slope; the axes have different scales.

MATH 203: derivative rules, implicit/logarithmic/exponential derivatives, L Hopital, graphing and extrema, optimization, Rolle and mean-value theorems, integration and the fundamental theorem. The full lecture syllabus is the main source.

Subject map

MATH 203: derivative rules, implicit/logarithmic/exponential derivatives, L Hopital, graphing and extrema, optimization, Rolle and mean-value theorems, integration and the fundamental theorem. The full lecture syllabus is the main source.

Follow this reading route from foundations to applications. Each line links to its lesson; revisit the stated prerequisites whenever a step feels unfamiliar.

01 Limits, continuity and the derivative definition p.187

02 Basic derivatives and tangent lines p.190

03 Product and quotient rules p.193

04 Chain rule and trigonometric functions p.196

05 Implicit differentiation p.200

06 Logarithmic, exponential and inverse-trig derivatives p.203

07 L Hopital and indeterminate forms p.207

08 Increasing, decreasing and concavity p.210

09 Graphing, asymptotes and absolute extrema p.213

10 Optimization and mean-value theorems p.216

11 Antiderivatives and substitution p.219

12 Definite integrals, signed area and the fundamental theorem p.222

Limits, continuity and the derivative definition

C1 / 01

- Distinguish a limit from a value at the point.
- Find a polynomial derivative from the difference quotient.

You need: Expanding a square; factoring and cancelling a nonzero common factor.

01 Idea

A limit describes nearby values, not necessarily the function value at the point. Continuity requires the function value and the two-sided limit to exist and agree. The derivative is the limiting slope of secant lines and gives instantaneous rate of change.

A secant line measures change across separated inputs. As those inputs approach one another, its slope can approach a single instantaneous slope: the derivative. The increment must remain nonzero while it is a denominator, even though the limiting process eventually removes it.

- Factoring and cancelling the increment can remove the apparent zero denominator.
- The nearby simplified slopes determine the limit, rather than an undefined quotient at zero increment.

02 Rule and symbols

Rule

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

h small nonzero input increment before the limit / same unit as x

$f'(x)$ instantaneous rate of change / output unit per input unit

03 Method

1. Write f(x+h) and subtract the complete original f(x).
2. Expand brackets carefully and collect terms.

3. Factor out h and cancel only while h is nonzero.

4. Take the limit h approaching zero in the simplified expression.

If the question says use the definition, construct the difference quotient instead of citing the power rule.

04 Worked example

For $f(x) = x^2$, the difference quotient becomes $\frac{2xh+h^2}{h} = 2x + h$.

01 For f(x) = x^2, the incremented value is $(x+h)^2$ = x^2 + 2xh + h^2.

02 Subtracting f(x) leaves 2xh + h^2.

03 Dividing by nonzero h gives 2x + h.

04 As h approaches zero, the secant slope approaches 2x, so f′(x) = 2x.

Watch out

Zero over zero is an indeterminate form, not a final answer of zero.

Exam habit

Place brackets around f(x+h) and f(x) before subtraction, especially when either contains negative terms.

05 Your turn

Use the definition to find the derivative of f(x)=3x squared.

If you get stuck, uncover one hint at a time:

1. Write $f(x+h) = 3(x+h)^2$, then subtract the complete original function.

2. Expand and factor h from the numerator of the difference quotient.

3. Cancel nonzero h, then let it approach zero. Keep the coefficient 3 throughout.

06 Check your understanding

Why should h not be set to zero before simplifying the difference quotient?

Why can the increment be cancelled before taking the derivative limit?

1. It is nonzero while approaching zero
2. It equals zero at every stage
3. Any denominator may always be cancelled

Answers and explanations: page 226.

Basic derivatives and tangent lines

C1 / 02

- Differentiate sums and powers correctly.
- Build a tangent line from the original point and derivative slope.

You need: Power rule; substituting an x value; point-slope form of a line.

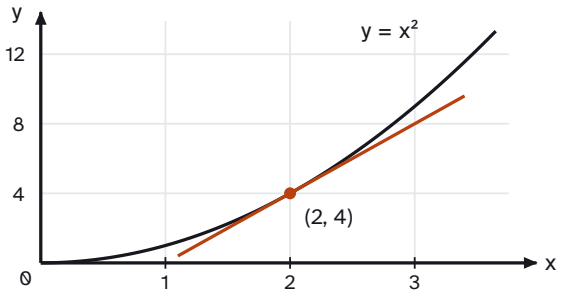
01 Idea

Differentiate a sum term by term. Constants disappear. The derivative gives the slope at a point, but the tangent line must also pass through the point on the original function. Rewrite roots and reciprocal powers before applying the power rule.

A derivative supplies a slope, not a whole line. Many parallel lines share that slope, so a tangent must also be anchored at its point on the original curve. Calculate the curve height from the original function and the slope from its derivative.

FIG. 04 / CALCULUS

A tangent measures the local rate of change



At x = 2

Slope = 2x = 4

Tangent equation:
y - 4 = 4(x - 2)

Takeaway: use coordinate changes for slope; the axes have different scales.

Figure 04. The tangent to $y = x^2$ at (2, 4) has slope 4 and equation y - 4 = 4(x - 2). Its plotted line touches the curve at the marked point. Read slope from coordinates, not the visual angle.

- The original function gives the tangent point.
- The derivative gives the local slope, and a point plus a finite slope determines the line.

02 Rule and symbols

Rule

$$\frac{d}{dx}x^n = nx^{n-1}$$

$$y - f(a) = f'(a)(x - a)$$

a input coordinate of tangent point / input unit

$f(a)$ output coordinate of tangent point / output unit

$f'(a)$ tangent slope / output unit per input unit

03 Method

1. Differentiate the original function term by term.
2. Substitute x = a into the derivative for the slope.
3. Substitute x = a into the original function for the point.
4. Use y - f(a) = f′(a)(x-a), then simplify only if requested.

A tangent-equation request needs both a point and a slope, even when differentiation itself is simple.

04 Worked example

For $f(x) = x^2 + 1$ at $x = 2$, the point is (2,5) and slope is 4.

01 For f(x) = x^2 + 1, f′(x) = 2x because the constant contributes zero.

02 At x = 2 the slope is 4, while the original function gives y = 5.

03 The tangent must therefore pass through (2,5) with slope 4.

04 Its equation is y - 5 = 4(x-2), which expands to y = 4x - 3.

Watch out

Do not use f'(a) as the y-coordinate.

Exam habit

Keep the unsimplified point-slope equation visible: it shows both required ingredients for method marks.

05 Your turn

Find the tangent to f(x)=x cubed at $x = 2$.

If you get stuck, uncover one hint at a time:

1. Evaluate x^3 at $x = 2$ to find the point on the original curve.
2. Differentiate x^3 and evaluate the derivative at the same input.
3. Use the point and slope in $y - y_0 = m(x - x_0)$, then check that the point satisfies your line.

06 Check your understanding

Does substituting x = 2 into f′ give the y coordinate of the tangent point?

Two points have the same finite derivative value. Their tangent lines must be:

1. Exactly the same line in every case
2. Parallel or coincident
3. Perpendicular

Answers and explanations: page 226.

Product and quotient rules

C1 / 03

- Differentiate a product using both changing factors.
- Preserve term order and the squared denominator in the quotient rule.

You need: Basic polynomial and sine derivatives; multiplication and fractions.

01 Idea

A product has two changing factors, so both contribute to the derivative. A quotient rule numerator must preserve its order; reversing it changes the sign. Do not differentiate numerator and denominator separately to differentiate a fraction.

When two factors change, a product changes through both of them. A quotient also has two changing parts, but its denominator contribution enters with a subtraction. Writing the full rule before simplifying preserves the separate contributions and protects the sign.

- The product rule includes one contribution from each changing factor.
- Reversing the quotient-rule subtraction reverses the final derivative sign.

02 Rule and symbols

Rule

$$(uv)' = u'v + uv'$$
$$(u/v)' = (u'v - uv')/v^2$$

u, v the two factor functions / problem dependent

u', v' their derivatives with respect to x / corresponding output per input unit

03 Method

1. Identify whether the expression is a product or quotient at the outermost level.
2. Name the component functions u and v and calculate their derivatives.
3. Insert all terms into the appropriate rule before simplifying.
4. For a quotient retain the original domain restriction and check the numerator subtraction order.

Inspect the outer operation: multiplication or division of functions signals the corresponding rule.

04 Worked example

For $y = x^2 \sin x$, derivative is $2x \sin x + x^2 \cos x$.

01 For y = x^2 sin x, choose u = x^2 and v = sin x.

02 Their derivatives are u′ = 2x and v′ = cos x.

03 The product rule keeps each factor once unchanged: u′v + uv′.

04 Thus y′ = 2x sin x + x^2 cos x; multiplying the two derivatives alone would miss both contributions.

Watch out

The quotient denominator is the square of the original denominator.

Exam habit

Write the entire quotient-rule numerator in brackets before expanding; the minus sign applies to the whole second product.

05 Your turn

Differentiate y=(x+1)/(x-1), for x not equal to 1.

If you get stuck, uncover one hint at a time:

1. Choose $u = x + 1$ and $v = x - 1$, then find both derivatives.
2. Form $(u'v - uv')/v^2$ with brackets around both numerator terms.
3. Simplify $(x - 1) - (x + 1)$ carefully and preserve $x \neq 1$.

06 Check your understanding

For y = (x+1)/(x-1), what is the derivative for x $\neq$ 1?

Why is separately differentiating top and bottom generally wrong for the derivative of u/v?

1. The denominator must always be constant
2. Fractions cannot be differentiated
3. It omits the complete effect of denominator change

Answers and explanations: page 226.

Chain rule and trigonometric functions

C1 / 04

- Identify inner and outer operations in a composite function.
- Include the inner derivative in trigonometric and power derivatives.

You need: Power rule; sine and cosine derivatives; radians.

01 Idea

A composite function has an outer operation acting on an inner expression. Differentiate the outside while preserving the inside, then multiply by the derivative of the inside. Trigonometric derivative formulas use radians.

A composite function is a machine inside a machine. The outer function responds to its inner input, while that inner input changes with the original variable. The chain rule multiplies these rates to describe the complete change.

- An outer derivative alone measures change per unit of inner expression.
- The inner derivative converts that rate into change per unit of the original variable.

02 Rule and symbols

Rule

$$\frac{d}{dx} f(g(x)) = f'(g(x))g'(x)$$

$$(\sin x)' = \cos x$$

$$(\cos x)' = -\sin x$$

$u = g(x)$ inner expression / problem dependent

$g'(x)$ inner derivative / inner-expression unit per input unit

03 Method

1. Identify the last operation applied to the expression: this is the outer function.
2. Differentiate that outer function while leaving the inner expression in place.
3. Multiply by the derivative of the inner expression.

4. Check the sign for cosine derivatives and use radians for trigonometric rules.

An operation surrounding a nontrivial expression, such as cosine surrounding a square, is a chain-rule cue.

04 Worked example

For $y = (3x + 1)^4$, derivative = $4(3x + 1)^3 \cdot 3 = 12(3x + 1)^3$.

01 In y = (3x+1)^4, the inner expression is 3x+1 and the outer operation is raising to the fourth power.

02 The outer derivative gives 4(3x+1)^3.

03 The inner derivative is 3, so it must multiply the outer result.

04 Combining the constants gives y′ = 12(3x+1)^3.

Watch out

Forgetting the inner derivative is the most common lost mark.

Exam habit

Before finishing, circle the inner expression and verify that its derivative appears as a factor.

05 Your turn

Differentiate y=cos(x^2).

If you get stuck, uncover one hint at a time:

1. Identify cosine as the outer function and x^2 as the inner expression.
2. Differentiate cosine while leaving x^2 inside, and retain the negative sign.
3. Multiply by the derivative of x^2. Check that both rates appear.

06 Check your understanding

What is the derivative of cos(x^2)?

In differentiating $\sin(7x)$, what does the extra factor 7 represent?

1. The rate of change of the inner expression
2. A second sine factor
3. A degrees-to-radians conversion

Answers and explanations: page 226.

Four more trigonometric derivatives

The supplied final asks for a cotangent derivative. Learn these rules together with sine, cosine and the chain rule.

- Use radians. Here u is a function of x, and u' means its derivative.
- Apply the outside rule, then multiply by the inside derivative.
- Use a rule only where the original function is defined. Remember: $\sec u = 1/\cos u$, $\csc u = 1/\sin u$, and $\cot u = \cos u/\sin u$.
- $\dfrac{d}{dx}\tan u = \sec^2 u\, u'$
- $\dfrac{d}{dx}\cot u = -\csc^2 u\, u'$
- $\dfrac{d}{dx}\sec u = \sec u \tan u\, u'$
- $\dfrac{d}{dx}\csc u = -\csc u \cot u\, u'$

Differentiate $y = \cot(5x)$.

1. The inside is $u = 5x$, so $u' = 5$.
2. The cotangent rule gives $y' = -\csc^2(5x) \times 5$.

$y' = -5\csc^2(5x)$. The minus sign and the factor 5 both matter.

Implicit differentiation

C1 / 05

- Differentiate y terms while treating y as a function of x.
- Collect and isolate all terms containing y′.

You need: Chain and product rules; algebraic rearrangement.

01 Idea

When x and y appear in one equation, y is treated as a function of x. Differentiating a y term therefore introduces y'. Collect all y-prime terms and solve algebraically. This can describe curves that are not single-valued functions globally.

An implicit equation links two changing coordinates without isolating one globally. As one coordinate changes, the other adjusts to stay on the curve. Differentiation records this linked movement, so differentiating a function of the dependent coordinate requires its own rate of change.

- A y-expression is a composite function of x, so chain-rule factors appear.
- Collecting derivative terms reveals a shared factor that can be isolated to find the slope.

02 Rule and symbols

Rule

$$\frac{d}{dx}(y^2) = 2yy'$$

$$\frac{d}{dx}(xy) = y + xy'$$

y' dy/dx, the local slope of the curve / y unit per x unit

03 Method

1. Differentiate both sides with respect to x.
2. Attach y′ when differentiating a function of y, and use the product rule for xy.
3. Move all terms containing y′ to one side.
4. Factor out y′, divide by its coefficient and state where that division is valid.

Use implicit differentiation when x and y are mixed and explicit rearrangement would be awkward or lose branches.

04 Worked example

From $x^2 + y^2 = 25$, differentiation gives $2x + 2yy' = 0$.

01 The curve $x^2 + y^2 = 25$ need not be solved globally for a single y value.

02 Differentiating x^2 gives 2x; differentiating y^2 gives 2y y′ by the chain rule.

03 The constant 25 differentiates to zero, so 2x + 2y y′ = 0.

04 Rearranging gives y′ = -x/y wherever $y \neq 0$; division by y requires this restriction.

Watch out

Do not treat y as a constant or omit the product rule on xy.

Exam habit

For an xy term write y + xy′, not simply y or xy′; both factors depend on x.

05 Your turn

Find dy/dx for xy+y squared=6.

If you get stuck, uncover one hint at a time:

1. Differentiate both sides of $xy + y^2 = 6$, using the product rule for xy.
2. Include y' when differentiating y^2, and collect all derivative terms.
3. Factor out y', move the remaining term and divide by its coefficient where nonzero.

06 Check your understanding

Why does the derivative of y^2 contain y'?

When y depends on x, the derivative of xy is:

1. xy'
2. $y + xy'$
3. y

Answers and explanations: page 226.

Logarithmic, exponential and inverse-trig derivatives

C1 / 06

- Differentiate exponential and logarithmic composites.
- Use logarithmic differentiation for a positive variable base and exponent.

You need: Chain rule; natural logarithms; product rule; positive-domain restrictions.

01 Idea

Exponential functions reproduce themselves up to an inner derivative. Natural logarithms give a reciprocal factor. Logarithmic differentiation helps when variable powers or products are awkward. Inverse-trigonometric functions are different from reciprocal trigonometric functions.

A variable power changes through both its base and its exponent. A logarithm moves the exponent into a product, where familiar differentiation rules track those two changes. The final step restores the original function after differentiating the logarithmic equation.

- Logarithms turn a positive power expression into a product involving its exponent.
- The derivative of the logarithm includes the original derivative divided by the original function, so that function must be restored.

02 Rule and symbols

Rule

$$(e^u)' = e^u u'$$

$$(\ln|u|)' = u'/u$$

$$(\arctan x)' = 1/(1+x^2)$$

e base of the natural exponential / dimensionless

$\ln u$ natural logarithm of a positive dimensionless argument / dimensionless

03 Method

1. Identify the outer function and check its real domain.
2. For a logarithm use inner derivative divided by the inner expression.
3. For an exponential retain the exponential and multiply by the inner derivative.
4. For y = x^x with x > 0, take logs first, differentiate implicitly, then restore y.

A variable in both base and exponent suggests logarithmic differentiation, after checking a suitable positive domain.

04 Worked example

For $y = \ln(x^2 + 1)$, derivative is $\frac{2x}{x^2+1}$.

01 For ln(x^2+1), the inner derivative is 2x and the inner expression is always positive.

02 The logarithmic chain rule therefore gives 2x/(x^2+1).

03 For the separate function e^(3x), the exponential is retained unchanged.

04 Multiplication by the inner derivative 3 gives 3e^(3x).

Watch out

The derivative of ln u is u prime/u, not 1/ln u.

Exam habit

The power rule nx^(n-1) assumes a constant exponent; x^x needs another method.

05 Your turn

Differentiate y=x to the power x for x greater than zero.

If you get stuck, uncover one hint at a time:

1. For $x > 0$, set $y = x^x$ and take natural logarithms.

2. Rewrite the right side as $x \ln x$; use the product rule while the left derivative is y'/y.
3. Multiply by y, then replace it with the original x^x.

06 Check your understanding

For x > 0, what is d(x^x)/dx?

Why is the constant-exponent power rule insufficient for x^x?

1. The expression is constant
2. Logarithms are invalid for positive x
3. The exponent changes as well as the base

Answers and explanations: page 226.

Other exponential, logarithm and inverse-trig rules

The supplied final includes 3^x and $\log_3 x$. Their derivatives need a factor involving $\ln 3$.

- The base a is a fixed positive number, with $a \neq 1$. For the logarithm, require $u > 0$.
- For the displayed arcsine and arccosine derivatives, require $|u| < 1$. Angles are in radians.
- Find the inside derivative u' first. Inverse sine means $\arcsin$; it is different from $1/\sin$.
- $\dfrac{d}{dx} a^u = a^u \ln(a)\, u'$
- $\dfrac{d}{dx} \log_a u = \dfrac{u'}{u \ln a}$
- $\dfrac{d}{dx} \arcsin u = \dfrac{u'}{\sqrt{1 - u^2}}$

- $\dfrac{d}{dx}\arccos u = -\dfrac{u'}{\sqrt{1-u^2}}$

Differentiate 3^x and $\log_3 x$.

1. In both cases the inside expression is $u = x$, so $u' = 1$.
2. For the exponential, multiply by $\ln 3$. For the logarithm, divide by $\ln 3$.

$(3^x)' = 3^x \ln 3$, and $(\log_3 x)' = 1/(x \ln 3)$ for $x > 0$.

L Hopital and indeterminate forms

C1 / 07

- Check for a permitted indeterminate quotient before using L Hopital.
- Evaluate a simple limit by differentiating numerator and denominator.

You need: Limits; exponential and trigonometric derivatives; chain rule.

01 Idea

L Hopital rule can evaluate a quotient limit with the form $\frac{0}{0}$ or $\frac{\infty}{\infty}$ under its differentiability and limit conditions. Differentiate numerator and denominator separately for this limit procedure; this is not the quotient rule. Other indeterminate forms must first be rewritten.

An indeterminate quotient describes competition between quantities approaching zero or growing without bound. When its conditions hold, L Hopital compares their rates of change. It is a limit procedure, so it does not replace the quotient rule for the derivative of a fraction.

- The initial limiting form must be checked before applying the rule.
- Differentiating a numerator containing an inner expression still requires the chain rule.

02 Rule and symbols

Rule

$$\lim \frac{f(x)}{g(x)} = \lim \frac{f'(x)}{g'(x)}$$

when the rule applies

$0/0$ indeterminate limiting form, not an arithmetic value / not a measured quantity

∞/∞ indeterminate quotient form / not a measured quantity

Use this when:

The numerator and denominator satisfy the differentiability and denominator-derivative conditions near the limiting point.

03 Method

1. Substitute the limiting input to identify the quotient form.
2. Confirm 0/0 or infinity/infinity and the rule conditions before differentiating.
3. Differentiate numerator and denominator separately, not with the quotient rule.
4. Evaluate the new limit; if still indeterminate, check conditions again before repeating.

Check the quotient form first; an ordinary fraction alone is not a reason to use L Hopital.

04 Worked example

As x approaches zero, $\frac{e^x-1}{x}$ has form $\frac{0}{0}$.

01 As x approaches zero, e^x - 1 approaches zero and x approaches zero.

02 The quotient therefore has the required 0/0 form.

03 Differentiate numerator and denominator separately to obtain e^x/1.

04 Its limit is e^0 = 1, so the original limit is 1.

Watch out

Do not use the rule on a quotient whose substitution gives an ordinary finite value.

Exam habit

Show the initial indeterminate form; applying the rule without checking its conditions can invalidate the method.

05 Your turn

Evaluate the limit of sin(3x)/x as x approaches zero.

If you get stuck, uncover one hint at a time:

1. Check the numerator and denominator as $x \to 0$ and identify the form.
2. Differentiate $\sin(3x)$ with its inner derivative and differentiate the denominator separately.
3. Evaluate the resulting derivative quotient at the limiting input; do not use a quotient-rule numerator.

06 Check your understanding

What is the limit of sin(3x)/x as x approaches zero?

A numerator approaches 2 and its denominator approaches 4. The appropriate first method is:

1. Use the ordinary quotient limit
2. Apply L Hopital because it is a fraction
3. Declare the form indeterminate

Answers and explanations: page 226.

Increasing, decreasing and concavity

C1 / 08

- Use a first-derivative sign chart to find monotonic intervals.
- Distinguish a stationary point from a proven extremum or inflection.

You need: Factoring quadratics; sign tests; first and second derivatives.

01 Idea

The sign of the first derivative controls increasing and decreasing intervals. The sign of the second controls concavity. A critical number lies in the domain where the first derivative is zero or undefined. An inflection point requires a concavity change, not merely a zero second derivative.

The derivative records the local slope of the graph. A negative derivative makes the graph fall as input increases; a positive derivative makes it rise. A change from falling to rising identifies a minimum, whereas a zero derivative without a sign change may not be an extremum.

- A zero derivative gives a candidate rather than a complete extremum proof.
- Signs on both sides tell whether increasing or decreasing behaviour actually changes.

02 Rule and symbols

Rule

$$f' > 0 : \text{ increasing;}$$
$$f'' > 0 : \text{ concave up}$$

f' first derivative, controlling increase or decrease / output per input unit

f'' second derivative, controlling concavity / output per input unit squared

03 Method

1. Find where f′ is zero or undefined within the original domain.
2. Split the domain at those critical numbers and test the sign on each interval.
3. Use sign changes to classify local maxima and minima.

4. Use f″ signs separately for concavity and require a concavity change for an inflection point.

Use a derivative sign chart for monotonic intervals and local extrema; use a separate concavity test for inflections.

04 Worked example

For $f = x^3 - 3x$, derivative is $3(x-1)(x+1)$.

01 For f(x) = x³ - 3x, f′ = 3x² - 3 = 3(x-1)(x+1).

02 The critical numbers are -1 and 1.

03 Testing x = -2, 0 and 2 gives derivative signs positive, negative and positive.

04 The function increases on (-∞,-1) and (1,∞), and decreases on (-1,1); the sign changes identify a local maximum at -1 and minimum at 1.

Watch out

A derivative equal to zero does not automatically mean a maximum.

Exam habit

Give open intervals for increasing and decreasing behaviour, then state extrema as points if asked.

05 Your turn

For $f(x) = x^2$-4x, find increasing and decreasing intervals and the minimum point.

If you get stuck, uncover one hint at a time:

1. Differentiate $x^2 - 4x$ and find where its derivative is zero.
2. Test derivative signs using one input on each side of that critical number.
3. Report the resulting intervals and evaluate the original function at the critical input to give the minimum point.

06 Check your understanding

Is f″(c) = 0 alone enough to prove an inflection point?

A first derivative changes from negative to positive at an interior critical point. This gives:

1. An inflection point guaranteed
2. A local minimum
3. A local maximum

Answers and explanations: page 226.

Graphing, asymptotes and absolute extrema

C1 / 09

- Find absolute extrema by comparing all candidates on a closed interval.
- Distinguish a removable hole from a vertical asymptote.

You need: Domain restrictions; critical numbers; evaluating a function at endpoints.

01 Idea

Graphing combines domain, intercepts, derivative signs and limiting behaviour. A cancelled factor can create a hole rather than an asymptote. On a closed interval a continuous function attains absolute extrema; compare endpoint and interior critical values.

An absolute extremum compares the entire allowed graph. On a closed interval, an endpoint can be highest or lowest without having a horizontal tangent. Comparing a short list of original function values turns this global problem into a finite check.

- Interior critical numbers supply possible internal extrema.
- Endpoints must also be considered because the allowed domain ends there.

02 Rule and symbols

Rule

Candidates = endpoints plus interior critical numbers

$[a, b]$ closed interval including both endpoints / input unit

$f(c)$ function value at a candidate point / output unit

03 Method

1. Check the domain and whether the function is continuous on the stated closed interval.
2. Find all interior critical numbers.
3. Evaluate the original function at those numbers and at both endpoints.

4. Compare the values and report each extremum value together with where it occurs.

Absolute maximum or minimum on a closed interval calls for endpoint-plus-critical-value comparison.

04 Worked example

For x^2 on $[-1, 2]$, candidates are -1,0,2.

01 For f(x) = x^2 on [-1,2], the derivative 2x is zero at x = 0.

02 The candidate inputs are the two endpoints -1 and 2 and the interior critical number 0.

03 The original function values are f(-1)=1, f(0)=0 and f(2)=4.

04 The smallest is 0 at x=0; the largest is 4 at x=2.

Watch out

Do not omit endpoints or compare derivative values instead of function values.

Exam habit

Use the original function for the comparison table; comparing derivative values does not locate the highest graph point.

05 Your turn

Find absolute maximum and minimum of $f(x) = x^2$-2x on [0,3].

If you get stuck, uncover one hint at a time:

1. Find the derivative of $x^2 - 2x$ and its critical input inside $[0, 3]$.
2. List that input and both endpoints as candidates.
3. Evaluate the original function at each candidate; report both the largest and smallest values and where each occurs.

06 Check your understanding

Can an endpoint be the absolute maximum even when its derivative is not zero?

A continuous function is strictly increasing on a closed interval. Its absolute maximum occurs:

1. Only where its derivative is zero
2. Nowhere because it keeps increasing
3. At the right endpoint

Answers and explanations: page 226.

Optimization and mean-value theorems

C1 / 10

- Translate a simple optimization constraint into one variable.
- Check mean-value-theorem assumptions and solve for the interior input.

You need: Area and perimeter; derivatives; continuity and differentiability.

01 Idea

Optimization translates a word problem into a one-variable function and an allowed domain. Rolle theorem requires equal endpoint values as well as continuity and interior differentiability. The mean-value theorem matches some instantaneous slope to the average slope.

The mean-value theorem links average change across an interval with instantaneous change somewhere inside. A secant joins endpoint heights, and at least one interior tangent must be parallel under the required conditions. The theorem matches slopes rather than automatically selecting a midpoint.

- Continuity and interior differentiability exclude breaks and corners that could defeat the guarantee.
- The matching input lies strictly inside the interval but need not be its midpoint.

02 Rule and symbols

Rule

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$a < c < b$$

x one rectangle side length / length unit

A rectangle area / square length unit

c interior input guaranteed by the mean-value theorem / input unit

03 Method

1. For optimization name the objective, write the constraint and identify the feasible domain.
2. Eliminate one variable, differentiate and find interior critical candidates.
3. Check endpoints or derivative signs to justify the optimum.
4. For a theorem question verify its assumptions before setting instantaneous slope equal to average slope.

Use the theorem when a function and interval are supplied and an interior slope-matching input is requested.

04 Worked example

A rectangle with perimeter 20 has area $x(10-x)$.

01 Perimeter 20 gives 2x + 2y = 20, so the other side is y = 10 - x.

02 Area becomes A(x) = x(10-x) with 0 < x < 10 for a nondegenerate rectangle.

03 Its derivative is 10 - 2x, which vanishes at x = 5.

04 The second derivative is -2, confirming a maximum; both sides are 5 and area is 25 square units.

Watch out

A candidate outside the physical domain is not a valid optimization answer.

Exam habit

A critical number alone is not a maximum proof; include a sign change, second derivative or candidate comparison.

05 Your turn

Use the mean-value theorem for $f(x)=x^2$ on [1,3] to find c.

If you get stuck, uncover one hint at a time:

1. Check that x^2 is continuous on $[1, 3]$ and differentiable inside.
2. Compute the endpoint secant slope from function values and interval length.
3. Set $f'(c)$ equal to that slope, solve and verify $1 < c < 3$.

06 Check your understanding

For f(x)=x^2 on [1,3], what c satisfies the mean-value theorem?

Must the input guaranteed by the mean-value theorem be the midpoint?

1. No; it solves the slope-matching equation
2. Yes; every function uses the midpoint
3. It must be an endpoint

Answers and explanations: page 226.

Antiderivatives and substitution

C1 / 11

- Choose a substitution whose differential matches the integrand.
- Return the antiderivative to x and include the integration constant.

You need: Chain rule; power integration; constant factors.

01 Idea

An indefinite integral represents a family of antiderivatives, so include an arbitrary constant. Substitution reverses the chain rule: select an inner expression whose derivative is present. Differentiate your answer as a fast correctness check.

Substitution recognizes the chain rule running backward. A temporary variable replaces an inner expression, while its differential accounts for a matching factor elsewhere. A leftover constant is harmless if tracked carefully rather than dropped.

- A useful substitution replaces the nested expression and its differential factor together.
- Differentiating the final result checks both the power and its multiplier.

02 Rule and symbols

Rule

$$\int x^n\,dx = x^{n+1}/(n+1) + C\ (n \neq -1)$$

$$\int dx/x = \ln|x| + C$$

u chosen inner expression / problem dependent

du differential of the inner expression / same unit as u

C arbitrary additive integration constant / antiderivative unit

03 Method

1. Choose an inner expression whose derivative is also present as a factor.
2. Write u and du and replace every part of the integral consistently.
3. Integrate in u, preserving any numerical factors.

4. Substitute back to x, add C and differentiate once to check.

Look for an inner expression accompanied by its derivative or a constant multiple of that derivative.

04 Worked example

For the integral of $2x(x^2+1)^3$, set $u = x^2 + 1$ and $du = 2x\,dx$.

01 For the integral of $2x(x^2+1)^3$, choose $u = x^2+1$.

02 Then $du = 2x\,dx$, so the complete integral becomes the integral of u^3 du.

03 The antiderivative is $u^4/4 + C$.

04 Substitute back to get $(x^2+1)^4/4 + C$; differentiation restores $2x(x^2+1)^3$.

Watch out

The power rule exception is n=-1; do not divide by zero.

Exam habit

Do not leave a mixture of u and x in the substituted integral; it hides missing differential factors.

05 Your turn

Evaluate the integral of $6x(x^2+2)$ squared with respect to x.

If you get stuck, uncover one hint at a time:

1. Choose $u = x^2 + 2$, calculate du and compare it with $6x\,dx$.
2. Rewrite the full integral in u, retaining any constant multiplier.
3. Integrate the power, return to x, include the arbitrary constant and differentiate to check.

06 Check your understanding

What is the integral of 6x(x^2+2)^2 dx?

Which substitution most directly simplifies $6x(x^2+2)^2$?

1. Rename the whole integrand without computing a differential
2. $u = x^2 + 2$
3. $u = 6x$

Answers and explanations: page 226.

Integrate sine, cosine and an exponential

These short rules reverse the chain rule. A sine integral appears in the supplied final.

- Here a and b are constants and $a \neq 0$. Trigonometric angles use radians.
- Divide by the coefficient a; differentiation would multiply by it.
- Add C for an indefinite integral. Differentiate your answer to check it.

- $\int \sin(ax+b)\,dx = -\dfrac{\cos(ax+b)}{a} + C$
- $\int \cos(ax+b)\,dx = \dfrac{\sin(ax+b)}{a} + C$
- $\int e^{ax+b}\,dx = \dfrac{e^{ax+b}}{a} + C$

Find $\int \sin(3x)\,dx$.

1. The sine antiderivative is negative cosine.
2. The inside derivative is 3, so divide by 3.
3. Check: $\dfrac{d}{dx}[-\cos(3x)/3] = \sin(3x)$.

$-\dfrac{1}{3}\cos(3x) + C$.

Definite integrals, signed area and the fundamental theorem

C1 / 12

- Distinguish signed integral from total geometrical area.
- Apply the fundamental theorem, including a variable-upper-limit chain factor.

You need: Antiderivatives; positive and negative regions; substituting bounds.

01 Idea

A definite integral gives signed accumulation: contributions below the axis are negative. Geometrical area requires positive contributions from each region. The fundamental theorem connects integration and differentiation; for a variable upper limit, include its derivative.

A definite integral combines signed contributions over fixed bounds. An antiderivative provides a shortcut through its change between endpoints. If the integrand is nonnegative throughout, the signed accumulation is also geometric area; otherwise positive and negative regions can cancel.

- The fundamental theorem requires upper antiderivative value minus lower value.
- Substitution changes the endpoint labels unless the antiderivative is returned to the original variable before evaluation.

02 Rule and symbols

Rule

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

$$\frac{d}{dx}\int_a^{g(x)} f(t)\,dt = f(g(x))g'(x)$$

a, b lower and upper integration bounds / input unit

F an antiderivative of f / f unit times input unit

t dummy integration variable / input unit

03 Method

1. Identify whether the question asks for signed accumulation or total area.
2. Find an antiderivative and calculate upper value minus lower value.
3. For total area split at sign changes and make each region contribution positive.
4. For differentiation with upper bound g(x), evaluate f at g(x) and multiply by g′(x).

With fixed limits, keep endpoint values consistent with the integration variable and evaluate their difference.

04 Worked example

The $\int_{-1}^{1} x\,dx$ is zero by cancellation, but total geometrical area is 1.

01 For y = x from -1 to 1, the region below the axis contributes negative signed area.

02 An antiderivative is $x^2/2$; evaluation at 1 and -1 gives 1/2 - 1/2 = 0.

03 Geometrically each half is a triangle with area 1/2.

04 Total geometrical area is 1/2 + 1/2 = 1, while the signed integral remains zero.

Watch out

Never keep x-limits on an integral that has been fully changed to u.

Exam habit

For a definite integral evaluate both bounds; no arbitrary +C is needed in the final numerical answer.

05 Your turn

Evaluate the integral from 0 to 1 of 2x(x^2+1) dx.

If you get stuck, uncover one hint at a time:

1. Notice $x^2 + 1$ and its derivative factor $2x\,dx$.
2. Substitute with transformed bounds, or find an antiderivative and return it to x.
3. Evaluate upper minus lower between 0 and 1. Since the integrand is nonnegative there, a negative result would indicate an error.

06 Check your understanding

Does a zero definite integral prove the function is zero throughout the interval?

After changing variable in a definite integral, what should happen to the bounds?

1. Always keep the old bound numbers with the new variable
2. Remove them permanently
3. Transform them, or return to the original variable before evaluation

Answers and explanations: page 226.

Read a Riemann sum as rectangles

The course syllabus includes sigma notation and area as a limit. The symbol $\sum$ simply means add the terms.

- Split the interval from a to b into n equal widths.
- Choose a sample point x_i^* in each small interval. Its function value is that rectangle's signed height.
- Add height times width. More, narrower rectangles approach the definite integral when the function is continuous.
- Below-axis heights contribute negative signed area.
- $\Delta x = \dfrac{b - a}{n}$
- rectangle sum $= \sum_{i=1}^{n} f(x_i^*)\,\Delta x$

Use two right-end rectangles to estimate $\int_0^1 x\,dx$.

1. Each width is $\Delta x = (1-0)/2 = 1/2$.
2. The right endpoints are $1/2$ and 1, giving heights $1/2$ and 1.
3. The estimate is $(1/2)(1/2) + (1)(1/2) = 3/4$.

Estimate: $3/4$. Exact integral: $1/2$. The right-end estimate is too large because the function increases.

Lesson review and answers

Limits, continuity and the derivative definition

Return to lesson: page 187.

Your turn: It would produce division by zero; the limit concerns nonzero h approaching zero.

Concept check: 1. Correct. Simplification is valid at nonzero increments, and those nearby values determine the limit.

Written method: [3(x+h) squared -3x squared]/h = [6xh+3h squared]/h = 6x+3h. Taking the limit gives 6x. One mark each for setup, expansion, cancellation and limit.

Basic derivatives and tangent lines

Return to lesson: page 190.

Your turn: No. It gives the slope; use f(2) for the y coordinate.

Concept check: 2. Correct. Equal slopes give parallel lines, which can coincide in a special case.

Written method: f(2)=8; derivative=3x squared; slope at 2 is 12. Tangent y-8=12(x-2), equivalently y=12x-16. One mark per step.

Product and quotient rules

Return to lesson: page 193.

Your turn: -2/(x-1)^2: the numerator is (x-1) - (x+1) = -2.

Concept check: 3. Correct. The quotient rule accounts for both changes and includes the squared denominator.

Written method: Derivative = [(x-1)-(x+1)]/(x-1) squared = -2/(x-1) squared. Award marks for quotient rule, numerator order, denominator square and simplification.

Chain rule and trigonometric functions

Return to lesson: page 196.

Your turn: -2x sin(x^2), combining the cosine derivative with the inner derivative 2x.

Concept check: 1. Correct. The inner expression changes seven units per unit increase in x.

Written method: Outer derivative is -sin(x^2); inner derivative is 2x. Multiply to

obtain -2x sin(x^2). Credit both layers, sign and product.

Implicit differentiation

Return to lesson: page 200.

Your turn: Because y depends on x; the chain rule multiplies 2y by dy/dx.

Concept check: 2. Correct. Both factors change, so the product rule gives both terms.

Written method: Differentiate: y+x y'+2y y'=0. Collect: (x+2y)y'=-y. Hence y'=-y/(x+2y), where the denominator is nonzero. Four marks across these steps.

Logarithmic, exponential and inverse-trig derivatives

Return to lesson: page 203.

Your turn: x^x(ln x + 1). Take ln y = x ln x, then y′/y = ln x + 1.

Concept check: 3. Correct. Treating the exponent as fixed misses part of the change.

Written method: Take logarithms: ln y=x ln x. Differentiate: y'/y=ln x+1. Multiply by y: y'=x to the x times (ln x+1). Four marks for these stages.

L Hopital and indeterminate forms

Return to lesson: page 207.

Your turn: 3, since the derivative quotient is 3 cos(3x)/1 and cos 0 = 1.

Concept check: 1. Correct. A nonzero finite denominator limit gives a directly determined quotient.

Written method: The form is $\frac{0}{0}$. L Hopital gives limit 3 cos(3x)/1 = 3. Alternatively use the standard sine limit. Credit form, derivative, evaluation and answer.

Increasing, decreasing and concavity

Return to lesson: page 210.

Your turn: No. Concavity must change across c.

Concept check: 2. Correct. The graph falls toward the point and rises away.

Written method: Derivative = 2x-4, zero at 2. It is negative for x<2 and positive for x>2. Minimum is (2,-4). One mark each for derivative, intervals and point.

Graphing, asymptotes and absolute extrema

Return to lesson: page 213.

Your turn: Yes. Absolute extrema on a closed interval must include endpoint checks.

Concept check: 3. Correct. The last allowed input has the largest output.

Written method: Critical point x=1. Values: f(0)=0, f(1)=-1, f(3)=3. Minimum -1 at x=1; maximum 3 at $x = 3$. Credit candidates, evaluations and both extrema.

Optimization and mean-value theorems

Return to lesson: page 216.

Your turn: c=2: the average slope is (9-1)/(3-1)=4 and f′(c)=2c=4.

Concept check: 1. Correct. A midpoint works for some functions but is not the general theorem conclusion.

Written method: The polynomial is continuous and differentiable. Average slope = (9-1)/(3-1)=4. Solve 2c=4 to get c=2, which is inside (1,3). Four marks for hypotheses, slope, solution and interval check.

Antiderivatives and substitution

Return to lesson: page 219.

Your turn: $(x^2+2)^3$ + C. With u=x^2+2, the integral becomes 3 times the integral of u^2 du.

Concept check: 2. Correct. Its differential matches the remaining x factor up to a constant.

Written method: Set u=x^2+2, $du = 2x\,dx$. Integral becomes 3 times integral u squared du = u cubed+C = (x^2+2) cubed+C. Credit substitution, factor, integration and constant.

Definite integrals, signed area and the fundamental theorem

Return to lesson: page 222.

Your turn: No. Positive and negative contributions can cancel.

Concept check: 3. Correct. The endpoints must still describe the same original inputs.

Written method: Set $u = x^2 + 1$; limits become 1 and 2. Integral of u du from 1 to 2 = (4-1)/2 = 3/2. Credit substitution, new limits, antiderivative and subtraction.

Calculus I / Test A

30 marks / Suggested time: 30 minutes. MCQs: 1 mark each. Written problems: 4 marks each. Attempt without notes.

Part I / Choose one answer

1. The derivative of x^2 at $x = 3$ is:

   A. 3

   B. 6

   C. 9

   D. 0

2. The derivative of a constant 7 is:

   A. 7

   B. 1

   C. x

   D. 0

3. The derivative of xe^x is:

   A. $e^x(1 + x)$

   B. 0

   C. e^x

   D. xe^x

4. The derivative of $\sin(5x)$ is:

   A. $-5\sin(5x)$

   B. $\sin(5)$

   C. $\cos(5x)$

   D. $5\cos(5x)$

5. For $x^2 + y^2 = 25$, the slope at (3,4) is:

   A. 0

   B. $\frac{3}{4}$

   C. $-\frac{3}{4}$

D. $\frac{4}{3}$

6. The derivative of $\arctan x$ is:

A. $\arctan(1)$

B. $\frac{1}{1+x^2}$

C. $\frac{1}{\cos^2 x}$

D. $-\frac{1}{1+x^2}$

Part II / Show your reasoning

1. Use the definition to find the derivative of f(x)=3x squared.

2. Find the tangent to f(x)=x cubed at $x = 2$.

3. Differentiate y=(x+1)/(x-1), for x not equal to 1.

4. Differentiate y=cos(x^2).

5. Find dy/dx for xy+y squared=6.

6. Differentiate y=x to the power x for x greater than zero.

Calculus I / Test B

30 marks / Suggested time: 30 minutes. MCQs: 1 mark each. Written problems: 4 marks each. Attempt without notes.

Part I / Choose one answer

1. Before applying L Hopital directly, the quotient should have form:

   A. $\frac{0}{0}$ or $\frac{\infty}{\infty}$

   B. $\frac{1}{1}$ only

   C. $\frac{0}{5}$ only

   D. Any fraction

2. If f' is negative on an interval, f is:

   A. Decreasing

   B. Constant

   C. Always concave down

   D. Increasing

3. To find absolute extrema on a closed interval, check:

   A. Only where f' is zero

   B. Only endpoints

   C. Endpoints and interior critical numbers

   D. Only where f is zero

4. Rolle theorem additionally requires:

   A. An infinite interval

   B. Equal endpoint function values

   C. A discontinuity

   D. A nonzero derivative everywhere

5. An indefinite integral must generally include:

   A. Only a numerical answer

   B. A fixed upper limit

   C. No variable

D. An arbitrary constant

6. The $\int_{-2}^{2} x\,dx$ equals:

A. 8

B. -4

C. 0

D. 4

Part II / Show your reasoning

1. Evaluate the limit of sin(3x)/x as x approaches zero.

2. For $f(x) = x^2$-4x, find increasing and decreasing intervals and the minimum point.

3. Find absolute maximum and minimum of $f(x) = x^2$-2x on [0,3].

4. Use the mean-value theorem for $f(x) = x^2$ on [1,3] to find c.

5. Evaluate the integral of 6x(x^2+2) squared with respect to x.

6. Evaluate the integral from 0 to 1 of 2x(x^2+1) dx.

Calculus I / Test A / Answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1

MCQ: B. The derivative is 2x, so its value at 3 is 6.

Written solution: [3(x+h) squared -3x squared]/h = [6xh+3h squared]/h = 6x+3h. Taking the limit gives 6x. One mark each for setup, expansion, cancellation and limit.

Return to lesson: page 187.

Question 2

MCQ: D. A constant function has zero rate of change.

Written solution: f(2)=8; derivative=3x squared; slope at 2 is 12. Tangent y-8=12(x-2), equivalently y=12x-16. One mark per step.

Return to lesson: page 190.

Question 3

MCQ: A. The product rule differentiates each factor in turn.

Written solution: Derivative = [(x-1)-(x+1)]/(x-1) squared = -2/(x-1) squared. Award marks for quotient rule, numerator order, denominator square and simplification.

Return to lesson: page 193.

Question 4

MCQ: D. The inner derivative contributes a factor of 5.

Written solution: Outer derivative is -sin(x^2); inner derivative is 2x. Multiply to obtain -2x sin(x^2). Credit both layers, sign and product.

Return to lesson: page 196.

Question 5

MCQ: C. Implicit differentiation gives slope $-\frac{x}{y}$.

Written solution: Differentiate: y+x y'+2y y'=0. Collect: (x+2y)y'=-y. Hence y'=-

y/(x+2y), where the denominator is nonzero. Four marks across these steps.

Return to lesson: page 200.

Question 6

MCQ: B. This is the standard inverse-tangent derivative.

Written solution: Take logarithms: ln y=x ln x. Differentiate: y'/y=ln x+1. Multiply by y: y'=x to the x times (ln x+1). Four marks for these stages.

Return to lesson: page 203.

Calculus I / Test B / Answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1

MCQ: A. The rule concerns specified indeterminate quotient forms.

Written solution: The form is $\frac{0}{0}$. L Hopital gives limit 3 cos(3x)/1 = 3. Alternatively use the standard sine limit. Credit form, derivative, evaluation and answer.

Return to lesson: page 207.

Question 2

MCQ: A. Negative slope means the function decreases as x increases.

Written solution: Derivative = 2x-4, zero at 2. It is negative for x<2 and positive for x>2. Minimum is (2,-4). One mark each for derivative, intervals and point.

Return to lesson: page 210.

Question 3

MCQ: C. The largest or smallest value may occur at either kind of candidate.

Written solution: Critical point x=1. Values: f(0)=0, f(1)=-1, f(3)=3. Minimum -1 at x=1; maximum 3 at $x = 3$. Credit candidates, evaluations and both extrema.

Return to lesson: page 213.

Question 4

MCQ: B. Equal endpoint heights make the average slope zero.

Written solution: The polynomial is continuous and differentiable. Average slope = (9-1)/(3-1)=4. Solve 2c=4 to get c=2, which is inside (1,3). Four marks for hypotheses, slope, solution and interval check.

Return to lesson: page 216.

Question 5

MCQ: D. Differentiation loses constants, so all antiderivatives differ by a constant on an interval.

Written solution: Set u=x^2+2, $du = 2x\,dx$. Integral becomes 3 times integral u squared du = u cubed+C = (x^2+2) cubed+C. Credit substitution, factor, integration and constant.

Return to lesson: page 219.

Question 6

MCQ: C. The odd integrand has equal and opposite signed areas.

Written solution: Set $u = x^2 + 1$; limits become 1 and 2. Integral of u du from 1 to 2 = (4-1)/2 = 3/2. Credit substitution, new limits, antiderivative and subtraction.

Return to lesson: page 222.

Formula sheet

Limits, continuity and the derivative definition

p.187

Rule

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Basic derivatives and tangent lines

p.190

Rule

$$\frac{d}{dx} x^n = n x^{n-1}$$

$$y - f(a) = f'(a)(x - a)$$

Product and quotient rules

p.193

Rule

$$(uv)' = u'v + uv'$$

$$(u/v)' = (u'v - uv')/v^2$$

Chain rule and trigonometric functions

p.196

Rule

$$\frac{d}{dx} f(g(x)) = f'(g(x))g'(x)$$

$$(\sin x)' = \cos x$$

$$(\cos x)' = -\sin x$$

Implicit differentiation

p.200

Rule

$$\frac{d}{dx}(y^2) = 2yy'$$

$$\frac{d}{dx}(xy) = y + xy'$$

Logarithmic, exponential and inverse-trig derivatives

p.203

Rule

$$(e^u)' = e^u u'$$

$$(\ln|u|)' = u'/u$$

$$(\arctan x)' = 1/(1 + x^2)$$

L Hopital and indeterminate forms p.207

Rule

$$\lim \frac{f(x)}{g(x)} = \lim \frac{f'(x)}{g'(x)}$$

when the rule applies

Increasing, decreasing and concavity p.210

Rule

$$f' > 0 : \text{ increasing;}$$
$$f'' > 0 : \text{ concave up}$$

Graphing, asymptotes and absolute extrema p.213

Rule

Candidates = endpoints plus interior critical numbers

Optimization and mean-value theorems p.216

Rule

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$
$$a < c < b$$

Antiderivatives and substitution p.219

Rule

$$\int x^n \, dx = x^{n+1}/(n+1) + C \ (n \neq -1)$$
$$\int dx/x = \ln |x| + C$$

Definite integrals, signed area and the fundamental theorem p.222

Rule

$$\int_a^b f(x) \, dx = F(b) - F(a)$$
$$\frac{d}{dx} \int_a^{g(x)} f(t) \, dt = f(g(x))g'(x)$$

Symbol glossary

Symbols are listed with their lesson context. A symbol may represent different quantities in different rules.

Limits, continuity and the derivative definition

h small nonzero input increment before the limit / same unit as x

$f'(x)$ instantaneous rate of change / output unit per input unit

Basic derivatives and tangent lines

a input coordinate of tangent point / input unit

$f(a)$ output coordinate of tangent point / output unit

$f'(a)$ tangent slope / output unit per input unit

Product and quotient rules

u, v the two factor functions / problem dependent

u', v' their derivatives with respect to x / corresponding output per input unit

Chain rule and trigonometric functions

$u = g(x)$ inner expression / problem dependent

$g'(x)$ inner derivative / inner-expression unit per input unit

Implicit differentiation

y' dy/dx, the local slope of the curve / y unit per x unit

Logarithmic, exponential and inverse-trig derivatives

e base of the natural exponential / dimensionless

$\ln u$ natural logarithm of a positive dimensionless argument / dimensionless

L Hopital and indeterminate forms

$0/0$ indeterminate limiting form, not an arithmetic value / not a measured quantity

∞/∞ indeterminate quotient form / not a measured quantity

Increasing, decreasing and concavity

f' first derivative, controlling increase or decrease / output per input unit

f'' second derivative, controlling concavity / output per input unit squared

Graphing, asymptotes and absolute extrema

$[a, b]$ closed interval including both endpoints / input unit

$f(c)$ function value at a candidate point / output unit

Optimization and mean-value theorems

x one rectangle side length / length unit

A rectangle area / square length unit

c interior input guaranteed by the mean-value theorem / input unit

Antiderivatives and substitution

u chosen inner expression / problem dependent

du differential of the inner expression / same unit as u

C arbitrary additive integration constant / antiderivative unit

Definite integrals, signed area and the fundamental theorem

a, b lower and upper integration bounds / input unit

F an antiderivative of f / f unit times input unit

t dummy integration variable / input unit

Source and coverage notes

MATH 203: derivative rules, implicit/logarithmic/exponential derivatives, L Hopital, graphing and extrema, optimization, Rolle and mean-value theorems, integration and the fundamental theorem. The full lecture syllabus is the main source.

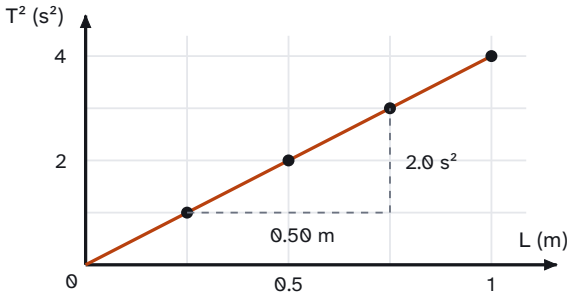
Open original course folder

The source audit inventories 159 non-programming files; 95 yielded substantial extracted text. Scanned diagrams and handwritten or image-only material were not fully transcribed. Check lecturer-specific derivations and conventions. The practice tests are original study material, not predictions of an examination.

Mechanics and Heat Lab

FIG. 05 / LABORATORY

Choose axes that turn the pendulum law into a line



Slope = rise / run
= 2.0 / 0.50 = 4.0 s²/m

g = 4 pi² / slope

Here g is about 9.87 m/s².
Keep the swing angle small.

Takeaway: T² goes vertically; L goes horizontally. The slope is not g.

All distinct experiments in the folder: pendulum, free fall, static and dynamic springs, force table, cooling, Boyle law, expansion, specific heat, latent heat and Joule equivalent. Duplicate versions are consolidated.

Subject map

All distinct experiments in the folder: pendulum, free fall, static and dynamic springs, force table, cooling, Boyle law, expansion, specific heat, latent heat and Joule equivalent. Duplicate versions are consolidated.

Follow this reading route from foundations to applications. Each line links to its lesson; revisit the stated prerequisites whenever a step feels unfamiliar.

01 Earn marks from tables, graphs and uncertainty p.248

02 Simple pendulum p.252

03 Free fall and axis choice p.256

04 Static Hooke law p.259

05 Dynamic spring and gravity p.262

06 Force table and equilibrium p.265

07 Newton cooling law p.268

08 Boyle law p.271

09 Linear thermal expansion p.274

10 Specific heat, latent heat and Joule equivalent p.277

11 Calorimetry: determine a solid specific heat p.280

12 Latent heat and electrical-to-thermal conversion p.283

Earn marks from tables, graphs and uncertainty

ML / 01

- Calculate a fitted-line slope with the correct axis units.
- Distinguish percentage error from measurement uncertainty.

You need: Coordinate differences; division; raw and derived measurements.

01 Idea

A lab result is a measured quantity with units and a justified method. Record raw observations before calculating derived columns. A slope is change in the vertical variable divided by change in the horizontal variable. Its units are essential and determine how to recover the desired constant.

An experimental result combines a number, units and a defensible measurement method. Repeated readings reveal spread, but they may all share the same calibration offset. Percentage error compares a result with a reference; it does not by itself explain whether the discrepancy comes from random variation or systematic bias.

- Repeating a measurement can reduce random scatter in the mean.
- A shared calibration offset remains in repeated readings unless it is identified and corrected.

02 Rule and symbols

Rule

$$m = \Delta y / \Delta x$$

$$\%\ \text{error} = 100 |x_{\text{measured}} - x_{\text{reference}}| / |x_{\text{reference}}|$$

m slope of the selected vertical variable against horizontal variable / vertical unit / horizontal unit

x_{measured} experimental estimate / same as reference

$x_{\text{reference}}$ accepted comparison value / same as measurement

03 Method

1. Record raw readings with units before calculating derived quantities.
2. Label axes with quantities and units and draw a suitable best-fit line.
3. Choose two widely separated points on that fitted line and calculate rise over run.
4. Use axis units to interpret the slope; compare with a reference using the stated percentage-error formula.

Use percentage error for a reference comparison, and discuss instrument or method bias separately from random spread.

04 Worked example

A fitted line rises by 0.8 square seconds over 0.2 m.

01 The example has vertical change 0.8 s^2 and horizontal change 0.2 m.

02 Slope is rise over run: 0.8/0.2 = 4 s^2/m.

03 The reciprocal, 0.25 m/s^2, belongs to a graph with reversed axes and is a different slope.

04 A slope should come from the fitted line, not necessarily from two adjacent raw observations.

Watch out

Do not force a line through the origin unless the model and measurement offsets justify it.

Exam habit

Write slope units next to the numerical calculation; they help reveal an inverted axis choice.

05 Your turn

A measured g is 9.6 $\mathrm{m/s^2}$ and reference g is 9.8 $\mathrm{m/s^2}$. Calculate percentage error and name one possible systematic error.

If you get stuck, uncover one hint at a time:

1. Compare the measured and reference values using an absolute difference; the denominator must be the reference.
2. Divide that difference by 9.8 m/s^2 and multiply by 100 percent.
3. For the second part name a specific fixed bias, such as a timing calibration offset, and explain why it would affect readings consistently.

06 Check your understanding

Can percentage error alone describe repeatability of a measurement?

A ruler has a fixed zero offset. Taking many readings and averaging them will:

1. Retain the systematic offset unless corrected
2. Automatically remove the offset
3. Prove the ruler is accurate if the readings agree

Answers and explanations: page 286.

Uncertainty is different from percentage error

An uncertainty describes measurement quality. Percentage error compares a result with a reference value.

- Use the uncertainty method specified by your experiment or instrument; do not invent a precision from extra calculator digits.
- Give the uncertainty the same unit as the value. Round the value to the same decimal place as the reported uncertainty.
- Repeated measurements help estimate and reduce the effect of random scatter. They do not remove a consistent zero offset.
- An uncertainty range is not a guarantee that every possible error is inside it.
- reported result $=$ value $\pm$ uncertainty
- $\%$ error $= 100\dfrac{|x_{\text{measured}} - x_{\text{reference}}|}{|x_{\text{reference}}|}$

A result is $(9.6 \pm 0.2)\,\mathrm{m/s^2}$, and the reference is $9.8\,\mathrm{m/s^2}$. State the given uncertainty and calculate percentage error.

1. The stated uncertainty is $0.2\,\mathrm{m/s^2}$.
2. Percentage error is $100|9.6 - 9.8|/9.8 \approx 2.04\%$.

Uncertainty: $\pm 0.2\,\mathrm{m/s^2}$. Percentage error: about 2.04%. They answer different questions.

Simple pendulum

ML / 02

- Measure pendulum length and period consistently.
- Recover g from one trial or a T^2-versus-L graph.

You need: Complete oscillation; squaring; slope interpretation.

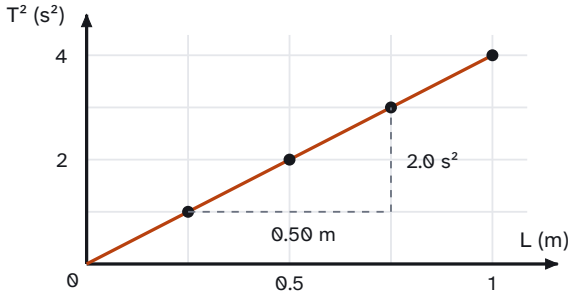
01 Idea

For small angles a pendulum period depends mainly on length and gravity, not bob mass. Measure length from pivot to the centre of the bob. Timing several complete oscillations reduces the relative impact of reaction time. Keep the starting angle small and avoid pushing the bob.

A pendulum period is the time for a complete return, not merely a trip to the other side. Timing many cycles spreads the start-stop reaction uncertainty over a longer total duration. Its length is a geometric quantity from pivot to bob centre, and a consistent length definition is as important as careful timing.

FIG. 05 / LABORATORY

Choose axes that turn the pendulum law into a line



Slope = rise / run
= 2.0 / 0.50 = 4.0 s²/m

g = 4 pi² / slope

Here g is about 9.87 m/s².
Keep the swing angle small.

Takeaway: T² goes vertically; L goes horizontally. The slope is not g.

Figure 05. For small pendulum oscillations, a plot of T^2 against L has slope $4\pi^2/g$. The illustrated slope is 4.0 s^2/m, giving g about 9.87 m/s^2. Points are ideal model values, not experimental data.

- Dividing a multi-cycle time by cycle count gives the mean single-cycle period.
- For small angles, period depends on length and gravity rather than bob mass.

02 Rule and symbols

Rule

$$T = t/N$$

$$T^2 = (4\pi^2/g)L$$

$g = 4\pi^2/m$ for T^2 versus L

L pivot-to-bob-centre length / m

t time for N complete oscillations / s

T period of one oscillation / s

g gravitational acceleration / m/s^2

Use this when:

The simple model assumes small angular displacement, negligible damping and a light inextensible suspension.

03 Method

1. Measure length from pivot to the bob centre and release from a small angle without a push.
2. Time several complete oscillations and divide total time by their count.
3. Calculate T^2 and repeat at several lengths if graphing.
4. For T^2 on the vertical axis and L horizontally, use g = 4π^2/slope.

A pendulum time for several cycles first requires conversion to period before applying the gravity relation.

04 Worked example

For L=1 m and T=2 s, g = $4\pi^2$ times 1/4 = 9.87 $\mathrm{m/s^2}$.

01 For the example L = 1 m and T = 2 s.

02 Square the measured period: T^2 = 4 s^2.

03 Rearranging T^2 = 4π^2L/g gives g = 4π^2L/T^2.

04 Substitution gives g = 4π^2 times 1/4 $\approx$ 9.87 m/s^2.

Watch out

Using total timing t instead of the single-cycle period T makes g dramatically wrong.

Exam habit

State which variables are on which axes before converting a slope into g.

05 Your turn

A pendulum takes 30 s for 15 cycles at length 0.99 m. Calculate period and g.

If you get stuck, uncover one hint at a time:

1. Use the 30 s reading as the total for 15 complete cycles, not the single period.
2. Divide total time by cycle count and square the resulting period.
3. Use length 0.99 m in the rearranged pendulum relation for gravity. Check that the squared period is in seconds squared.

06 Check your understanding

A bob crosses equilibrium twice in one full cycle. Should each crossing be counted as a cycle?

The pendulum bob is replaced by a heavier bob at the same effective length and small angle. The ideal period:

1. Halves whenever mass doubles
2. Stays approximately the same

3. Increases in proportion to mass

Answers and explanations: page 286.

Free fall and axis choice

ML / 03

- Linearize free fall with a time-squared graph.
- Select the correct g conversion for either axis orientation.

You need: Constant acceleration from rest; square of time; fitted-line slope.

01 Idea

For release from rest with negligible air drag, fall distance grows with time squared. The course sheets mention both graph orientations; the equation must decide the slope conversion. A light sensor or electronic timer avoids large manual timing errors.

Free fall from rest is quadratic in time, so plotting distance directly against time gives a curve. Squaring time creates a straight-line relationship. Which variable is vertical then determines whether slope represents half of gravity or the reciprocal form; the graph label is part of the calculation.

- For height plotted vertically against time squared, slope has acceleration units.
- Reversing the axes reverses the ideal slope and changes the conversion needed for gravity.

02 Rule and symbols

Rule

$$h = \tfrac{1}{2} g t^2;$$

$$m_{h \text{ vs } t^2} = g/2;$$

$$m_{t^2 \text{ vs } h} = 2/g$$

h vertical fall distance / m

t fall time / s

g gravitational acceleration / m/s^2

m graph slope / depends on axes

Use this when:

The object starts from rest and air resistance is negligible over the measured fall.

03 Method

1. Release from rest and measure fall time with the stated apparatus.
2. Calculate t^2 for each height.
3. State the graph axes and compare $h = (g/2)t^2$ with $y = mx$.
4. For h against t^2 use $g = 2m$; for t^2 against h use $g = 2/m$.

When a free-fall graph is given, read the axes before recalling any slope-to-gravity rule.

04 Worked example

A plot of h against t^2 has slope 4.9 m per square second, so g=9.8 $\mathrm{m/s^2}$.

01 With h on the vertical axis, the fitted slope is 4.9 $\mathrm{m/s^2}$.

02 The equation $h = (g/2)t^2$ shows this slope is half g, so $g = 9.8\ \mathrm{m/s^2}$.

03 Reversing the axes makes slope 2/g, about 0.204 $\mathrm{s^2/m}$.

04 The reciprocal slopes describe the same ideal relationship; neither slope can be interpreted without its axes.

Watch out

Do not use g=2 times slope for a t-squared-versus-height graph.

Exam habit

Do not memorize slope = g/2 without the graph orientation; use the displayed equation to decide.

05 Your turn

The slope of t^2 against height is 0.20 square seconds per metre. Find g.

If you get stuck, uncover one hint at a time:

1. The stated graph puts time squared vertically and height horizontally. Write the fall equation with that vertical quantity isolated.
2. Compare it with a straight-line equation to identify the slope as the reciprocal-gravity form.
3. Use the slope 0.20 s^2/m to solve for gravity. Verify that the resulting units are m/s^2.

06 Check your understanding

A t^2-versus-h slope is 0.20 s^2/m. What is g?

For release from rest without drag, why is height plotted against time squared?

1. It removes gravity from the experiment
2. It makes height proportional to time
3. It makes the ideal relationship linear

Answers and explanations: page 286.

Static Hooke law

ML / 04

- Distinguish spring extension from loaded length.
- Find spring stiffness from a load or a force-extension slope.

You need: Weight mg; length differences; centimetres to metres.

01 Idea

Within the proportional range, spring extension is proportional to the applied load. Extension is loaded length minus unloaded length. The restoring spring force points opposite the extension, while the applied load magnitude is k times extension. An elastic spring returns to its original length after unloading, but that alone does not prove a straight-line force-extension relation.

A spring experiment measures extension relative to the unloaded state. The hanging mass supplies a weight, which is a force rather than a length. Within the proportional range, plotting load against extension reveals stiffness: a steep line means more force is needed for each extra metre of stretch.

- Using total loaded length would include an irrelevant original-length offset.
- Weight must be calculated from mass before dividing by extension to obtain stiffness.

02 Rule and symbols

Rule

$$F = k\Delta L$$

$$F = mg$$

$$k = m_{F \text{ vs } \Delta L}$$

F applied load magnitude / N

ΔL loaded minus unloaded spring length / m

k spring stiffness / N/m

m total suspended mass / kg

Use this when:

Use Hooke law only over the proportional linear range; elastic recovery alone does not guarantee strict proportionality.

03 Method

1. Record the unloaded spring length first.
2. Add the load safely and calculate extension as loaded minus unloaded length.
3. Convert extension to metres and load mass to force using mg.
4. Find k from F/ΔL or the slope of F against ΔL; check the graph remains linear.

For static spring problems, convert mass to force and loaded-minus-unloaded length to extension.

04 Worked example

A 2 N load produces 0.04 m extension, so k=50 N/m.

01 The applied load is 2 N and the extension is 0.04 m, not the total spring length.

02 Using F = kΔL gives k = 2/0.04 = 50 N/m.

03 This means an additional 1 m extension would require 50 N only if the same linear model remained valid.

04 If unloading leaves a changed length, the spring has not returned elastically; avoid extrapolating the original model.

Watch out

Centimetres must be converted to metres before reporting k in N/m.

Exam habit

Include the hanger in total suspended mass if the apparatus instructions require it; otherwise the force scale can acquire an offset.

05 Your turn

A 0.30 kg mass extends a spring by 6 cm. Use g=10 $\mathrm{m/s^2}$. Find k.

If you get stuck, uncover one hint at a time:

1. The 0.30 kg mass supplies a weight; the 6 cm measurement is its extension.
2. Convert extension to metres and multiply mass by the specified gravity.
3. Divide the applied force by extension to obtain stiffness. Retain N/m and assume the spring remains in its proportional range.

06 Check your understanding

A 0.30 kg total load extends the spring 6 cm with g = 10 m/s^2. What is k?

A spring original length is 10 cm and its loaded length is 14 cm. Which length belongs in the Hooke-law extension term?

1. The 4 cm difference
2. The 14 cm loaded length
3. The 10 cm original length

Answers and explanations: page 286.

Dynamic spring and gravity

ML / 05

- Relate mass-spring period to load and spring stiffness.
- Use graph axes to distinguish a k experiment from a g experiment.

You need: Static spring extension; period measurement; slope units.

01 Idea

An oscillating mass-spring system has a period that increases with mass and decreases with stiffness. Combining static extension with dynamic period allows gravity to be estimated. Small oscillations and the same effective load model are needed for consistency.

Static stretch and dynamic oscillation probe the same spring in different ways. A larger oscillating mass produces a longer period, while a stiffer spring produces a shorter one. Squaring period makes its relationship with mass linear, so the graph slope contains the stiffness information.

- For period squared against mass, a larger slope corresponds to a smaller spring stiffness.
- A period-squared-versus-extension graph instead contains gravity, so the two slope interpretations cannot be exchanged.

02 Rule and symbols

Rule

$$T^2 = 4\pi^2 m/k$$

$$T^2 = (4\pi^2/g)\Delta L$$

T oscillation period / s

m oscillating mass in the stated model / kg

k spring stiffness / N/m

ΔL static equilibrium extension / m

g gravitational acceleration / m/s^2

Use this when:

Small oscillations stay in the spring proportional range; the static and dynamic models must treat hanger and effective spring mass consistently.

03 Method

1. Measure small vertical oscillations about equilibrium and obtain period from several cycles.
2. For T^2 against mass compare the slope with $4\pi^2/k$.
3. For T^2 against static extension compare the slope with $4\pi^2/g$.
4. Apply the matching rearrangement and retain the effective-mass assumptions used by the apparatus.

Inspect whether the horizontal axis is mass or static extension before converting a spring-oscillation slope.

04 Worked example

A T-squared-versus-extension slope of 4.0 square seconds per metre gives $g=4\pi^2/4.0=9.87\ \mathrm{m/s^2}$.

01 The example plots T^2 vertically against static extension horizontally.

02 Its slope is 4.0 s^2/m, corresponding to $4\pi^2/g$.

03 Rearrangement gives $g = 4\pi^2/4.0 \approx 9.87\ m/s^2$.

04 A T^2-versus-mass slope would instead determine k; substituting it here would mix two different experiments.

Watch out

The symbols for mass and graph slope can both be m; label them explicitly to avoid confusion.

Exam habit

Use slope units to identify the requested constant: s^2/kg for stiffness, s^2/m for gravity in these two plots.

05 Your turn

A T-squared-versus-mass graph has slope 0.80 square seconds per kg. Find spring constant.

If you get stuck, uncover one hint at a time:

1. The horizontal axis is mass, so choose the stiffness relation rather than the extension-based gravity relation.
2. Write the slope coefficient of mass in the squared-period equation.
3. Rearrange to make stiffness the subject and substitute 0.80 s^2/kg. Check that the result has N/m units.

06 Check your understanding

A T^2-versus-mass slope is 0.80 s^2/kg. What is k?

Two springs give period-squared-versus-mass lines. The stiffer spring has:

1. The same slope regardless of stiffness
2. The smaller slope
3. The larger slope

Answers and explanations: page 286.

Force table and equilibrium

ML / 06

- Calculate the equilibrant from perpendicular forces.
- Explain how the force-table ring indicates approximate equilibrium.

You need: Vector components; Pythagoras; angles from positive x.

01 Idea

The force table demonstrates vector addition. The resultant of two forces is their vector sum. The equilibrant has equal magnitude and opposite direction. A centred ring that does not press on the pin indicates approximate balance, subject to pulley friction.

The force table balances vector directions, not just force magnitudes. Two perpendicular pulls create a resultant pointing between them; a third force balances that resultant only if it points exactly opposite. A centred ring provides experimental evidence of approximate balance when it is not held by the pin.

- Equal resultant and equilibrant magnitudes are not enough without opposite directions.
- A force pushing the ring against the central pin can be balanced partly by the pin, hiding string-force imbalance.

02 Rule and symbols

Rule

$$\sum F_x = 0$$

$$\sum F_y = 0$$

$$F_i = m_i g$$

F_i tension force from hanging mass in the ideal setup / N

m_i total hanging mass for a string / kg

R resultant force magnitude / N

Use this when:

Mass-to-tension conversion neglects pulley friction and string mass; actual balance is approximate.

03 Method

1. Convert each hanging mass to a force using the chosen g.
2. Resolve all forces into signed horizontal and vertical components.
3. Find the resultant, then reverse both components for the equilibrant.
4. Adjust the balancing force and check the ring is centred without pressing on the pin.

When equilibrium is requested, first find the resultant, then reverse its components for the equilibrant.

04 Worked example

For 3 N east and 4 N north, the resultant is 5 N northeast.

01 The 3 N east and 4 N north forces give resultant components (+3,+4) N.

02 Its magnitude is $\sqrt{(3^2+4^2)}$ = 5 N.

03 Its precise direction is 53.1° north of east; northeast here describes its quadrant.

04 The equilibrant has components (-3,-4) N and magnitude 5 N, directed 53.1° south of west.

Watch out

The equilibrant is not the resultant in the same direction.

Exam habit

If an angle is requested, northeast alone is insufficient; give a numerical angle and reference axis.

05 Your turn

Two forces are 6 N east and 8 N north. Find equilibrant magnitude and direction.

If you get stuck, uncover one hint at a time:

1. Draw 6 N east and 8 N north as perpendicular components.
2. Find their resultant magnitude and its reference angle from the component ratio.
3. Reverse both components to obtain the equilibrant. Give a numerical angle with a compass reference, not merely southwest.

06 Check your understanding

Does the equilibrant have the same direction as the resultant?

Why should the force-table ring be free of contact with the central pin?

1. The pin removes all pulley friction
2. A ring touching a pin has no weight
3. The pin could supply an extra balancing force

Answers and explanations: page 286.

Newton cooling law

ML / 07

- Calculate temperature excess above the surroundings.
- Interpret an exponential cooling trend without confusing it with Celsius temperature.

You need: Subtracting temperatures; halving; logarithm recognition.

01 Idea

A hot object approaches room temperature as its temperature excess decays. The simple model assumes approximately constant surroundings and an effective cooling constant over the measured range. The temperature excess, not the absolute temperature, is the decaying quantity.

Cooling depends on how much hotter the object is than its surroundings. An object at 50°C in a 20°C room has an excess of 30°C, not 50°C. The simple exponential model shrinks this excess, then room temperature must be added back to recover the actual object temperature.

- Equal temperature excesses in otherwise equivalent conditions give equal cooling-rate magnitudes in the model.
- As excess falls, the cooling rate becomes smaller and the object approaches room temperature.

02 Rule and symbols

Rule

$$T - T_r = (T_0 - T_r)e^{-kt}$$

$$\ln[(T - T_r)/(T_0 - T_r)] = -kt$$

T object temperature / °C or K consistently

T_r room temperature / same unit as T

k effective cooling constant / inverse time

t elapsed time / s or min consistently with k

Use this when:

Room temperature is approximately constant and the effective cooling constant applies over the measured range.

03 Method

1. Measure room temperature and record it as the reference.
2. Subtract room temperature from each object temperature.
3. Compare excesses over time or use the stated logarithmic plot.
4. Add the room temperature back when converting a final excess into an object temperature.

A statement that temperature excess halves means halve object-minus-room temperature, not the Celsius reading itself.

04 Worked example

If room temperature is 20 degrees C and the object cools from 80 to 50, the excess falls from 60 to 30 degrees, so it halves.

01 The room is at 20°C, so the initial 80°C object has excess 80 - 20 = 60°C.

02 At 50°C its excess is 50 - 20 = 30°C.

03 The excess has halved, even though 50 is not half of 80.

04 This distinction explains why the model approaches room temperature rather than cooling toward 0°C.

Watch out

The object approaches room temperature, not zero degrees C.

Exam habit

When logging data, use a dimensionless excess ratio such as (T-Tr)/(T0-Tr), with temperatures consistently measured.

05 Your turn

Room temperature is 25 degrees C. A body starts at 85 degrees C and its temperature excess halves in 5 minutes. What is its temperature then?

If you get stuck, uncover one hint at a time:

1. Subtract room temperature 25°C from the initial 85°C reading to obtain excess.
2. Halve that excess for the stated 5-minute interval.
3. Add the unchanged room temperature back. Check that the result lies between initial object temperature and room temperature.

06 Check your understanding

At room temperature 25°C, an object starts at 85°C and its excess halves. What is its new temperature?

A 60°C object is placed in a 20°C room or a 40°C room under otherwise identical conditions. It initially cools faster in:

1. The 20°C room
2. The 40°C room
3. Both rooms equally because the object starts at 60°C

Answers and explanations: page 286.

Boyle law

ML / 08

- Use absolute pressure in a fixed-temperature gas comparison.
- Predict the inverse pressure-volume trend and check thermal settling.

You need: Inverse proportion; multiplying and dividing; consistent pressure and volume units.

01 Idea

At fixed temperature and fixed amount of gas, absolute pressure is inversely proportional to volume. A narrow uniform tube may use gas-column length as a proxy for volume. Let the gas settle thermally after changing pressure.

Boyle law follows the pressure-volume trade-off for a fixed amount of gas at constant temperature. The gas can occupy less space only while exerting greater absolute pressure. Rapid compression can warm the gas, so thermal settling is needed before treating readings as one constant-temperature comparison.

- The pressure-volume product stays constant only while amount and temperature stay fixed.
- Using gauge pressure instead of absolute pressure changes the comparison unless converted appropriately.

02 Rule and symbols

Rule

$$P_1V_1 = P_2V_2$$

$$P \propto 1/V$$

P absolute gas pressure / Pa or consistent pressure unit

V gas volume / m³ or consistent volume unit

Use this when:

Gas amount and temperature remain constant; the ideal Boyle-law approximation is suitable for the measured range.

03 Method

1. Confirm the gas amount stays fixed and readings are taken at the same temperature.
2. Use absolute pressure rather than gauge pressure.
3. Write P1V1 = P2V2 and isolate the requested volume or pressure.
4. Check that larger pressure gives smaller volume and allow time for thermal settling between trials.

Use Boyle law when gas amount is fixed and two states are explicitly at the same temperature.

04 Worked example

A gas at 100 kPa and 60 cubic centimetres has volume 30 cubic centimetres at 200 kPa, if temperature stays fixed.

01 Initially the gas has pressure 100 kPa and volume 60 cm^3.

02 At fixed temperature and amount, their product stays constant.

03 Doubling pressure to 200 kPa requires half the volume: V2 = 100 times 60/200 = 30 cm^3.

04 Keeping kPa and cm^3 is valid for this ratio because the corresponding units match on both sides.

Watch out

Compressing quickly can warm the gas and violate the constant-temperature assumption.

Exam habit

Do not convert a gauge reading to absolute pressure by ignoring atmospheric pressure; use the apparatus definition.

05 Your turn

A trapped gas has volume 40 cubic centimetres at absolute pressure 120 kPa. Find its volume at 160 kPa at the same temperature.

If you get stuck, uncover one hint at a time:

1. Check that the supplied 120 and 160 kPa are absolute pressures and temperatures match.
2. Set the initial pressure-volume product equal to the final product, then isolate final volume.
3. Use 120 kPa, 40 cm^3 and 160 kPa consistently. The larger final pressure should give a smaller final volume.

06 Check your understanding

If volume is compressed quickly, why wait before recording a Boyle-law reading?

A gas is compressed rapidly and its temperature rises. Can the original constant-temperature Boyle comparison be applied immediately?

1. Only if gauge pressure replaces absolute pressure
2. No; the temperature condition is not yet satisfied
3. Yes; any gas compression follows the same constant product

Answers and explanations: page 286.

Linear thermal expansion

ML / 09

- Calculate a linear expansion coefficient with compatible length units.
- Extract the coefficient from a length-change-versus-temperature graph.

You need: Small length changes; temperature differences; mm-to-m conversion.

01 Idea

Heating changes the length of a rod slightly. The coefficient of expansion describes fractional length change per degree. A dial gauge often reads millimetres while the rod length is recorded in metres, so consistent units are critical.

An expansion coefficient describes fractional length change per degree, so it should not depend on choosing millimetres or metres for the apparatus readout. Problems arise when the tiny displacement is entered in millimetres while the original rod remains in metres. Consistent length units must precede division.

- The same displacement has different numerical values in metres and millimetres.
- Dividing length change by original length removes the length unit, leaving inverse temperature.

02 Rule and symbols

Rule

$$\Delta L = \alpha L_0 \Delta T$$

$$\alpha = m_{\Delta L \text{ vs } \Delta T} / L_0$$

α linear expansion coefficient / K^{-1}

L_0 initial rod length / m

ΔL length increase / m

ΔT temperature increase / K

Use this when:

The temperature range is small enough to treat α as approximately constant and the rod temperature as sufficiently uniform.

03 Method

1. Record original rod length and the initial and final temperatures.
2. Convert gauge displacement to the same length unit as the rod.
3. Rearrange $\Delta L = \alpha L 0 \Delta T$ to solve for α.
4. For a graph of ΔL against ΔT divide its slope by L0.

When a dial displacement and rod length use different units, convert them before calculating the coefficient.

04 Worked example

A 1 m rod expands 0.5 mm over 50 K, giving alpha = 0.0005/(1 times 50)=0.00001 per K.

01 The 1 m rod expands by 0.5 mm, which is 0.0005 m.

02 The temperature rise is 50 K.

03 Divide extension by original length and temperature rise: α = 0.0005/(1 times 50).

04 The result is 0.00001 K^{-1}, or 1.0 × 10^{-5} K^{-1}.

Watch out

A graph against absolute temperature may have an intercept; do not confuse it with a change-of-temperature plot.

Exam habit

Write the displacement conversion explicitly; the small numerical size of α is physically expected.

05 Your turn

A 0.50 m rod expands 0.30 mm after warming by 40 K. Find its expansion coefficient.

If you get stuck, uncover one hint at a time:

1. The 0.30 mm reading is a length increase, not final rod length.
2. Convert it to metres, then divide by original length 0.50 m and temperature rise 40 K.
3. Report an inverse-kelvin coefficient. Its small magnitude is expected for a typical solid.

06 Check your understanding

What conversion error occurs if 0.5 mm is entered as 0.5 m?

A student enters a millimetre displacement number as metres while keeping rod length in metres. The coefficient is:

1. 1000 times too small
2. Unaffected because coefficients have no length unit
3. 1000 times too large

Answers and explanations: page 286.

Specific heat, latent heat and Joule equivalent

ML / 10

- Separate sensible heating, phase-change energy and electrical input.
- Keep the joule-per-calorie conversion dimensional.

You need: Energy balance; electrical power; grams-to-kilograms conversion.

01 Idea

Mixing calorimetry balances heat lost by the hot sample against heat gained by water and calorimeter. Latent heat accounts for a phase change. Electrical heating provides IVt joules; comparing it with heat measured in calories gives joules per calorie, not a dimensionless ratio.

Electrical heating supplies energy over a measured time. If a sample already at its melting point uses all that input only for melting, dividing by melted mass gives latent heat per kilogram. If temperature changes or losses also occur, the same division would misattribute energy to melting.

- Voltage times current gives power, so multiplying by time gives total supplied energy.
- Latent heat is energy per mass, not total energy and not a temperature rise.

02 Rule and symbols

Rule

$$m_s c_s (T_s - T_f) = (m_w c_w + C_{\text{cal}})(T_f - T_w)$$

$$Q = mL$$

$$E = IVt$$

Q thermal energy transferred / J

L specific latent heat / J/kg

I heater current / A

V heater voltage / V

t heating duration / s

Use this when:

The latent-heat example assumes the sample starts at its melting point and all stated energy is used for melting.

03 Method

1. Identify whether temperature changes, phase changes or both occur.
2. Calculate electrical input as IVt when heater readings are supplied.
3. Use mcΔT for temperature change and mL for a phase change.
4. If comparing joules with calories, divide joules by calories and state J/cal.

Use the electrical-input and latent-heat relations together only after identifying where the supplied energy goes.

04 Worked example

If 500 J melts 0.002 kg at its melting point with no other heating, latent heat is 250000 J/kg.

01 The sample receives 500 J only for melting and the melted mass is 0.002 kg.

02 Rearranging Q = mL gives L = 500/0.002 = 250000 J/kg.

03 In the separate conversion example, 418.4 J is compared with 100 cal.

04 Their ratio is 418.4/100 = 4.184 J/cal; the unit is essential because the two energy scales differ.

Watch out

Ignoring calorimeter heat capacity or heat loss can bias the inferred material property.

Exam habit

State where the energy goes before dividing by mass; not all heating energy necessarily produces the phase change.

05 Your turn

A heater at 6 V carries 2 A for 100 s. If all energy melts a sample already at its melting point and the melted mass is 0.004 kg, find supplied energy and latent heat.

If you get stuck, uncover one hint at a time:

1. Find heater power from 6 V and 2 A before using the 100 s duration.
2. Multiply power by duration to obtain input energy in joules.
3. Because all input melts 0.004 kg at the melting point, divide that energy by melted mass for latent heat. Keep the two answers separately labelled.

06 Check your understanding

If both electrical input and thermal gain are already in joules, is their ratio a J/cal conversion?

Why must the sample already be at its melting point in the simple electrical melting calculation?

1. Otherwise some input may first warm the sample
2. Otherwise electrical energy cannot enter matter
3. Because latent heat means temperature rise per second

Answers and explanations: page 286.

Calorimetry: determine a solid specific heat

ML / 11

- Balance heat lost by a solid against heat gained by water and calorimeter.
- Solve for specific heat while using the correct temperature changes.

You need: Heat capacity; mass in kg; a common final equilibrium temperature.

01 Idea

A hot solid placed in cooler water transfers energy to the water and calorimeter. At equilibrium they share a final temperature. The ideal model assumes negligible heat loss to surroundings and no phase change. The calorimeter contribution must be included if it is specified.

Calorimetry matches energy leaving a hot sample with energy entering cooler water and, if relevant, its container. Each body has its own temperature change despite reaching one shared final temperature. Writing those two changes explicitly is often the most important step.

- The solid temperature drop determines how much energy it loses for each unit of its specific heat.
- The water temperature rise and known specific heat determine the energy gained in the ideal balance.

02 Rule and symbols

Rule

$$c_s = \frac{(m_w c_w + C_{\text{cal}})(T_f - T_w)}{m_s(T_s - T_f)}$$

c_s solid specific heat capacity / J/(kg K)

m_s, m_w solid and water masses / kg

C_{cal} calorimeter heat capacity / J/K

T_f final equilibrium temperature / °C or K

Use this when:

The example neglects surroundings losses and calorimeter heat capacity and assumes no phase change.

03 Method

1. Identify the initially hot body, initially cold water and common final temperature.
2. Write heat lost by the solid equal to heat gained by water plus calorimeter.
3. Use positive temperature drops and rises on opposite sides.
4. Solve for cs and state the heat-loss and calorimeter assumptions.

Use a heat-lost equals heat-gained balance when a hot solid and cooler water reach one final temperature.

04 Worked example

With a negligible calorimeter, 0.10 kg solid cooling from 80 to 30 degrees C heats 0.20 kg water from 25 to 30 degrees C.

01 Water warms from 25°C to 30°C, a rise of 5 K.

02 Its heat gain is 0.20 times 4200 times 5 = 4200 J.

03 The 0.10 kg solid cools from 80°C to 30°C, a drop of 50 K.

04 With negligible calorimeter heat capacity and losses, 0.10 times cs times 50 = 4200, giving cs = 840 J/(kg K).

Watch out

Using the same temperature change for hot solid and water is generally wrong.

Exam habit

Use the calorimeter heat-capacity term when supplied; discarding it undercounts heat gained by the cold side.

05 Your turn

A 0.20 kg solid cools from 90 to 40 degrees C and warms 0.10 kg water from 30 to 40 degrees C. Ignore calorimeter heat capacity and losses; use water c=4200 J/(kg K). Find the solid specific heat.

If you get stuck, uncover one hint at a time:

1. Write the solid temperature drop and water temperature rise as separate positive differences.
2. Calculate water heat gain using 0.10 kg and its given specific heat; ignore the calorimeter only because the question says so.
3. Equate that heat to solid mass 0.20 kg times unknown specific heat times its cooling drop, then isolate specific heat.

06 Check your understanding

Why is the solid temperature change not the same as the water temperature change?

If a calorimeter warms but its heat gain is ignored, what happens to the inferred solid specific heat under this calculation?

1. It cannot change because water mass is unchanged
2. It is underestimated
3. It is overestimated

Answers and explanations: page 286.

Latent heat and electrical-to-thermal conversion

ML / 12

- List all energy stages for ice melting and subsequent warming.
- Interpret how lost heat biases a measured electrical-to-calorie ratio.

You need: Specific heat versus latent heat; phase-change temperatures; units of energy.

01 Idea

A phase-change experiment separates melting from subsequent warming of the melted material. In the Joule-equivalent experiment electrical energy is compared with heat gained by water and calorimeter. If both quantities are already in joules, their ratio tests energy accounting; it is not a joule-per-calorie conversion.

A joule-per-calorie experiment compares the same intended energy transfer measured on two unit scales. If some supplied energy escapes without appearing in measured heat gain, the denominator becomes too small. Distinguishing units from efficiency prevents treating every energy ratio as a unit conversion.

- Joules divided by calories has a conversion unit.
- Joules divided by joules is dimensionless and describes energy accounting rather than a joule-per-calorie conversion.

02 Rule and symbols

Rule

$$Q_{\text{ice}} = mL_f + mc_w(T_f - 0^\circ\text{C})$$

$$J_{\text{eq}} = E_{\text{joules}}/Q_{\text{calories}}$$

m ice mass / kg

L_f specific latent heat of fusion / J/kg

c_w liquid water specific heat / J/(kg K)

J_{eq} joules per calorie conversion estimate / J/cal

Use this when:

The ice formulas use an ordinary-pressure melting point of approximately 0°C and

account for the stated final phase.

03 Method

1. Identify whether ice starts below or at its melting point.
2. Include warming of ice if needed, then melting, then warming of meltwater to the final temperature.
3. For conversion, keep electrical energy in joules and measured thermal gain in calories.
4. Assess a loss by asking whether the recorded heat gain is too small or too large for the supplied energy.

Check the denominator units before interpreting any electrical-to-thermal energy ratio.

04 Worked example

If ice begins below zero degrees C, an additional warming-of-ice term is required before melting.

01 Ice below 0°C first needs energy to warm to 0°C; that stage uses ice specific heat.

02 At 0°C, energy mLf melts it without a temperature rise in the ideal fixed-pressure model.

03 If the final state is liquid above 0°C, add mcw(Tf - 0°C) for the meltwater.

04 When ice already starts at 0°C the first stage is absent; the melting and later warming stages remain if required.

Watch out

Do not divide joules by joules and label the dimensionless answer J/cal.

Exam habit

Do not use the water specific heat for warming solid ice; the two phases have different heat capacities.

05 Your turn

An electrical experiment supplies 836.8 J and records heat gain of 200 cal. Find the conversion factor. If unaccounted heat escapes, does E/Q measured tend to be too high or too low?

If you get stuck, uncover one hint at a time:

1. Use electrical energy as the numerator and thermal gain in calories as the denominator.
2. Divide 836.8 J by 200 cal and retain the unit J/cal.
3. For the loss question, keep input fixed and ask what happens to the ratio if the measured heat-gain denominator becomes too small.

06 Check your understanding

If 836.8 J supplies a recorded heat gain of 200 cal, what is E/Q, and how would unrecorded heat loss bias it?

Both electrical input and measured thermal gain are stated in joules. Their ratio has:

1. Units of J/cal automatically
2. Units of watts
3. No unit, so it is not directly a J/cal conversion

Answers and explanations: page 286.

Lesson review and answers

Earn marks from tables, graphs and uncertainty

Return to lesson: page 248.

Your turn: No. It compares with a reference; repeated-reading spread or an uncertainty estimate is needed to discuss repeatability.

Concept check: 1. Correct. Every reading carries the same bias, so averaging alone preserves it.

Written method: Percentage error = 100 times 0.2/9.8 = 2.04 percent. A consistent height-scale zero offset is one possible systematic error. Two marks for the calculation and two for a specific error with its consistency explained.

Simple pendulum

Return to lesson: page 252.

Your turn: No. Count a return to the same position moving in the same direction as one full cycle.

Concept check: 2. Correct. Bob mass cancels in the small-angle simple-pendulum model.

Written method: T=30/15=2 s. g=$4\pi^2$ times 0.99/4=9.77 $\mathrm{m/s^2}$. Two marks per calculation.

Free fall and axis choice

Return to lesson: page 256.

Your turn: 10 m/s^2, using g = 2/0.20.

Concept check: 3. Correct. Distance from rest is proportional to time squared.

Written method: g = 2/0.20 = 10 $\mathrm{m/s^2}$. Four marks for axis identification, rearrangement, substitution and unit.

Static Hooke law

Return to lesson: page 259.

Your turn: 50 N/m: the force is 3 N and extension is 0.06 m.

Concept check: 1. Correct. Extension is the change from the unloaded length.

Written method: Force = 3 N; extension = 0.06 m; k=3/0.06=50 N/m. Award marks for force, conversion, ratio and unit.

Dynamic spring and gravity

Return to lesson: page 262.

Your turn: About 49.35 N/m, from $4\pi^2$/0.80.

Concept check: 2. Correct. Stiffness appears inversely in this slope.

Written method: Slope = $4\pi^2$/k, so k=$4\pi^2$/0.80=49.35 N/m. Credit the selected axes, rearrangement, numerical value and units.

Force table and equilibrium

Return to lesson: page 265.

Your turn: No. It has equal magnitude and opposite direction.

Concept check: 3. Correct. Contact can hide an imbalance among the intended string forces.

Written method: Resultant magnitude is 10 N at 53.1 degrees north of east. Equilibrant is 10 N at 233.1 degrees from positive x, or 53.1 degrees south of west. Two marks for magnitude and two for direction.

Newton cooling law

Return to lesson: page 268.

Your turn: 55°C: initial excess 60°C becomes 30°C, then add 25°C.

Concept check: 1. Correct. Its initial temperature excess is larger in the colder room.

Written method: Initial excess = 60 degrees C. Halved excess = 30 degrees C. Add room temperature: 25+30=55 degrees C. Credit both differences and final interpretation.

Boyle law

Return to lesson: page 271.

Your turn: Compression can change temperature; waiting helps the gas return to the intended common temperature.

Concept check: 2. Correct. Wait for the specified thermal condition or use an appropriate different model.

Written method: V2 = 120 times 40/160 = 30 cubic centimetres. Credit controlled conditions, equation, rearrangement and unit.

Linear thermal expansion

Return to lesson: page 274.

Your turn: The calculated expansion coefficient becomes 1000 times too large.

Concept check: 3. Correct. The entered displacement is 1000 times larger than the true metre value.

Written method: Convert expansion to 0.00030 m. Alpha=0.00030/(0.50 times 40)=0.000015 per K. Credit conversion, formula, substitution and result.

Specific heat, latent heat and Joule equivalent

Return to lesson: page 277.

Your turn: No. It is dimensionless and tests energy accounting or an efficiency interpretation under the model.

Concept check: 1. Correct. Pre-melting warming requires a separate energy term.

Written method: Energy=6 times 2 times 100=1200 J. Latent heat=$\frac{1200}{0.004} = 300000\ \mathrm{J/kg}$. Two marks per calculation; the no-loss and initial-phase-temperature assumptions matter.

Calorimetry: determine a solid specific heat

Return to lesson: page 280.

Your turn: They have different starting temperatures, though they reach the same final equilibrium temperature.

Concept check: 2. Correct. Ignoring the container undercounts energy lost by the solid, giving too small a specific heat.

Written method: Water gains 0.10 times 4200 times 10=4200 J. Solid loses 0.20 times cs times 50=10cs. Thus cs=420 J/(kg K). Two marks for each side and result.

Latent heat and electrical-to-thermal conversion

Return to lesson: page 283.

Your turn: 4.184 J/cal. If heat escapes and measured Q is too small, E/Q tends to be too high.

Concept check: 3. Correct. Like energy units cancel.

Written method: Factor=836.8/200=4.184 J/cal. Heat escaping reduces measured Q for the supplied E, tending to make the ratio too high. Two marks for each part.

Mechanics and Heat Lab / Test A

30 marks / Suggested time: 30 minutes. MCQs: 1 mark each. Written problems: 4 marks each. Attempt without notes.

Part I / Choose one answer

1. If y is force in N and x is extension in m, slope units are:

   A. N/m

   B. m/N

   C. N times m

   D. No units

2. Pendulum length should be measured to:

   A. The top of the table

   B. The lowest swing point

   C. The bottom of the bob

   D. The centre of the bob

3. For t^2 vertically versus h horizontally, g equals:

   A. Slope squared

   B. Slope/2

   C. 2 times slope

   D. 2/slope

4. The spring constant is the slope of:

   A. Time versus temperature

   B. Extension versus force

   C. Force versus extension

   D. Mass versus time

5. For fixed k, multiplying the oscillating mass by four makes the period:

   A. Four times

   B. Half

   C. Twice

D. Unchanged

6. The equilibrant of a 7 N eastward resultant is:

A. 7 N east

B. 14 N west

C. Zero

D. 7 N west

Part II / Show your reasoning

1. A measured g is 9.6 $\mathrm{m/s^2}$ and reference g is 9.8 $\mathrm{m/s^2}$. Calculate percentage error and name one possible systematic error.

2. A pendulum takes 30 s for 15 cycles at length 0.99 m. Calculate period and g.

3. The slope of t^2 against height is 0.20 square seconds per metre. Find g.

4. A 0.30 kg mass extends a spring by 6 cm. Use g=10 $\mathrm{m/s^2}$. Find k.

5. A T-squared-versus-mass graph has slope 0.80 square seconds per kg. Find spring constant.

6. Two forces are 6 N east and 8 N north. Find equilibrant magnitude and direction.

Mechanics and Heat Lab / Test B

30 marks / Suggested time: 30 minutes. MCQs: 1 mark each. Written problems: 4 marks each. Attempt without notes.

Part I / Choose one answer

1. In the simple cooling model, the cooling rate is proportional to:

   A. Mass only

   B. Zero

   C. Temperature excess above surroundings

   D. Absolute time only

2. Boyle law requires:

   A. Fixed temperature and gas amount

   B. Changing gas amount

   C. Fixed pressure

   D. Fixed volume

3. Two rods are made of the same material and have the same temperature rise. A rod twice as long has expansion:

   A. Twice as large

   B. Unchanged

   C. Four times as large

   D. Half as large

4. Electrical energy supplied by constant voltage and current for time t is:

   A. V/I

   B. IVt

   C. I/t

   D. Vt/I

5. In an ideal mixing experiment, the final equilibrium temperature lies:

   A. Below both

   B. Between the initial temperatures

C. Always at zero

D. Above both initial temperatures

6. To determine joules per calorie, divide electrical energy in joules by heat measured in:

A. Watts

B. Calories

C. Kelvin

D. Joules

Part II / Show your reasoning

1. Room temperature is 25 degrees C. A body starts at 85 degrees C and its temperature excess halves in 5 minutes. What is its temperature then?

2. A trapped gas has volume 40 cubic centimetres at absolute pressure 120 kPa. Find its volume at 160 kPa at the same temperature.

3. A 0.50 m rod expands 0.30 mm after warming by 40 K. Find its expansion coefficient.

4. A heater at 6 V carries 2 A for 100 s. If all energy melts a sample already at its melting point and the melted mass is 0.004 kg, find supplied energy and latent heat.

5. A 0.20 kg solid cools from 90 to 40 degrees C and warms 0.10 kg water from 30 to 40 degrees C. Ignore calorimeter heat capacity and losses; use water c=4200 J/(kg K). Find the solid specific heat.

6. An electrical experiment supplies 836.8 J and records heat gain of 200 cal. Find the conversion factor. If unaccounted heat escapes, does E/Q measured tend to be too high or too low?

Mechanics and Heat Lab / Test A / Answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1

MCQ: A. Slope units are vertical units divided by horizontal units.

Written solution: Percentage error = 100 times 0.2/9.8 = 2.04 percent. A consistent height-scale zero offset is one possible systematic error. Two marks for the calculation and two for a specific error with its consistency explained.

Return to lesson: page 248.

Question 2

MCQ: D. The ideal mass is represented at the bob centre.

Written solution: T=30/15=2 s. g=$4\pi^2$ times 0.99/4=9.77 $\mathrm{m/s^2}$. Two marks per calculation.

Return to lesson: page 252.

Question 3

MCQ: D. Rearranging h = gt squared/2 gives t^2 = (2/g)h.

Written solution: g = 2/0.20 = 10 $\mathrm{m/s^2}$. Four marks for axis identification, rearrangement, substitution and unit.

Return to lesson: page 256.

Question 4

MCQ: C. Force = k times extension matches y = mx.

Written solution: Force = 3 N; extension = 0.06 m; k=3/0.06=50 N/m. Award marks for force, conversion, ratio and unit.

Return to lesson: page 259.

Question 5

MCQ: C. Period is proportional to the square root of mass.

Written solution: Slope = $4\pi^2$/k, so k=$4\pi^2$/0.80=49.35 N/m. Credit the selected axes, rearrangement, numerical value and units.

Return to lesson: page 262.

Question 6

MCQ: D. Equal magnitude and opposite direction produce zero net force.

Written solution: Resultant magnitude is 10 N at 53.1 degrees north of east. Equilibrant is 10 N at 233.1 degrees from positive x, or 53.1 degrees south of west. Two marks for magnitude and two for direction.

Return to lesson: page 265.

Mechanics and Heat Lab / Test B / Answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1

MCQ: C. Newton cooling uses the difference between object and surrounding temperatures.

Written solution: Initial excess = 60 degrees C. Halved excess = 30 degrees C. Add room temperature: 25+30=55 degrees C. Credit both differences and final interpretation.

Return to lesson: page 268.

Question 2

MCQ: A. Only pressure and volume are varied in the ideal Boyle experiment.

Written solution: V2 = 120 times 40/160 = 30 cubic centimetres. Credit controlled conditions, equation, rearrangement and unit.

Return to lesson: page 271.

Question 3

MCQ: A. Expansion is proportional to original length.

Written solution: Convert expansion to 0.00030 m. Alpha=0.00030/(0.50 times 40)=0.000015 per K. Credit conversion, formula, substitution and result.

Return to lesson: page 274.

Question 4

MCQ: B. Electrical power is VI; multiplying by time gives energy.

Written solution: Energy=6 times 2 times 100=1200 J. Latent heat=$\frac{1200}{0.004} = 300000\ \mathrm{J/kg}$. Two marks per calculation; the no-loss and initial-phase-temperature assumptions matter.

Return to lesson: page 277.

Question 5

MCQ: B. Energy flows from the hotter body to the cooler one until equilibrium.

Written solution: Water gains 0.10 times 4200 times 10=4200 J. Solid loses 0.20 times cs times 50=10cs. Thus cs=420 J/(kg K). Two marks for each side and result.

Return to lesson: page 280.

Question 6

MCQ: B. The denominator determines the per-calorie unit.

Written solution: Factor=836.8/200=4.184 J/cal. Heat escaping reduces measured Q for the supplied E, tending to make the ratio too high. Two marks for each part.

Return to lesson: page 283.

Formula sheet

Earn marks from tables, graphs and uncertainty

p.248

Rule

$$m = \Delta y / \Delta x$$

$$\% \text{ error} = 100 |x_{\text{measured}} - x_{\text{reference}}| / |x_{\text{reference}}|$$

Simple pendulum

p.252

Rule

$$T = t / N$$

$$T^2 = (4\pi^2 / g) L$$

$$g = 4\pi^2 / m \text{ for } T^2 \text{ versus } L$$

Free fall and axis choice

p.256

Rule

$$h = \tfrac{1}{2} g t^2;$$

$$m_{h \text{ vs } t^2} = g/2;$$

$$m_{t^2 \text{ vs } h} = 2/g$$

Static Hooke law

p.259

Rule

$$F = k \Delta L$$

$$F = mg$$

$$k = m_{F \text{ vs } \Delta L}$$

Dynamic spring and gravity

p.262

Rule

$$T^2 = 4\pi^2 m / k$$

$$T^2 = (4\pi^2 / g) \Delta L$$

Force table and equilibrium

p.265

Rule

$$\sum F_x = 0$$

$$\sum F_y = 0$$

$$F_i = m_i g$$

Newton cooling law

p.268

Rule

$$T - T_r = (T_0 - T_r)e^{-kt}$$

$$\ln[(T - T_r)/(T_0 - T_r)] = -kt$$

Boyle law

p.271

Rule

$$P_1 V_1 = P_2 V_2$$

$$P \propto 1/V$$

Linear thermal expansion

p.274

Rule

$$\Delta L = \alpha L_0 \Delta T$$

$$\alpha = m_{\Delta L \text{ vs } \Delta T}/L_0$$

Specific heat, latent heat and Joule equivalent

p.277

Rule

$$m_s c_s (T_s - T_f) = (m_w c_w + C_{\text{cal}})(T_f - T_w)$$

$$Q = mL$$

$$E = IVt$$

Calorimetry: determine a solid specific heat

p.280

Rule

$$c_s = \frac{(m_w c_w + C_{\text{cal}})(T_f - T_w)}{m_s (T_s - T_f)}$$

Latent heat and electrical-to-thermal conversion

p.283

Rule

$$Q_{\text{ice}} = mL_f + mc_w(T_f - 0^\circ\text{C})$$

$$J_{\text{eq}} = E_{\text{joules}}/Q_{\text{calories}}$$

Symbol glossary

Symbols are listed with their lesson context. A symbol may represent different quantities in different rules.

Earn marks from tables, graphs and uncertainty

m slope of the selected vertical variable against horizontal variable / vertical unit / horizontal unit

x_{measured} experimental estimate / same as reference

$x_{\text{reference}}$ accepted comparison value / same as measurement

Simple pendulum

L pivot-to-bob-centre length / m

t time for N complete oscillations / s

T period of one oscillation / s

g gravitational acceleration / m/s^2

Free fall and axis choice

h vertical fall distance / m

t fall time / s

g gravitational acceleration / m/s^2

m graph slope / depends on axes

Static Hooke law

F applied load magnitude / N

ΔL loaded minus unloaded spring length / m

k spring stiffness / N/m

m total suspended mass / kg

Dynamic spring and gravity

T oscillation period / s

m oscillating mass in the stated model / kg

k spring stiffness / N/m

ΔL static equilibrium extension / m

g gravitational acceleration / m/s^2

Force table and equilibrium

F_i tension force from hanging mass in the ideal setup / N

m_i total hanging mass for a string / kg

R resultant force magnitude / N

Newton cooling law

T object temperature / °C or K consistently

T_r room temperature / same unit as T

k effective cooling constant / inverse time

t elapsed time / s or min consistently with k

Boyle law

P absolute gas pressure / Pa or consistent pressure unit

V gas volume / m³ or consistent volume unit

Linear thermal expansion

α linear expansion coefficient / K^{-1}

L_0 initial rod length / m

ΔL length increase / m

ΔT temperature increase / K

Specific heat, latent heat and Joule equivalent

Q thermal energy transferred / J

L specific latent heat / J/kg

I heater current / A

V heater voltage / V

t heating duration / s

Calorimetry: determine a solid specific heat

c_s solid specific heat capacity / J/(kg K)

m_s, m_w solid and water masses / kg

C_{cal} calorimeter heat capacity / J/K

T_f final equilibrium temperature / °C or K

Latent heat and electrical-to-thermal conversion

m ice mass / kg

L_f specific latent heat of fusion / J/kg

c_w liquid water specific heat / J/(kg K)

J_{eq} joules per calorie conversion estimate / J/cal

Source and coverage notes

All distinct experiments in the folder: pendulum, free fall, static and dynamic springs, force table, cooling, Boyle law, expansion, specific heat, latent heat and Joule equivalent. Duplicate versions are consolidated.

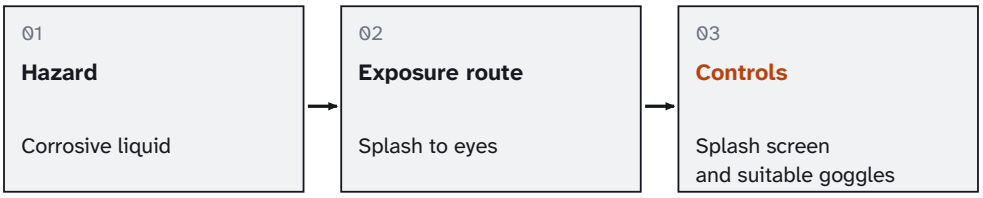
Open original course folder

The source audit inventories 159 non-programming files; 95 yielded substantial extracted text. Scanned diagrams and handwritten or image-only material were not fully transcribed. Check lecturer-specific derivations and conventions. The practice tests are original study material, not predictions of an examination.

Instruments and Lab Safety

FIG. 06 / SAFETY

Connect the hazard, exposure route and control



Match controls to the assessed task and the approved procedure.

Takeaway: eye protection adds a layer; first eliminate or reduce the hazard.

SCI 101 lecture chapters 1-3: accident prevention, chemical hazards, laboratory techniques, emergency principles and biosafety. This is exam study material; real laboratory work follows the current instructor procedure and chemical-specific safety information.

Subject map

SCI 101 lecture chapters 1-3: accident prevention, chemical hazards, laboratory techniques, emergency principles and biosafety. This is exam study material; real laboratory work follows the current instructor procedure and chemical-specific safety information.

Follow this reading route from foundations to applications. Each line links to its lesson; revisit the stated prerequisites whenever a step feels unfamiliar.

01 Hazard, risk and prevention p.309

02 Personal protective equipment and contamination p.312

03 Exposure routes and toxic effects p.315

04 Labels, SDS and chemical properties p.318

05 Glassware, measurement and safe apparatus p.321

06 Hoods, heating, cooling and pressure p.324

07 Waste, housekeeping and emergency priorities p.327

08 Biosafety and the control hierarchy p.330

Hazard, risk and prevention

LS / 01

- Distinguish a hazard from the risk of harm in a particular situation.
- Choose a control that interrupts the actual exposure route.

You need: Hazard means potential for harm; exposure means contact with the harmful agent.

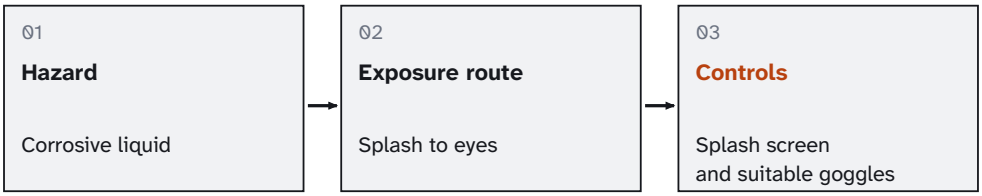
01 Idea

A hazard is a source of possible harm. Risk depends on exposure and the chance and severity of harm. Accident prevention is shared: prepare, follow approved procedures and report unsafe conditions. A familiar chemical or routine experiment can still present hazards.

A substance hazard is its potential to cause harm; the practical risk depends on how it is used and who can be exposed. Identifying the chemical is therefore only the start. A useful assessment links the harmful property to an activity, an exposure route and a suitable control.

FIG. 06 / SAFETY

Connect the hazard, exposure route and control



Match controls to the assessed task and the approved procedure.

Takeaway: eye protection adds a layer; first eliminate or reduce the hazard.

Figure 06. A useful safety answer connects a specific hazard to a route of exposure and suitable controls. A splash screen and appropriate goggles can limit splash exposure; they do not replace reducing or removing the hazard where possible.

- The same hazard can create different risks under different exposure conditions.
- A control is relevant only if it reduces the source, exposure likelihood or harmful consequence.

02 Rule and symbols

Rule

Good answer: hazard → exposure route → control

03 Method

1. Name the harmful property and the activity creating exposure.
2. Identify who could be exposed, by which route and under what conditions.
3. Select a control that reduces the source or interrupts the route.
4. Explain the remaining need for the approved procedure and required personal protection.

For an unfamiliar reagent, preparation means identifying hazards and the approved method before beginning work.

04 Worked example

A volatile harmful liquid on an open bench creates an inhalation hazard.

01 The example liquid is volatile and harmful, so vapour can leave an open container.

02 A person nearby may inhale that vapour; this connects the chemical hazard to an exposure route.

03 An approved fume-hood procedure controls vapour at its source when suitable for that substance and operation.

04 Goggles protect eyes from appropriate hazards but do not remove vapour from inhaled air.

Watch out

Do not write only be careful; name a concrete control.

Exam habit

For a scenario answer write a linked chain: hazard, exposure route, suitable control and reason.

05 Your turn

A student uses an unfamiliar reagent without reading its safety information. Identify the problem and two appropriate preparatory actions.

If you get stuck, uncover one hint at a time:

1. Identify what the student has not established before starting: hazards and suitable controls.
2. Choose one preparatory action that obtains reagent-specific safety information.
3. Choose another that confirms the approved procedure or required supervision. Explain how each prevents an uninformed exposure.

06 Check your understanding

Why is naming only a hazardous chemical insufficient to describe a risk scenario?

The same harmful volatile liquid is sealed in approved storage or poured openly during a task. Why can risk differ?

1. The opportunity for exposure differs
2. Its chemical identity necessarily changes
3. A sealed container makes the liquid chemically harmless

Answers and explanations: page 334.

Personal protective equipment and contamination

LS / 02

- Trace how contaminated gloves transfer material to clean surfaces.
- Select PPE for the task and remove it without spreading contamination.

You need: Contamination transfer; clean and contaminated work zones; task-specific protection.

01 Idea

Splash goggles protect against chemical splashes; ordinary spectacles do not provide equivalent coverage. Gloves must be compatible with the chemical and inspected. No glove protects indefinitely. Contaminated gloves can carry chemicals to phones, door handles and notebooks.

PPE is selected for the route and type of exposure, not simply worn as a general safety symbol. Ordinary prescription glasses leave openings around the eyes, while a coat mainly protects covered skin and clothing. Protection for a splash task must match the actual eye and body hazards specified by the method.

- A barrier cannot protect an area it does not cover adequately.
- Different materials and tasks can require different PPE, so everyday glasses are not automatically splash-rated eye protection.

02 Rule and symbols

Rule
Select PPE for the specific hazard and task

03 Method

1. Identify the contaminant and which surfaces or body areas need protection.
2. Use PPE specified for the material and task rather than assuming one glove type protects against everything.
3. Keep contaminated gloves away from phones, door handles and other clean shared items.

4. Remove and dispose of PPE according to the approved procedure, then perform required hand hygiene.

When asked whether clothing is sufficient, identify the exposed area and required protection for the particular task.

04 Worked example

A student touches a phone with contaminated gloves.

01 The student glove is already contaminated before touching the phone.

02 Contact transfers material from glove to phone, creating a new contaminated surface.

03 Later bare-hand contact can move that material to skin, food or the face.

04 Removing gloves correctly before touching clean items breaks that transfer chain; the phone may still need the approved decontamination procedure.

Watch out

More layers of unsuitable gloves do not establish chemical compatibility.

Exam habit

Explain what is transferred and to where; simply saying wear gloves does not solve a contamination scenario.

05 Your turn

Explain why ordinary prescription glasses and a lab coat alone are insufficient for a splash-risk task.

If you get stuck, uncover one hint at a time:

1. List which areas a lab coat covers and which areas remain exposed.
2. Consider whether ordinary prescription glasses provide the task-required splash protection around the eyes.

3. Explain the coverage gaps and identify the need for approved splash-rated protection and any other task-specific PPE, following the laboratory method.

06 Check your understanding

Do gloves make it acceptable to touch your face during laboratory work?

Why should glove material be selected for the chemical and task?

1. Thicker-looking gloves always remove all exposure routes
2. Different glove materials resist different chemicals differently
3. Every glove blocks every chemical equally

Answers and explanations: page 334.

Exposure routes and toxic effects

LS / 03

- Identify inhalation, ingestion, skin or eye contact and injection exposures.
- Distinguish route of exposure from the timing of a toxic effect.

You need: Contact with an agent can occur by several routes; visible vapour is not required.

01 Idea

Chemicals can enter by inhalation, ingestion, skin or eye absorption, or injection through contaminated sharps. Acute exposure is short-term; chronic exposure is repeated or long-term. Effects can be local or systemic, immediate or delayed, reversible or irreversible.

Exposure route describes how material reaches the body, not whether the person notices it. Vapour can be inhaled, while a contaminated sharp can introduce material through skin. Odour is a sensory property with limits, so absence of smell cannot establish absence of a hazardous exposure.

- Different routes require different preventive controls.
- Some hazards have little warning odour, and smell may not reliably indicate a safe concentration.

02 Rule and symbols

Rule
Dose, duration, route and substance all influence harm

03 Method

1. Read the activity and locate the point where the agent could enter or contact the body.
2. Name the route using the specific evidence in the scenario.
3. Distinguish a rapid acute effect from an effect associated with longer or repeated exposure.
4. Match prevention to the identified route and follow the material-specific procedure.

Classify the physical entry mechanism first, then explain why sensory impressions cannot replace safety information.

04 Worked example

Eating in a laboratory can lead to ingestion of contamination.

01 Eating in the laboratory can move contamination from hands or surfaces to food.

02 Swallowing that material is ingestion, even when the contaminant is not visible.

03 A contaminated needle can breach the skin and introduce material directly: this is injection exposure.

04 The absence of visible vapour does not remove either risk because neither route depends on inhalation.

Watch out

A lack of smell does not prove safety.

Exam habit

Name both the route and the evidence, for example ingestion because contaminated food is swallowed.

05 Your turn

Classify inhaling vapour and a contaminated needle puncture by exposure route. Explain why no smell does not prove no hazard.

If you get stuck, uncover one hint at a time:

1. For each event, identify how the material crosses into or contacts the body.
2. Airborne vapour enters through breathing; a contaminated puncture breaches skin directly. Name the route for each.
3. For the smell explanation, state that detectability and hazardous exposure are

different questions; refer to substance-specific safety information.

06 Check your understanding

A droplet splashes into an eye. Is the route inhalation?

A hazardous substance has no noticeable odour during use. What can be concluded?

1. There is no inhalation hazard
2. PPE and handling controls are unnecessary
3. Odour alone cannot establish safety

Answers and explanations: page 334.

Labels, SDS and chemical properties

LS / 04

- Use a label and SDS to connect chemical properties with controls.
- Distinguish corrosive and oxidizing hazards in an explanation.

You need: Chemical identity; label symbols; an SDS gives substance-specific hazard and handling information.

01 Idea

Read the identity and hazard information before use. Safety data sheets describe hazards, handling, storage, exposure controls, first aid and other substance-specific information. Flammable, oxidizing, corrosive and toxic refer to different hazards and can apply to one substance simultaneously.

One chemical can have several hazards that require separate reasoning. Flammability concerns ignition and combustion, whereas corrosivity concerns damaging contact. An answer that names one generic precaution may miss the second hazard, even when it sounds broadly cautious.

- An ignition control addresses a fire pathway but does not by itself prevent corrosive contact.
- Splash/contact controls address exposed tissues but do not make a flammable material nonflammable.

02 Rule and symbols

Rule

$$\text{Identity} + \text{hazard} + \text{controls} + \text{incompatibilities}$$

03 Method

1. Confirm the chemical identity from its label before selecting a procedure.
2. Read relevant SDS information for hazards, incompatibilities and required handling controls.
3. Explain how the named property causes harm in the scenario.

4. Use the approved storage or handling arrangement rather than inferring compatibility from appearance.

When a label carries more than one hazard, give a distinct risk-and-control link for each.

04 Worked example

A corrosive liquid can damage tissue, while an oxidizer may intensify fire even when it is not itself a fuel.

01 A corrosive liquid can damage living tissue or certain materials by contact.

02 An oxidizer can support or intensify combustion even when it is not itself the fuel.

03 The mechanisms differ, so the same general statement handle carefully does not explain either control.

04 A good answer links splash/contact controls to corrosivity and incompatibility or fire-prevention controls to oxidizing behaviour as specified by the SDS.

Watch out

Do not infer that all colourless liquids are harmless or that one symbol captures every hazard.

Exam habit

Pair each property with its consequence; listing pictogram names alone gives less evidence of understanding.

05 Your turn

A bottle has both flammable and corrosive hazards. State two distinct risks and a control principle for each.

If you get stuck, uncover one hint at a time:

1. Separate the two label properties before suggesting controls.

2. For flammability, explain an ignition/fire risk and a suitable approved ignition-control principle.
3. For corrosivity, explain tissue-damage risk and an approved exposure-prevention principle. Do not treat one control as covering both mechanisms.

06 Check your understanding

Does a chemical being nonflammable prove it cannot intensify a fire?

A bottle is both flammable and corrosive. Which statement is best?

1. Both fire and damaging-contact risks need suitable controls
2. Removing ignition sources alone resolves every hazard
3. Eye protection alone makes ignition impossible

Answers and explanations: page 334.

Glassware, measurement and safe apparatus

LS / 05

- Select apparatus according to measurement purpose and safe operating limits.
- Explain why volumetric preparation and heating require different vessels.

You need: Volume; calibration mark; damaged glassware can fail during use.

01 Idea

Inspect glassware for chips and cracks and apparatus for damaged wiring before use. A beaker is useful for holding and mixing; a graduated cylinder gives more useful volume estimates; a volumetric pipette or flask is designed for a specified accurate volume. Read liquid levels at eye height using the appropriate meniscus.

Accurate liquid handling depends on both choosing the right instrument and using it as calibrated. A fixed-volume volumetric pipette is intended to deliver one specified volume under its stated conditions. Safe pipetting and proper reading matter as much as the printed volume.

- A vessel marked for approximate volume may not deliver the required fixed volume accurately.
- The calibration method determines how the instrument should be filled, read and drained.

02 Rule and symbols

Rule	Choose the instrument for required precision, not convenience

03 Method

1. State whether the task requires approximate volume, accurate delivery or preparation to a final volume.
2. Choose the apparatus designed for that purpose and check its condition.
3. Follow the specified filling, meniscus and operating procedure.
4. Before heating, verify that the vessel and setup are approved for the process.

For an accurate fixed transfer, distinguish a volumetric pipette from a flask that contains a final prepared volume.

04 Worked example

A volumetric flask prepares a solution to its marked final volume.

01 The volumetric flask is calibrated to contain a stated final volume at its reference conditions.

02 The solution is brought to the mark as part of an approved preparation procedure.

03 Its main role is volume accuracy, not serving as a directly heated reaction vessel.

04 For heating a reaction, use the vessel and support specified by the laboratory method rather than substituting the flask.

Watch out

Extra decimal places do not make an imprecise instrument accurate.

Exam habit

In apparatus questions state the purpose and limitation together: prepares final volume, unsuitable for directly heating the reaction.

05 Your turn

Name a suitable device to transfer an accurate fixed 10 mL volume and explain two good practices.

If you get stuck, uncover one hint at a time:

1. The task asks to transfer a fixed accurate volume, not prepare a solution to a final flask mark.
2. Select an appropriate fixed-volume volumetric pipette and approved pipette aid.

3. For good practices, explain correct meniscus reading and use according to its calibration/draining instructions. Mouth pipetting is not acceptable.

06 Check your understanding

Does having a calibration mark mean a flask is suitable for direct heating?

A volumetric flask and a volumetric pipette both show 10 mL. Why are they not automatically interchangeable?

1. The same printed number guarantees identical delivery
2. They are designed for different volume-handling purposes
3. Only a flask ever has calibration

Answers and explanations: page 334.

Hoods, heating, cooling and pressure

LS / 06

- Explain why heating a closed unsuitable vessel can create a pressure hazard.
- Recognize when an apparatus setup requires instructor review.

You need: Heating can increase vapour production and gas pressure; equipment has specific operating limits.

01 Idea

A fume hood controls airborne chemical exposure when used correctly. It is not a storage cabinet or a waste-evaporation device. Heating and cooling systems can introduce burns, pressure changes and material failure. Approved setups account for both temperature and pressure.

A heated closed vessel can store pressure as well as thermal energy. A volatile liquid supplies vapour, and heating can increase that vapour pressure. If the vessel or closure is unsuitable, failure can release hot hazardous contents suddenly; a hood does not turn it into pressure-rated equipment.

- Heating an enclosed volatile liquid can raise internal pressure.
- Container failure can eject hot liquid or vapour, adding exposure and burn consequences.

02 Rule and symbols

Rule	Control vapour, ignition, pressure and thermal contact separately

03 Method

1. Identify heat source, volatile materials and any sealed volume.
2. Check the approved procedure and apparatus pressure and temperature limits.
3. Use the specified containment, heating and ventilation arrangements.
4. If the setup differs or becomes unsafe, stop and alert the responsible instructor instead of modifying it ad hoc.

A tightly sealed volatile-liquid heating setup requires checking the approved apparatus rather than improvising a modification.

04 Worked example

A volatile solvent warmed in an unsuitable sealed container can build pressure.

01 Warming the volatile solvent increases vapour formation.

02 In an unsuitable sealed container, expanding gas and vapour can increase internal pressure.

03 The container or closure may fail because it was not designed for that operation.

04 The example therefore calls for stopping and consulting the approved setup; improvising a vent is not a validated correction.

Watch out

A hood does not remove every fire, pressure or contact hazard.

Exam habit

Explain the mechanism, pressure buildup from heating, before giving the required stop-and-consult response.

05 Your turn

Give two reasons why heating a tightly sealed vessel containing a volatile liquid is hazardous.

If you get stuck, uncover one hint at a time:

1. Trace how heating affects vapour inside a tightly sealed space.
2. Connect the resulting pressure to possible vessel or closure failure.
3. State a second consequence involving hot released contents, such as burns or hazardous exposure, and relate it to the need for the approved setup.

06 Check your understanding

Does placing an unsuitable sealed heated vessel in a fume hood solve its pressure hazard?

Does placing an unsuitable sealed heated vessel in a fume hood make the pressure setup acceptable?

1. Yes; all hoods prevent pressure buildup
2. Yes; a hood removes liquid volatility
3. No; ventilation does not make a vessel pressure-rated

Answers and explanations: page 334.

Waste, housekeeping and emergency priorities

LS / 07

- Use identification and approved procedures to choose a waste or spill response.
- State safe priorities when a spill material is unknown.

You need: Chemical incompatibility; designated waste streams; laboratory emergency instructions.

01 Idea

Waste must follow the laboratory segregation and disposal procedure. Broken glass, sharps and chemical waste are different streams. Good housekeeping keeps exits and emergency equipment accessible. In an incident, alert others and follow the established response; do not improvise a hazardous cleanup.

Emergency equipment is useful only when people can reach and operate it promptly. A storage box can create risk without being chemically hazardous itself because it delays the response to another event. Housekeeping therefore protects the reliability of the laboratory safety system.

- A blocked eyewash can delay response to an eye exposure.
- Keeping the access route clear reduces that delay before an emergency occurs.

02 Rule and symbols

Rule

Recognize → alert → follow trained response

03 Method

1. Identify the material only if that can be done safely from available records or labels.
2. Keep others clear and alert the instructor or responsible responder.
3. Use the laboratory response plan for the identified hazard and your training level.

4. For routine waste, use the designated compatible container and labelling procedure.

An obstructed emergency station is a prevention and access problem, even when no spill or exposure is currently happening.

04 Worked example

An unknown spill is not an invitation to mix in a neutralizer.

01 An unknown spill has uncertain properties, so an assumed neutralizer may be incompatible.

02 Adding another chemical could cause heat, gas formation or another hazardous reaction.

03 Alerting the instructor and keeping people clear limits exposure while the material and response are assessed.

04 Cleanup then follows the approved spill procedure for the identified material and trained personnel.

Watch out

Do not recommend that an untrained student fight every fire or clean every spill.

Exam habit

In an emergency scenario prioritize alerting and exposure prevention; do not invent a chemical cleanup recipe.

05 Your turn

A box blocks access to the eyewash station. Explain the risk and an appropriate preventive action.

If you get stuck, uncover one hint at a time:

1. Identify the purpose of an eyewash and when rapid access matters.
2. Explain how a box could delay a person trying to respond to an eye exposure.

3. Propose promptly clearing and maintaining the route through the responsible housekeeping procedure. Link the action to access rather than claiming the box itself is corrosive.

06 Check your understanding

Should an unknown laboratory liquid automatically be washed into a sink?

Why can a harmless empty box near emergency equipment still create risk?

1. It can delay access when the equipment is needed
2. It must release toxic vapour
3. Emergency equipment works only when boxes are present

Answers and explanations: page 334.

Biosafety and the control hierarchy

LS / 08

- Classify controls within the hazard-control hierarchy.
- Distinguish engineering containment from training, work practices and PPE.

You need: A biological hazard may be spread through aerosols, contact or sharps depending on the activity.

01 Idea

Biosafety aims to prevent unintended exposure to biological agents and their release. The general control hierarchy prefers removing a hazard, then using a safer substitute, then engineering controls, administrative controls and work practices, and PPE. The suitable controls depend on the task: removing or replacing the biological material may not be possible. Training and procedures are administrative controls; suitable containment equipment is an engineering control. Safe work practices include hand hygiene and avoiding aerosol generation.

Controls work at different places in the hazard pathway. Equipment containment can isolate the source, procedures guide behaviour, and PPE forms a barrier worn by a person. Classifying a control means explaining its mechanism; it does not mean one category makes every other category unnecessary.

- Engineering controls act through equipment or physical containment.
- Training and handwashing procedures guide work practices, while goggles are worn personal protection.

02 Rule and symbols

Rule

Eliminate → substitute → engineering → administrative/work practices → PPE

03 Method

1. Identify the biological material and activity-specific exposure route.
2. Consider elimination or substitution where feasible before downstream controls.

3. Classify equipment containment as engineering, training and procedures as administrative or work-practice controls, and worn barriers as PPE.
4. Explain how the selected combination addresses the task and follows institutional biosafety procedures.

When asked to classify controls, describe what each does before assigning its hierarchy category.

04 Worked example

Training is administrative.

01 Training changes how people work, so it is an administrative control.

02 A suitable biological safety cabinet provides engineering containment for its approved uses.

03 Gloves form a worn barrier and are PPE.

04 A safe handling technique is a work practice, generally grouped with administrative controls; the categories describe different protective mechanisms.

Watch out

A device with airflow is not automatically suitable for every chemical or biological hazard.

Exam habit

Do not label every cabinet or hood as interchangeable containment; state suitable equipment for the defined biological task.

05 Your turn

Classify goggles, safety training, a suitable containment cabinet and correct handwashing.

If you get stuck, uncover one hint at a time:

1. Sort the examples into equipment, behaviour/procedure and worn protection.

2. Classify the suitable cabinet as engineering containment and goggles as PPE.
3. Place training under administrative controls and correct handwashing under work practices, commonly grouped with administrative controls. Explain the basis rather than just listing labels.

06 Check your understanding

What is the general hierarchy from source-focused to last-line protection?

A suitable containment cabinet and goggles are used together. Why can both be needed?

1. PPE always eliminates the source hazard
2. They interrupt exposure through different protective mechanisms
3. They belong to exactly the same control category

Answers and explanations: page 334.

The control hierarchy in the correct order

The hierarchy ranks ways to control a hazard. A list of control types is not automatically in the preferred order.

- Eliminate: remove the hazard.
- Substitute: use a safer alternative after checking its risks.
- Engineering controls: separate people from the hazard using suitable equipment or containment.
- Administrative controls and work practices: use training, procedures and safe methods.
- PPE: use suitable worn protection. It adds protection but depends on correct selection and use.

- In a laboratory, eliminating or replacing the material may not be possible. The assessed task determines the suitable combination of controls.

Classify a suitable containment cabinet, safety training, gloves and correct handwashing.

1. The cabinet is an engineering control.
2. Training is administrative; gloves are PPE.
3. Handwashing is a safe work practice, often grouped with administrative controls.

Engineering; administrative; PPE; work practice, respectively. Engineering controls come before administrative controls and PPE in the general hierarchy.

Lesson review and answers

Hazard, risk and prevention

Return to lesson: page 309.

Your turn: Risk also depends on how exposure can occur, how likely it is and how serious its consequences could be.

Concept check: 1. Correct. Handling conditions can change exposure while the underlying hazardous property remains.

Written method: Problem: unassessed chemical hazards. Actions: consult the reagent SDS and approved procedure; clarify required controls and waste handling with the instructor before use. One mark for hazard, one for exposure reasoning and one per specific action.

Personal protective equipment and contamination

Return to lesson: page 312.

Your turn: No. Gloves can carry contamination directly to the face.

Concept check: 2. Correct. Compatibility and performance depend on material, chemical and conditions.

Written method: Prescription glasses do not seal against splashes; appropriate splash goggles are needed. A coat does not protect hands, so compatible gloves may be required by the assessed procedure. Award one mark per protection gap and suitable control.

Exposure routes and toxic effects

Return to lesson: page 315.

Your turn: No. It is eye or mucous-membrane contact.

Concept check: 3. Correct. Safety depends on identified properties and exposure controls, not smell alone.

Written method: Vapour: inhalation. Needle: injection. Some harmful substances are odourless, or odour detection is unreliable. One mark per route and two for the explanation.

Labels, SDS and chemical properties

Return to lesson: page 318.

Your turn: No. An oxidizer may intensify combustion without being a fuel.

Concept check: 1. Correct. The properties create separate pathways requiring an appropriate combination of controls.

Written method: Fire risk: control ignition sources using the approved setup. Tissue damage from splashes: prevent contact using appropriate containment and compatible protection. One mark per risk and per linked control.

Glassware, measurement and safe apparatus

Return to lesson: page 321.

Your turn: No. Calibration and suitability for heating are different design requirements.

Concept check: 2. Correct. A flask contains a prepared volume; a pipette is designed to transfer a calibrated volume according to its type.

Written method: Use an appropriate 10 mL volumetric pipette with a pipette aid. Read the meniscus at eye level and follow the calibrated delivery procedure. Credit device, aid and two measurement practices.

Hoods, heating, cooling and pressure

Return to lesson: page 324.

Your turn: No. A hood does not make a vessel pressure-rated or validate an unsafe heating setup.

Concept check: 3. Correct. Engineering ventilation and pressure suitability are different requirements.

Written method: Vapour generation can raise pressure, and thermal expansion can further stress the vessel. Rupture can release hot liquid or vapour. Credit two mechanisms and their resulting consequences.

Waste, housekeeping and emergency priorities

Return to lesson: page 327.

Your turn: No. Its identity, compatibility and approved disposal route must be established.

Concept check: 1. Correct. Obstruction creates a response-time hazard even if the box material is harmless.

Written method: It can delay emergency irrigation after exposure. Keep access clear and report the obstruction so it is corrected promptly under laboratory procedures. Credit emergency delay, possible worsening of injury, removal/reporting and prevention.

Biosafety and the control hierarchy

Return to lesson: page 330.

Your turn: Elimination, substitution, engineering controls, administrative controls and work practices, then PPE.

Concept check: 2. Correct. Containment and worn barriers can form complementary protections specified for the task.

Written method: Goggles: PPE. Training: administrative control. Suitable containment cabinet: engineering control. Correct handwashing: safe work practice (often grouped with administrative controls). One mark each.

Instruments and Lab Safety / Test A

20 marks / Suggested time: 20 minutes. MCQs: 1 mark each. Written problems: 4 marks each. Attempt without notes.

Part I / Choose one answer

1. Which best describes risk?
   A. The mass of any bottle
   B. The same as smell
   C. The label colour alone
   D. The chance and severity of harm under the exposure conditions
2. Which is appropriate before touching a clean door handle?
   A. Keep contaminated gloves on
   B. Wipe gloves on clothing
   C. Remove contaminated gloves correctly
   D. Smell the gloves
3. Eating with contaminated hands is mainly which exposure route?
   A. Inhalation
   B. Injection
   C. Noise exposure
   D. Ingestion
4. An oxidizer is especially associated with:
   A. Intensifying combustion
   B. Preventing all fires
   C. Measuring temperature
   D. Making all chemicals inert

Part II / Show your reasoning

1. A student uses an unfamiliar reagent without reading its safety information. Identify the problem and two appropriate preparatory actions.

2. Explain why ordinary prescription glasses and a lab coat alone are insufficient for a splash-risk task.

3. Classify inhaling vapour and a contaminated needle puncture by exposure route. Explain why no smell does not prove no hazard.

4. A bottle has both flammable and corrosive hazards. State two distinct risks and a control principle for each.

Instruments and Lab Safety / Test B

20 marks / Suggested time: 20 minutes. MCQs: 1 mark each. Written problems: 4 marks each. Attempt without notes.

Part I / Choose one answer

1. Which is designed to prepare one accurately specified solution volume?

   A. A beaker

   B. A watch glass

   C. A volumetric flask

   D. A test-tube rack

2. A fume hood should be used as:

   A. Permanent chemical storage

   B. A controlled work enclosure for appropriate vapour hazards

   C. A way to evaporate all waste

   D. A food cupboard

3. An unknown chemical spill should first prompt a student to:

   A. Alert the instructor and keep people clear

   B. Mix it with another chemical

   C. Ignore it

   D. Taste it

4. Which is an administrative control?

   A. Gloves

   B. Training and written procedures

   C. A sealed rotor

   D. Splash goggles

Part II / Show your reasoning

1. Name a suitable device to transfer an accurate fixed 10 mL volume and explain two good practices.

2. Give two reasons why heating a tightly sealed vessel containing a volatile liquid is hazardous.

3. A box blocks access to the eyewash station. Explain the risk and an appropriate preventive action.

4. Classify goggles, safety training, a suitable containment cabinet and correct handwashing.

Instruments and Lab Safety / Test A / Answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1

MCQ: D. Risk depends on both the hazard and how people are exposed.

Written solution: Problem: unassessed chemical hazards. Actions: consult the reagent SDS and approved procedure; clarify required controls and waste handling with the instructor before use. One mark for hazard, one for exposure reasoning and one per specific action.

Return to lesson: page 309.

Question 2

MCQ: C. This prevents spreading contamination outside the work area.

Written solution: Prescription glasses do not seal against splashes; appropriate splash goggles are needed. A coat does not protect hands, so compatible gloves may be required by the assessed procedure. Award one mark per protection gap and suitable control.

Return to lesson: page 312.

Question 3

MCQ: D. Material can pass from the hands to food and the digestive system.

Written solution: Vapour: inhalation. Needle: injection. Some harmful substances are odourless, or odour detection is unreliable. One mark per route and two for the explanation.

Return to lesson: page 315.

Question 4

MCQ: A. Oxidizers can support or intensify combustion.

Written solution: Fire risk: control ignition sources using the approved setup. Tissue damage from splashes: prevent contact using appropriate containment and compatible protection. One mark per risk and per linked control.

Return to lesson: page 318.

Instruments and Lab Safety / Test B / Answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1

MCQ: C. The calibration mark defines the specified contained volume under its calibration conditions.

Written solution: Use an appropriate 10 mL volumetric pipette with a pipette aid. Read the meniscus at eye level and follow the calibrated delivery procedure. Credit device, aid and two measurement practices.

Return to lesson: page 321.

Question 2

MCQ: B. It is an engineering control for relevant airborne exposures.

Written solution: Vapour generation can raise pressure, and thermal expansion can further stress the vessel. Rupture can release hot liquid or vapour. Credit two mechanisms and their resulting consequences.

Return to lesson: page 324.

Question 3

MCQ: A. Unknown hazards require assessment and the established response.

Written solution: It can delay emergency irrigation after exposure. Keep access clear and report the obstruction so it is corrected promptly under laboratory procedures. Credit emergency delay, possible worsening of injury, removal/reporting and prevention.

Return to lesson: page 327.

Question 4

MCQ: B. Administrative controls organize how work is authorized and performed.

Written solution: Goggles: PPE. Training: administrative control. Suitable containment cabinet: engineering control. Correct handwashing: safe work practice (often grouped with administrative controls). One mark each.

Return to lesson: page 330.

Formula sheet

Hazard, risk and prevention p.309

Rule

Good answer: hazard → exposure route → control

Personal protective equipment and contamination p.312

Rule

Select PPE for the specific hazard and task

Exposure routes and toxic effects p.315

Rule

Dose, duration, route and substance all influence harm

Labels, SDS and chemical properties p.318

Rule

Identity + hazard + controls + incompatibilities

Glassware, measurement and safe apparatus p.321

Rule

Choose the instrument for required precision, not convenience

Hoods, heating, cooling and pressure p.324

Rule

Control vapour, ignition, pressure and thermal contact separately

Waste, housekeeping and emergency priorities p.327

Rule

Recognize → alert → follow trained response

Biosafety and the control hierarchy p.330

Rule

Eliminate → substitute → engineering → administrative/work practices → PPE

Symbol glossary

Symbols are listed with their lesson context. A symbol may represent different quantities in different rules.

Source and coverage notes

SCI 101 lecture chapters 1-3: accident prevention, chemical hazards, laboratory techniques, emergency principles and biosafety. This is exam study material; real laboratory work follows the current instructor procedure and chemical-specific safety information.

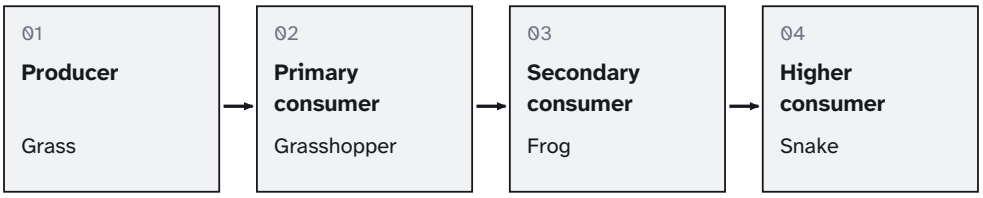
Open original course folder

The source audit inventories 159 non-programming files; 95 yielded substantial extracted text. Scanned diagrams and handwritten or image-only material were not fully transcribed. Check lecturer-specific derivations and conventions. The practice tests are original study material, not predictions of an examination.

Humans and the Environment

FIG. 07 / ENVIRONMENT

Food-chain arrows point from food to consumer



Energy transfer Some energy disperses as heat at every stage.

Takeaway: nutrients cycle; energy flows through the feeding relationships.

ENV101 elective: ecosystems, relationships among organisms, pollution, ecological footprint, waste management and sustainable development. The Arabic source lectures are explained here in English.

Subject map

ENV101 elective: ecosystems, relationships among organisms, pollution, ecological footprint, waste management and sustainable development. The Arabic source lectures are explained here in English.

Follow this reading route from foundations to applications. Each line links to its lesson; revisit the stated prerequisites whenever a step feels unfamiliar.

01 Environment and ecosystem components p.352

02 Energy flow, food chains and decomposers p.355

03 Population growth and limiting factors p.358

04 Relationships among organisms p.361

05 Pollution: sources, pathways and effects p.364

06 Ecological footprint and biocapacity p.367

07 Waste management and the hierarchy p.370

08 Sustainable development and integrated decisions p.373

Environment and ecosystem components

EN / 01

- Classify living and nonliving ecosystem components.
- Explain one interaction linking an organism with an abiotic factor.

You need: An ecosystem includes organisms and the physical conditions around them.

01 Idea

An ecosystem combines organisms with the physical and chemical surroundings with which they interact. Biotic factors are living components; abiotic factors include light, temperature, water and minerals. A population is one species in an area; a community includes interacting populations.

An ecosystem includes a living community and the conditions through which it interacts. Sunlight and minerals are not alive, but they affect what organisms can do. An abiotic factor is therefore not an unimportant factor; it is simply a nonliving part of the system.

- Living status determines the basic biotic/abiotic classification.
- Nonliving conditions can still strongly affect organism growth, survival and interactions.

02 Rule and symbols

Rule

Organism → population → community → ecosystem

03 Method

1. Name the ecosystem and define the area being considered.
2. Sort its organisms as biotic and its physical or chemical conditions as abiotic.
3. Choose one organism and explain how a named abiotic factor affects it.
4. Use a causal sentence rather than an unconnected list of examples.

For classification, ask whether the item is an organism or a physical/chemical environmental factor.

04 Worked example

Fish, algae and bacteria are biotic components of a pond.

01 Fish, algae and bacteria are organisms, so they are biotic pond components.

02 Water temperature and dissolved oxygen are abiotic conditions even though living things can affect them.

03 Fish depend on sufficient dissolved oxygen for respiration.

04 The example therefore shows both a classification and an interaction between the living community and its physical environment.

Watch out

A habitat is where an organism lives; an ecosystem includes the interacting system.

Exam habit

For explain questions add an interaction after classifying: name the factor, the organism and the effect.

05 Your turn

Classify sunlight, grass, soil mineral content and a rabbit as biotic or abiotic.

If you get stuck, uncover one hint at a time:

1. Use living organism versus nonliving environmental factor as the test, not whether the item is useful.
2. Consider sunlight and mineral content separately from grass and rabbit.
3. Assign each of the four items a category and give a short reason for any item you initially found ambiguous.

06 Check your understanding

Are bacteria abiotic because they are microscopic?

Soil mineral content affects plant growth. Why is it still abiotic?

1. It is a nonliving chemical condition
2. Anything affecting plants is biotic
3. All material found in soil is living

Answers and explanations: page 376.

Energy flow, food chains and decomposers

EN / 02

- Assign producer and consumer roles within a stated food chain.
- Distinguish energy flow from nutrient cycling.

You need: Producers capture energy; consumers obtain energy by feeding; decomposers break down organic material.

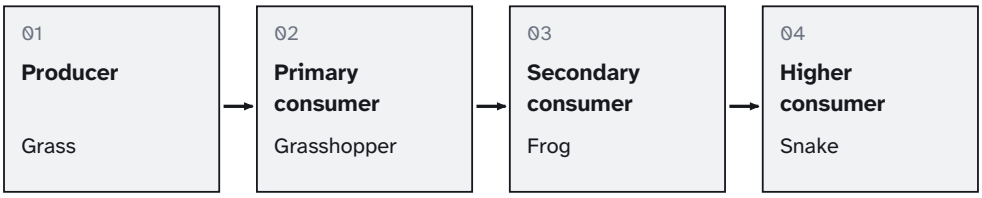
01 Idea

Producers use light or chemical energy to make organic material from inorganic substances. Consumers obtain organic material by feeding. Decomposers break down dead material and help recycle nutrients. Energy passes through feeding levels and much becomes dispersed heat. Nutrients cycle, but usable energy is not recycled in the same way.

Food-chain energy is a flow through successive roles. Only some energy at one trophic level becomes available to the next, because organisms use energy and not all material is eaten or assimilated. A specified transfer percentage is a model instruction for the calculation, not a universal ecological constant.

FIG. 07 / ENVIRONMENT

Food-chain arrows point from food to consumer



Energy transfer Some energy disperses as heat at every stage.

Takeaway: nutrients cycle; energy flows through the feeding relationships.

Figure 07. The arrow direction is food to consumer: grass to grasshopper to frog to snake. This is one simplified chain within a food web. Usable energy decreases along the chain; nutrients can be recycled.

- A percentage must be applied to the energy available at the immediately preceding level.
- Repeating a fractional transfer reduces energy successively rather than subtracting the same original amount each time.

02 Rule and symbols

Rule

Producer → primary consumer → secondary consumer

03 Method

1. Start the chain with its producer and draw arrows in the direction of food-energy transfer.
2. Assign each consumer level from what that organism eats in the stated chain.
3. Explain that energy is used and dissipated at each transfer.
4. Describe decomposers returning nutrients to the environment without claiming energy is recycled.

If a question explicitly gives transfer efficiency, apply it once for every trophic-level step.

04 Worked example

In grass to grasshopper to frog, grass is the producer, grasshopper the primary consumer and frog the secondary consumer.

01 Grass captures light energy and makes organic material, so it is the producer.

02 The grasshopper eats the producer and is the primary consumer.

03 The frog eats that consumer and is the secondary consumer.

04 The arrows grass → grasshopper → frog show transfer toward the eater; dead material from all levels can enter decomposer pathways.

Watch out

The 10 percent rule is a rough teaching approximation, not a universal exact constant.

Exam habit

Consumer levels are roles in the specified chain; do not assign a permanent level without looking at the feeding relationship.

05 Your turn

A question specifies 10 percent transfer efficiency at each step. Producers have 10000 energy units. Find energy available to primary and secondary consumers.

If you get stuck, uncover one hint at a time:

1. Identify the two steps: producers to primary consumers, then primary to secondary.
2. For the first transfer apply the stated 10 percent to 10000 energy units.
3. For the second transfer apply 10 percent to the first result, not to the original producer amount. Label the two consumer-level energies.

06 Check your understanding

Does energy cycle back through the ecosystem in the same way as mineral nutrients?

A question assumes 10 percent transfer at every step. What fraction of producer energy reaches secondary consumers?

1. 20 percent
2. 1 percent
3. 10 percent

Answers and explanations: page 376.

Population growth and limiting factors

EN / 03

- Calculate net population change from four demographic processes.
- Explain how a limiting factor can constrain population growth.

You need: Population means members of one species in a defined area and period; basic signed addition.

01 Idea

Population size changes through births, deaths, immigration and emigration. Rapid exponential growth can occur when resources are abundant. Limited food, space and other conditions constrain growth. Carrying capacity describes a population size an environment can sustain under specified conditions.

A population boundary makes gains and losses countable. Births and immigration add members inside the boundary; deaths and emigration remove them. Net change is the balance of those processes, while final population also includes the members already present at the start.

- Immigration and births have the same positive effect on local count despite different origins.
- A positive net change must be added to the initial population rather than reported as the final size.

02 Rule and symbols

Rule

$$\Delta N = \text{births} + \text{immigration} - \text{deaths} - \text{emigration}$$

03 Method

1. Set the population boundary and time interval.
2. Treat births and immigration as additions and deaths and emigration as removals.
3. Add the gains and subtract the losses to obtain net change.

4. If explaining growth limits, link a specific resource or condition to survival or reproduction.

A starting population plus four demographic counts calls for a change calculation followed by a final-size calculation.

04 Worked example

If births are 30, deaths 10, immigration 5 and emigration 8, the population increases by 17.

01 The example gives 30 births and 5 immigrants, so total additions are 35.

02 It gives 10 deaths and 8 emigrants, so total removals are 18.

03 Net change is 35 - 18 = +17 individuals.

04 This is the increase during the interval, not the final population; a starting population would be needed to calculate the latter.

Watch out

Carrying capacity is not an unchanging property independent of habitat conditions.

Exam habit

Label a positive result increase and a negative result decrease; do not report net change as the final population size.

05 Your turn

A population begins with 200 individuals. During a period there are 25 births, 12 deaths, 8 immigrants and 6 emigrants. Find the final size.

If you get stuck, uncover one hint at a time:

1. Separate the starting 200 individuals from changes during the period.
2. Add births and immigrants, then subtract deaths and emigrants to obtain net change.

3. Add that net change to 200. Report final size as individuals rather than confusing it with the smaller change value.

06 Check your understanding

If births equal deaths, must population size stay unchanged?

Births equal deaths, but immigration exceeds emigration. The local population:

1. Stays constant
2. Must decrease
3. Increases

Answers and explanations: page 376.

Relationships among organisms

EN / 04

- Identify ecological interactions from their effects on each partner.
- Support an interaction label with scenario evidence.

You need: Benefit, harm and little measurable effect are relative to the organisms in the interaction.

01 Idea

Competition occurs when organisms use a limited shared resource. Predation involves one organism consuming another. In mutualism both benefit; in commensalism one benefits without a substantial effect on the other; in parasitism one benefits at the host expense.

An interaction label summarizes what happens to each partner compared with the relevant situation without the interaction. A benefit for one partner does not automatically mean a benefit for the other. Writing the two effects separately is more reliable than guessing a label from the organisms involved.

- Mutualism requires a benefit to both partners.
- A benefit/harm pattern differs from a benefit/no-clear-effect pattern, so the second partner cannot be ignored.

02 Rule and symbols

Rule

Mutualism $(+, +)$

commensalism $(+, 0)$

parasitism $(+, -)$

03 Method

1. Identify the two organisms and the resource or activity linking them.
2. Describe the effect on the first organism and on the second separately.
3. Match the two effects to the appropriate interaction category.
4. Use evidence from the case rather than assuming all close associations are

mutualistic.

For relationship scenarios, identify the sign of the effect on each organism before naming the interaction.

04 Worked example

A parasite obtaining nutrients while harming its host is a plus-minus interaction.

01 The parasite obtains nutrients, so it benefits from the interaction.

02 The host is harmed, giving the benefit/harm pattern commonly written plus/minus.

03 The two competing plants both face restricted access to limited light.

04 Competition is therefore a minus/minus interaction in the comparison with unrestricted resource access, even if one competitor does better than the other.

Watch out

A relationship may be context-dependent; answer using the effects given in the question.

Exam habit

Write the effect on each partner before naming the interaction; this makes similar scenarios easier to distinguish.

05 Your turn

Identify a relationship where both benefit and another where one benefits while the other is harmed. Explain the signs.

If you get stuck, uncover one hint at a time:

1. Choose one relationship with a benefit for each partner and one with benefit for one and harm for the other.
2. Name the appropriate relationship category for each chosen example.

3. Explain the two signs in words, showing what each partner gains or loses rather than giving an unexplained plus/minus label.

06 Check your understanding

Does one competitor winning mean competition is automatically a benefit/harm interaction?

One organism benefits while the other has no clear benefit or harm. Is this necessarily parasitism?

1. No; parasitism requires harm to the host
2. Yes; any one-sided benefit is parasitism
3. Yes; no effect means harm

Answers and explanations: page 376.

Pollution: sources, pathways and effects

EN / 05

- Build a source-pathway-receptor explanation of pollution harm.
- Select prevention that interrupts a stated nutrient-runoff pathway.

You need: Nutrients support growth; decomposition can consume dissolved oxygen.

01 Idea

Pollution changes environmental conditions in ways that can cause harm or reduce usefulness. Sources can be natural or human-made. Air, water and soil pollution are connected through transport and deposition. A useful answer links a source to a pathway, an effect and a control.

Pollution explanations are strongest when each link shows how one change causes the next. Fertilizer does not have to poison fish directly to harm them: nutrients can change biological growth, and subsequent decomposition can alter oxygen conditions. Prevention can target an earlier link before ecological damage occurs.

- Runoff transports nutrients from a land source into water.
- Decomposition of increased organic material can use dissolved oxygen and harm oxygen-dependent organisms.

02 Rule and symbols

Rule

Source → pathway → receptor → effect

03 Method

1. Name the pollution source and the released material.
2. Trace how the material reaches the affected ecosystem.
3. Explain the intermediate process connecting exposure to the final effect.
4. Choose a preventive measure and state which link it reduces.

A source-to-effect question needs an intermediate pathway and mechanism, plus prevention linked to that chain.

04 Worked example

Nutrient-rich runoff can stimulate excessive algal growth.

01 Excess nutrients from fertilizer can be carried by runoff into a water body.

02 Nutrient enrichment can stimulate excessive algal growth.

03 When organic material is decomposed, microbial respiration can consume dissolved oxygen.

04 Low oxygen can harm fish; reducing excess fertilizer or intercepting runoff addresses the source or pathway before that effect occurs.

Watch out

Do not say algae always produce an immediate oxygen shortage without explaining decomposition and conditions.

Exam habit

Use connecting verbs such as carries, stimulates and consumes to make every step in the causal chain explicit.

05 Your turn

Explain a three-step route from excess fertilizer use to fish harm and name one preventive measure.

If you get stuck, uncover one hint at a time:

1. Begin with excess fertilizer and explain how its nutrients reach water.
2. Link nutrient enrichment to excess growth and then decomposition-related oxygen depletion under suitable conditions.
3. Finish with fish harm and one measure that reduces the source or intercepts the pathway. Use connected causal statements.

06 Check your understanding

Why is algae cause fish death an incomplete explanation?

Which measure most directly interrupts the pathway from fertilized land to a nearby stream?

1. Counting fish without changing the source or pathway
2. Intercepting nutrient-rich runoff before it reaches the stream
3. Adding more fertilizer whenever algae increase

Answers and explanations: page 376.

Ecological footprint and biocapacity

EN / 06

- Compare ecological demand with biocapacity on a common basis.
- Interpret a numerical balance as a deficit or reserve.

You need: Demand versus available capacity; subtraction; per-person and total quantities must not be mixed.

01 Idea

Ecological footprint estimates demand for biologically productive area needed to support consumption and absorb certain wastes. Biocapacity represents available biological production capacity. A deficit occurs when demand exceeds capacity. A global hectare is a productivity-adjusted unit, not simply any local hectare.

Footprint and biocapacity compare demand with biological supply within an accounting model. A deficit means the demand figure exceeds the available capacity figure on the same basis. The calculation only makes sense if both quantities describe the same population basis and productivity-adjusted unit.

- Subtracting capacity from demand tests whether demand exceeds capacity.
- A positive gap indicates deficit; the reverse comparison would indicate a reserve rather than a negative physical area.

02 Rule and symbols

Rule

$$\text{Ecological deficit} = \text{footprint} - \text{biocapacity}$$

if positive

03 Method

1. Identify footprint as demand and biocapacity as available biological production capacity.
2. Check that both figures use the same units, time basis and population basis.
3. Subtract biocapacity from footprint to test for a deficit.

4. Explain the result within the indicator scope rather than treating it as every environmental impact.

For a footprint balance, check units and per-person versus total basis before subtracting and interpreting.

04 Worked example

A population with footprint 3 global hectares per person and biocapacity 2 has a deficit of 1 per person.

01 The example footprint is 3 global hectares per person, describing demand.

02 Biocapacity is 2 global hectares per person, describing available capacity on the same basis.

03 Demand exceeds capacity by 3 - 2 = 1 global hectare per person.

04 The result is an ecological deficit, not an area measured in ordinary unadjusted local hectares.

Watch out

Ecological footprint and carbon footprint are related but not identical measures.

Exam habit

Write global hectares per person in a per-person result; omitting the population basis changes its meaning.

05 Your turn

A region has footprint 4.5 and biocapacity 2.8 global hectares per person. Identify and calculate the balance.

If you get stuck, uncover one hint at a time:

1. Identify which of 4.5 and 2.8 describes demand and which describes available capacity.
2. Compare them to decide deficit or reserve before calculating the gap.

3. Subtract the smaller from the larger for its magnitude and retain global hectares per person. State the type of balance explicitly.

06 Check your understanding

Can a region total footprint be compared directly with biocapacity per person?

A region has footprint lower than its biocapacity on the same basis. This indicates:

1. An ecological deficit
2. Proof that every environmental impact is zero
3. An ecological reserve in this accounting comparison

Answers and explanations: page 376.

Waste management and the hierarchy

EN / 07

- Distinguish prevention, reuse, recycling, recovery and disposal.
- Apply the waste hierarchy to a concrete material scenario.

You need: Reuse keeps an item in service; recycling reprocesses its material.

01 Idea

Waste management involves preventing, collecting, sorting, treating and disposing of waste. Prevention and reduction usually come before reuse, recycling, recovery and final disposal. Hazardous items require appropriate separate handling; not every household discard belongs in an ordinary bin.

The waste hierarchy separates avoiding a discard from keeping an item in use and from reprocessing its material. These actions can all reduce disposal, but they act at different stages. Correct classification follows what actually happens to the item, not a general impression that the action is environmentally helpful.

- Prevention avoids waste generation before later treatment is needed.
- Reuse keeps the item in service, whereas recycling reprocesses its material into a new product.

02 Rule and symbols

Rule

Prevent/reduce → reuse → recycle → recover → dispose

03 Method

1. Identify whether the action avoids waste, keeps the item in use or processes discarded material.
2. Classify the action using the general hierarchy.
3. For a proposed choice consider prevention or reduction first where feasible.
4. Check material hazards and the approved local collection stream before actual disposal.

Inspect the action verb in a scenario: avoid, use again, reprocess or dispose.

04 Worked example

Refilling a reusable bottle is reuse.

01 Refilling a durable bottle keeps the same item in service, so it is reuse.

02 Processing separated used glass into new glass changes the material into a new product, so it is recycling.

03 Avoiding unnecessary packaging prevents that material becoming waste at all.

04 These actions belong at different stages of the hierarchy and should not all be labelled recycling.

Watch out

Recycling does not erase all upstream resource use or justify unlimited consumption.

Exam habit

Use the action in the scenario as evidence: avoid means prevention, use again means reuse, reprocess means recycling.

05 Your turn

Classify avoiding excess packaging, reusing a jar, reprocessing aluminium cans and placing residual waste in an approved landfill.

If you get stuck, uncover one hint at a time:

1. Read each action separately rather than giving one category for the whole list.
2. Ask whether the action avoids material use, keeps the item in service, reprocesses material or places a residue in final disposal.
3. Match those distinctions to packaging, jar, cans and landfill in order. Use a short label and explanatory action for each.

06 Check your understanding

Is using a clean jar again to store suitable items recycling?

A metal drink container is melted and made into a new metal product. This is:

1. Recycling
2. Reuse of the same item
3. Waste prevention

Answers and explanations: page 376.

Sustainable development and integrated decisions

EN / 08

- Assess a project across environmental, social and economic dimensions.
- State measurable indicators and a trade-off for a sustainability decision.

You need: Present and future needs; a benefit in one dimension can have costs in another.

01 Idea

Sustainable development addresses present needs while preserving future ability to meet needs. Environmental, social and economic dimensions interact. A project can improve one dimension while creating costs in another, so evaluation should consider benefits, trade-offs and long-term effects.

Sustainability decisions combine outcomes that do not all move together. An energy-efficiency project may save electricity and money while requiring upfront spending or causing temporary disruption. Indicators make these claims testable; a trade-off explains how a benefit in one area comes with a cost or constraint elsewhere.

- Environmental, social and economic effects need distinct evidence.
- Comparing effects over time prevents an upfront cost or one immediate benefit from dominating the whole judgement.

02 Rule and symbols

Rule

$$\text{Environment} + \text{society} + \text{economy}$$

03 Method

1. Define the project and the needs it is intended to meet.
2. Give one environmental, one social and one economic effect.
3. Select an indicator that could test each effect.

4. Explain one trade-off over the project life rather than declaring the project sustainable from a single benefit.

When evaluating a project, choose one measurable indicator per dimension and connect a concrete benefit with a concrete cost.

04 Worked example

A public-transport project can reduce emissions, improve access and lower travel costs, but its construction and operation also require resources and funding.

01 The public-transport example can reduce transport emissions if its use changes travel patterns.

02 Improved access to education or work is a social benefit to evaluate.

03 Travel costs, operating costs and funding provide economic questions.

04 Construction also consumes resources and money, so a full assessment compares these costs with benefits over time.

Watch out

Do not equate sustainable development with only economic growth or only nature protection.

Exam habit

Replace vague words such as greener with a measurable indicator such as electricity use, emissions, access or cost over a stated period.

05 Your turn

Give one environmental, one social and one economic indicator for a campus energy-efficiency project, plus one trade-off to evaluate.

If you get stuck, uncover one hint at a time:

1. List the three dimensions and identify a campus outcome for each.

2. Choose measures such as electricity consumption, occupant comfort and financial savings or payback, giving a clear basis for comparison.
3. Add a trade-off, for example upfront spending versus later savings, and explain what decision needs evaluation rather than merely listing two unrelated facts.

06 Check your understanding

Is lower operating energy alone enough to prove a project is sustainable?

Which is a measurable social indicator for a campus energy-efficiency project?

1. Electricity use in kilowatt-hours alone
2. Student-reported thermal comfort using a defined survey
3. The project is very green

Answers and explanations: page 376.

Lesson review and answers

Environment and ecosystem components

Return to lesson: page 352.

Your turn: No. Bacteria are living organisms and therefore biotic components.

Concept check: 1. Correct. Influence on life does not itself make a factor alive.

Written method: Sunlight and mineral content are abiotic. Grass and rabbit are biotic. One mark each.

Energy flow, food chains and decomposers

Return to lesson: page 355.

Your turn: No. Energy flows through and is eventually dissipated as heat; nutrients can be recycled.

Concept check: 2. Correct. Two transfers multiply one tenth by one tenth.

Written method: Primary consumers receive 1000 units; secondary consumers receive 100 units. Two marks for each transfer. These values depend on the stated efficiency.

Population growth and limiting factors

Return to lesson: page 358.

Your turn: No. Immigration and emigration can still produce a net change.

Concept check: 3. Correct. The birth/death balance is zero, leaving a positive migration balance.

Written method: Net change =25-12+8-6=15. Final size=215. Credit signs of all four processes and final total.

Relationships among organisms

Return to lesson: page 361.

Your turn: No. Both incur the effects of competing for a limited resource; one may still outperform the other.

Concept check: 1. Correct. Benefit with little or no effect describes a different pattern from benefit with harm.

Written method: Mutualism is (+,+): both gain a benefit. Parasitism is (+,-): parasite benefits and host is harmed. One mark for each name and each explanation.

Pollution: sources, pathways and effects

Return to lesson: page 364.

Your turn: It omits the conditions and intermediate processes, especially decomposition and dissolved-oxygen depletion.

Concept check: 2. Correct. This acts on transport between source and receptor.

Written method: Runoff carries nutrients to water; algal growth increases; decomposition can lower dissolved oxygen and harm fish. Reduce excess nutrient application or intercept runoff. One mark for each causal step and one for prevention.

Ecological footprint and biocapacity

Return to lesson: page 367.

Your turn: No. Convert both to totals or both to per-person values first.

Concept check: 3. Correct. Capacity exceeds the counted demand.

Written method: It has an ecological deficit of 4.5-2.8=1.7 global hectares per person. Credit comparison, subtraction, units and deficit interpretation.

Waste management and the hierarchy

Return to lesson: page 370.

Your turn: It is reuse, because the same item is used again without material reprocessing.

Concept check: 1. Correct. The discarded material is reprocessed into a new product.

Written method: Prevention, reuse, recycling and disposal, respectively. One mark for each classification.

Sustainable development and integrated decisions

Return to lesson: page 373.

Your turn: No. Social access, economic viability, construction impacts and long-term effects also require assessment.

Concept check: 2. Correct. A defined comfort measure directly examines the learner experience.

Written method: Examples: electricity use/emissions; comfort or accessibility; cost savings/payback; upfront cost versus long-term savings. One mark for each relevant, measurable item.

Humans and the Environment / Test A

20 marks / Suggested time: 20 minutes. MCQs: 1 mark each. Written problems: 4 marks each. Attempt without notes.

Part I / Choose one answer

1. Which is an abiotic factor?

   A. Algae

   B. Bacteria

   C. Water temperature

   D. Fish

2. Decomposers mainly help by:

   A. Recycling nutrients from dead matter

   B. Creating sunlight

   C. Stopping all energy loss

   D. Removing all producers

3. Which directly reduces a local population?

   A. Reproduction

   B. Births

   C. Immigration

   D. Emigration

4. Both organisms benefit in:

   A. Parasitism

   B. Mutualism

   C. Predation

   D. Competition

Part II / Show your reasoning

1. Classify sunlight, grass, soil mineral content and a rabbit as biotic or abiotic.

2. A question specifies 10 percent transfer efficiency at each step. Producers have 10000 energy units. Find energy available to primary and secondary consumers.

3. A population begins with 200 individuals. During a period there are 25 births, 12 deaths, 8 immigrants and 6 emigrants. Find the final size.

4. Identify a relationship where both benefit and another where one benefits while the other is harmed. Explain the signs.

Humans and the Environment / Test B

20 marks / Suggested time: 20 minutes. MCQs: 1 mark each. Written problems: 4 marks each. Attempt without notes.

Part I / Choose one answer

1. Excess nutrient input into water is most closely associated with:

   A. Eutrophication

   B. Lens refraction

   C. Sterilization

   D. No biological change

2. A footprint larger than biocapacity indicates:

   A. Unlimited sustainability

   B. An ecological reserve

   C. An ecological deficit

   D. Zero resource use

3. Which is highest in the general waste hierarchy?

   A. Disposal after mixing

   B. Landfill

   C. Uncontrolled burning

   D. Waste prevention

4. A complete sustainability assessment considers:

   A. Only this month

   B. Environmental, social and economic effects

   C. Only profit

   D. Only tree numbers

Part II / Show your reasoning

1. Explain a three-step route from excess fertilizer use to fish harm and name one preventive measure.

2. A region has footprint 4.5 and biocapacity 2.8 global hectares per person. Identify and calculate the balance.

3. Classify avoiding excess packaging, reusing a jar, reprocessing aluminium cans and placing residual waste in an approved landfill.

4. Give one environmental, one social and one economic indicator for a campus energy-efficiency project, plus one trade-off to evaluate.

Humans and the Environment / Test A / Answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1

MCQ: C. Temperature is a physical condition rather than a living organism.

Written solution: Sunlight and mineral content are abiotic. Grass and rabbit are biotic. One mark each.

Return to lesson: page 352.

Question 2

MCQ: A. They break down organic matter and return nutrients to the system.

Written solution: Primary consumers receive 1000 units; secondary consumers receive 100 units. Two marks for each transfer. These values depend on the stated efficiency.

Return to lesson: page 355.

Question 3

MCQ: D. Emigration removes individuals from the local population.

Written solution: Net change =25-12+8-6=15. Final size=215. Credit signs of all four processes and final total.

Return to lesson: page 358.

Question 4

MCQ: B. Mutualism is defined by benefits to both participants.

Written solution: Mutualism is (+,+): both gain a benefit. Parasitism is (+,-): parasite benefits and host is harmed. One mark for each name and each explanation.

Return to lesson: page 361.

Humans and the Environment / Test B / Answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1

MCQ: A. Nutrient enrichment can drive excessive growth and subsequent oxygen depletion.

Written solution: Runoff carries nutrients to water; algal growth increases; decomposition can lower dissolved oxygen and harm fish. Reduce excess nutrient application or intercept runoff. One mark for each causal step and one for prevention.

Return to lesson: page 364.

Question 2

MCQ: C. Demand exceeds the available biological capacity under the accounting model.

Written solution: It has an ecological deficit of 4.5-2.8=1.7 global hectares per person. Credit comparison, subtraction, units and deficit interpretation.

Return to lesson: page 367.

Question 3

MCQ: D. Avoiding waste reduces the need for later treatment and disposal.

Written solution: Prevention, reuse, recycling and disposal, respectively. One mark for each classification.

Return to lesson: page 370.

Question 4

MCQ: B. The concept integrates all three dimensions and future consequences.

Written solution: Examples: electricity use/emissions; comfort or accessibility; cost savings/payback; upfront cost versus long-term savings. One mark for each relevant, measurable item.

Return to lesson: page 373.

Formula sheet

Environment and ecosystem components p.352

Rule

$$\text{Organism} \rightarrow \text{population} \rightarrow \text{community} \rightarrow \text{ecosystem}$$

Energy flow, food chains and decomposers p.355

Rule

$$\text{Producer} \rightarrow \text{primary consumer} \rightarrow \text{secondary consumer}$$

Population growth and limiting factors p.358

Rule

$$\Delta N = \text{births} + \text{immigration} - \text{deaths} - \text{emigration}$$

Relationships among organisms p.361

Rule

Mutualism $(+, +)$

commensalism $(+, 0)$

parasitism $(+, -)$

Pollution: sources, pathways and effects p.364

Rule

$$\text{Source} \rightarrow \text{pathway} \rightarrow \text{receptor} \rightarrow \text{effect}$$

Ecological footprint and biocapacity p.367

Rule

$$\text{Ecological deficit} = \text{footprint} - \text{biocapacity}$$

if positive

Waste management and the hierarchy p.370

Rule

$$\text{Prevent/reduce} \rightarrow \text{reuse} \rightarrow \text{recycle} \rightarrow \text{recover} \rightarrow \text{dispose}$$

Sustainable development and integrated decisions

p.373

Rule

Environment + society + economy

Symbol glossary

Symbols are listed with their lesson context. A symbol may represent different quantities in different rules.

Source and coverage notes

ENV101 elective: ecosystems, relationships among organisms, pollution, ecological footprint, waste management and sustainable development. The Arabic source lectures are explained here in English.

Open original course folder

The source audit inventories 159 non-programming files; 95 yielded substantial extracted text. Scanned diagrams and handwritten or image-only material were not fully transcribed. Check lecturer-specific derivations and conventions. The practice tests are original study material, not predictions of an examination.

Professional Development Seminar

FIG. 08 / PROFESSIONAL DEVELOPMENT

A useful portfolio leads from evidence to action



Select evidence that answers a question about learning.

Takeaway: explain what the evidence shows, then choose the next improvement.

Elective lecture themes: teaching as a profession, teacher formation, professional learning, portfolios, standards and educational technology. This follows the teacher-development focus of the supplied file; it is not a generic career-skills course.

Subject map

Elective lecture themes: teaching as a profession, teacher formation, professional learning, portfolios, standards and educational technology. This follows the teacher-development focus of the supplied file; it is not a generic career-skills course.

Follow this reading route from foundations to applications. Each line links to its lesson; revisit the stated prerequisites whenever a step feels unfamiliar.

01 Teaching as a profession p.392

02 Professionalization and the teacher role p.395

03 Teacher formation: before and during service p.398

04 Preparation, practice and professional standards p.401

05 Plan continuing professional development p.404

06 Portfolios and reflective evaluation p.407

07 Ethics, collaboration and learning communities p.410

08 Educational technology and evaluation of impact p.413

Teaching as a profession

PD / 01

- Explain two criteria that make teaching a profession.
- Support each criterion with a distinct classroom example.

You need: A professional criterion is a defining feature, not simply an employment benefit.

01 Idea

The source distinguishes a profession through specialized knowledge, extended preparation, social purpose, ethical responsibilities and organized practice. Teaching is more than delivering facts: it develops learners and contributes to society. A strong exam response connects each professional criterion to a teaching example.

A profession combines expertise with responsibility for how that expertise affects others. Teaching therefore involves both knowing a subject and making informed, ethical decisions about learners. An exam answer becomes convincing when it connects each professional feature to an observable teaching action.

- Specialized preparation explains why professional judgement requires more than familiarity with subject facts.
- Ethical responsibilities explain why competence must be used fairly and for a social purpose.

02 Rule and symbols

Rule

Criterion + explanation + teaching example

03 Method

1. Define profession using expertise, preparation, social purpose and ethical responsibility.
2. Select two distinct criteria relevant to the question.
3. Explain each criterion in plain language.
4. Attach a concrete teaching example to each explanation.

For reasons teaching is a profession, give distinct criterion-example pairs instead of repeated general praise.

04 Worked example

Specialized knowledge includes subject expertise and pedagogy; ethical responsibility includes fair assessment and respect for learners.

01 Subject expertise concerns what is taught; pedagogy concerns how learners are helped to understand it.

02 Together they illustrate specialized professional knowledge rather than information delivery alone.

03 Fair assessment and respect for learners illustrate ethical responsibility.

04 The examples support two different criteria, so they provide more evidence than repeating expertise in several forms.

Watch out

Do not assume being paid alone makes an activity a profession.

Exam habit

Use two complete criterion-explanation-example pairs when four marks are available, rather than a long unsupported list.

05 Your turn

Give two reasons teaching is considered a profession, each with an example.

If you get stuck, uncover one hint at a time:

1. Select two distinct criteria, such as specialized preparation and ethical responsibility.
2. Explain each criterion briefly in your own words.
3. Attach one teaching example to each: show what the teacher learns or does and why that action illustrates the chosen criterion.

06 Check your understanding

Does being paid alone establish that an activity is a profession?

Which pair best shows two different professional criteria?

1. Subject-and-pedagogy preparation, plus fair assessment
2. Subject knowledge, plus knowing many subject facts
3. Being busy, plus having a long working day

Answers and explanations: page 416.

Professionalization and the teacher role

PD / 02

- Describe teacher roles through observable professional actions.
- Explain how assessment evidence guides subsequent instruction.

You need: A role describes a function; an action shows how that function appears in practice.

01 Idea

The lecture describes development from pre-professional through semi-professional toward growing professionalization. Teacher roles extend from presenting knowledge to guiding learning, assessing progress, modelling values and supporting society. Historical labels describe development, not a reason to dismiss present expertise.

Teacher roles describe purposes expressed through actions. An assessor interprets evidence, a guide helps learners choose strategies, and a model demonstrates professional values. A role label alone leaves the reader guessing; an action and its effect show that the role is understood.

- The same classroom activity can serve different roles depending on its purpose and follow-up.
- Describing a learner effect connects a teacher action to its professional function.

02 Rule and symbols

Rule

$$\text{Role} \rightarrow \text{observable professional action}$$

03 Method

1. Identify the role asked about, such as assessor or learning guide.
2. Describe a specific action the teacher takes in that role.
3. Explain how the action supports learners.

4. For historical comparisons, discuss changes in preparation or organization rather than dismissing current expertise.

When asked for roles beyond presenting content, identify an action that changes how learning is supported.

04 Worked example

A teacher acting as an assessor uses evidence from student work to adapt instruction.

01 As an assessor, the teacher examines student work to identify what learners understand.

02 If several answers show the same error, that evidence indicates a learning gap.

03 Adapting instruction in response connects the assessor role to better teaching decisions.

04 As a guide, the teacher supports reasoning through questions or strategies instead of supplying every answer.

Watch out

A list of role names without explanation earns less evidence of understanding.

Exam habit

After every role name add a verb and object: diagnoses a misconception, gives targeted feedback or guides strategy choice.

05 Your turn

Describe two teacher roles beyond explaining subject content.

If you get stuck, uncover one hint at a time:

1. Choose two roles beyond delivering subject explanations, ensuring they are meaningfully different.

2. For each role, name one observable action, such as diagnosing a learning gap or supporting strategy choice.
3. Explain how that action helps learning. A title without an action is only a partial answer.

06 Check your understanding

Why is gives a test alone a weak explanation of the assessor role?

A teacher asks a learner to compare two solution strategies rather than immediately giving an answer. This most directly shows:

1. Avoiding all responsibility for learning
2. Guiding reasoning
3. Merely storing information

Answers and explanations: page 416.

Teacher formation: before and during service

PD / 03

- Distinguish pre-service preparation, induction and continuing development.
- Match a support activity to its career stage and purpose.

You need: Teachers develop before entering practice and throughout their careers.

01 Idea

Teacher formation is a continuing process. Pre-service preparation establishes knowledge and skills; induction supports entry to actual teaching; continuing professional development addresses changing needs and improves practice. Training is one method of development, not the whole process.

Teacher formation is a continuing sequence, but support has different purposes at different stages. Initial preparation builds foundations before entry; induction helps a new teacher use them in a real workplace. Comparing both timing and purpose makes the distinction clearer than treating one as simply more training.

- Pre-service learning builds foundations before entry to the profession.
- Induction bridges those foundations and the demands of early independent practice.

02 Rule and symbols

Rule

Pre-service → induction → continuing development

03 Method

1. Locate the person before employment, in early practice or in established service.
2. Name the matching formation stage.
3. Explain the specific knowledge or support needed at that stage.
4. Describe how the activity supports practice rather than treating attendance as proof of improvement.

Stage-comparison questions require a timing statement and a purpose statement for each stage.

04 Worked example

A mentor supporting a newly appointed teacher belongs to induction.

01 The example teacher is newly appointed and is entering real classroom practice.

02 Mentor support at this point is induction, bridging initial preparation and workplace demands.

03 Later peer observation and inquiry are examples of continuing development.

04 The method may look similar across stages, but the career context and purpose determine the classification.

Watch out

Graduation marks an entry point, not the end of professional learning.

Exam habit

In a comparison give both timing and purpose for each stage; stating before and after alone is incomplete.

05 Your turn

Compare pre-service preparation with induction in two clear points.

If you get stuck, uncover one hint at a time:

1. Set up two headings in your notes: before entry and early practice.
2. Under each, identify its main learning or support purpose.
3. Turn these into two parallel comparisons covering when support occurs and what it helps the teacher do. Do not classify solely by whether a workshop or mentor is present.

06 Check your understanding

Is every teacher workshop automatically pre-service preparation?

A newly appointed teacher receives workplace mentoring while beginning classroom duties. Why is this induction rather than pre-service?

1. Mentoring can never be used before employment
2. All learning stops after induction
3. It supports entry into actual professional practice

Answers and explanations: page 416.

Preparation, practice and professional standards

PD / 04

- Explain how supervised field experience links theory with practice.
- Construct a feedback cycle using observable evidence.

You need: Preparation includes subject knowledge, pedagogy and supervised teaching; standards describe expected practice.

01 Idea

Teacher preparation combines subject knowledge, pedagogical understanding and supervised practice. Field experience connects theory with real learners and classroom decisions. Professional standards make expectations explicit and support self-evaluation. Licensing is a recognition process whose current requirements must be checked with the responsible body.

Reflection becomes professional improvement when it leads to a tested change. An observed lesson supplies evidence, but evidence must be interpreted, used to revise practice and checked again. The next lesson is therefore part of the cycle, rather than a separate event after feedback ends.

- Specific evidence makes the improvement target clearer than a vague judgement.
- Checking the revised lesson reveals whether the chosen change actually helped.

02 Rule and symbols

Rule

$$\text{Knowledge} \rightarrow \text{supervised application} \rightarrow \text{feedback} \rightarrow \text{improvement}$$

03 Method

1. Identify the teaching expectation or intended learner outcome.
2. Collect relevant evidence from observation and student work.
3. Choose one specific change suggested by that evidence.

4. Apply the change and evaluate the next attempt against the same expectation.

An improvement-cycle question needs evidence, a focused need, a targeted action and a follow-up evaluation.

04 Worked example

After teaching a supervised lesson, a trainee reviews observations and student work, then revises the next lesson.

01 The trainee first teaches a supervised lesson, putting planned ideas into practice.

02 Observation notes and student work show what occurred and what learners achieved.

03 The trainee compares that evidence with the intended learning and revises the next lesson.

04 The cycle is completed by checking whether the revision improves the next observed outcome.

Watch out

Do not invent current licence requirements from an older lecture.

Exam habit

When standards are mentioned, link one expectation to evidence; do not invent current licensing rules from a lecture.

05 Your turn

Give a four-step improvement cycle for a trainee after an observed lesson.

If you get stuck, uncover one hint at a time:

1. Start with reviewing observation notes and student work rather than jumping directly to a new activity.
2. Identify one specific weakness and choose a change that addresses it.

3. Arrange the four-step cycle so the change is applied and then evaluated with new evidence. Keep every step linked to the same improvement need.

06 Check your understanding

Why does feedback without a subsequent change and evaluation leave the cycle incomplete?

After an observed lesson, a trainee changes the task but collects no follow-up evidence. What is missing?

1. Evaluation of whether the change helped
2. Any possibility of changing practice
3. A requirement to repeat the original lesson unchanged

Answers and explanations: page 416.

Plan continuing professional development

PD / 05

- Write a professional-development plan linking need, goal, activity and evidence.
- Distinguish participation evidence from evidence of impact.

You need: A development goal should address an identified teaching or learning need.

01 Idea

Professional development is purposeful and sustained. It begins with a learning need, sets a clear goal, chooses suitable activities and evaluates effects on practice and learners. Workshops, peer observation, professional reading and collaborative inquiry can all contribute.

A development plan is a chain from a real teaching need to evidence of improvement. A workshop can supply ideas, but the plan also needs classroom application and a way to judge results. A focused goal makes it possible to tell whether the activity was useful.

- An activity becomes relevant when it addresses the diagnosed need.
- Before-and-after learner evidence can show impact more directly than participation records alone.

02 Rule and symbols

Rule

Need → goal → activity → evidence of impact

03 Method

1. Use classroom evidence to name one specific development need.
2. Write a focused goal and a realistic review period.
3. Choose a learning activity and classroom application that address the goal.
4. Compare relevant evidence before and after the application to judge progress.

When asked for a development plan, link need, goal, activity and evidence within a defined review period.

04 Worked example

A teacher notices weak student explanations, studies questioning strategies, tries them for four weeks and compares the quality of student responses.

01 The teacher notices weak student explanations, identifying a specific need.

02 Studying questioning strategies is the development activity, but study alone does not show impact.

03 Trying the strategies for four weeks puts the learning into classroom practice.

04 Comparing response samples before and after can reveal whether explanations became clearer and better supported.

Watch out

Attendance certificates show participation, not necessarily improved teaching.

Exam habit

Keep all four plan parts linked to the same need; an unrelated workshop does not satisfy a coherent development plan.

05 Your turn

Write a short development plan for a teacher whose students struggle to explain answers.

If you get stuck, uncover one hint at a time:

1. Describe the need precisely: students give answers without sufficient explanation.
2. Set a focused goal and choose a relevant learning activity plus classroom application.
3. Specify a review period and compare response samples using clear explanation

criteria. Ensure the evidence measures the original need.

06 Check your understanding

What does a workshop attendance certificate demonstrate directly?

A teacher attends a questioning workshop. Which next step best tests its value for weak student explanations?

1. Collect another certificate without classroom application
2. Apply selected strategies and compare explanation samples over time
3. Count the slides shown in the workshop

Answers and explanations: page 416.

Portfolios and reflective evaluation

PD / 06

- Select portfolio evidence for a defined goal or standard.
- Explain how reflection turns an artifact into evidence of development.

You need: An artifact is a document or sample; evidence requires an explanation of what it demonstrates.

01 Idea

A professional portfolio is a purposeful selection of evidence linked to goals and standards, with reflection on what the evidence demonstrates. It can include plans, student work, feedback and records of improvement. A large unorganized collection is less useful than a selective, explained one.

A portfolio is an argument supported by selected artifacts. A lesson plan can show intended alignment, while student work can show something about the outcome. Commentary connects the evidence to a professional claim and acknowledges what one artifact alone cannot prove.

FIG. 08 / PROFESSIONAL DEVELOPMENT

A useful portfolio leads from evidence to action



Select evidence that answers a question about learning.

Takeaway: explain what the evidence shows, then choose the next improvement.

Figure 08. A portfolio connects selected evidence to reflection and a targeted next action. A student-work sample becomes more useful when the teacher explains what it shows and how teaching will change.

- An artifact becomes useful evidence when its relevance to a goal or standard is explained.
- Reflection connects intention, outcome and a next learning step rather than simply decorating a collection.

02 Rule and symbols

Rule

Evidence + reflection + link to a goal

03 Method

1. State the portfolio purpose or professional goal.
2. Select a small number of relevant artifacts and organize them clearly.
3. Explain what each artifact shows and what its limitations are.
4. Reflect on the outcome and identify a next improvement, protecting learner privacy.

When naming portfolio items, pair each with the specific aspect of practice it can demonstrate.

04 Worked example

A lesson plan paired with student work and a reflection on changes shows more than a lesson plan alone.

01 The lesson plan shows what the teacher intended learners to do.

02 Student work shows something about the resulting learning and where difficulties remained.

03 A reflection compares the plan with that outcome and explains why a change was made.

04 The three pieces together support a development claim more clearly than an unexplained plan alone.

Watch out

Do not include identifiable student information unnecessarily; follow the institution privacy rules.

Exam habit

For each item finish the sentence this demonstrates... and connect it to a goal or standard; anonymize learner material as required.

05 Your turn

Name two portfolio items and explain what each can demonstrate.

If you get stuck, uncover one hint at a time:

1. What aspects of teaching should your two items provide evidence for? Select two distinct aspects before choosing artifacts.
2. Consider a planning document and an anonymized learner-work sample; identify the different practice or outcome each can demonstrate.
3. Add a short explanation linking each artifact to its claim. Avoid claiming that a plan alone proves the intended learning occurred.

06 Check your understanding

Does adding more unrelated documents necessarily make a portfolio stronger?

A lesson plan alone most directly shows:

1. That every student achieved the objective
2. That no future improvement is needed
3. What learning and teaching were intended

Answers and explanations: page 416.

Ethics, collaboration and learning communities

PD / 07

- Rewrite personal criticism as evidence-based constructive feedback.
- Connect ethical collaboration with learner respect and improvement.

You need: Observation is what can be seen or recorded; interpretation and personal judgement are different.

01 Idea

Professional learning can be collaborative: colleagues discuss evidence, share practice and solve problems. Ethical practice includes respect, fairness, honesty and responsibility. Feedback should focus on observable practice and improvement, rather than personal attacks.

Constructive feedback describes a changeable practice rather than labelling a person. The recipient needs to know what was observed, how learners were affected and what could be tried next. Respect and specificity make challenge more useful without requiring uncritical agreement.

- Observable evidence gives the colleague something concrete to examine.
- An actionable suggestion turns criticism into a possible improvement step.

02 Rule and symbols

Rule

Observation + effect + constructive next step

03 Method

1. Identify an observable feature of teaching rather than a personal label.
2. Describe its apparent effect on learners with available evidence.
3. Suggest one feasible next step connected to that effect.
4. Use respectful language and follow confidentiality and reporting responsibilities.

To rewrite harsh feedback, replace personal judgement with observation, learner effect and a feasible next action.

04 Worked example

A colleague gives feedback that a task confused learners and suggests clearer success criteria, rather than calling the teacher incompetent.

01 Calling a teacher incompetent attacks the person and gives no specific route to improvement.

02 The example instead identifies that the task confused learners.

03 Suggesting clearer success criteria connects the feedback to a teaching action the colleague can try.

04 A later check of learner understanding can show whether that change helped, completing the improvement conversation.

Watch out

Collaboration is not simply agreeing with everyone or sharing private student details.

Exam habit

Structure scenario answers as observation, learner effect and next step; avoid unsupported claims about personal character.

05 Your turn

Rewrite this feedback constructively: Your lesson was bad.

If you get stuck, uncover one hint at a time:

1. Remove the personal or vague judgement from the original sentence.
2. Identify a specific observable difficulty and its effect on learners, using a clearly stated example rather than pretending to have evidence you did not collect.
3. Suggest one relevant next step in respectful language, such as clarifying suc-

cess criteria and checking understanding before independent work.

06 Check your understanding

Is constructive feedback the same as agreeing with every colleague decision?

Which statement is most useful as the observation part of feedback?

1. Several learners asked what the task goal was
2. You are a careless person
3. Your lesson was bad in every way

Answers and explanations: page 416.

Educational technology and evaluation of impact

PD / 08

- Choose an educational tool to serve a specific learning objective.
- Evaluate technology through diagnosis, teacher response and learner evidence.

You need: A tool supports an activity; learning depends on how the activity and follow-up are designed.

01 Idea

Educational technology should serve a learning goal. Teachers need to select suitable tools, design learning activities, assess outcomes and consider access and ethical use. Technical operation alone does not establish teaching value. The seminar mentions digital competencies; programming instruction is excluded from this pack as requested.

Technology helps learning through what it enables teachers and learners to do. A quiz can reveal an error pattern quickly, but learning benefit depends on how that pattern changes the next teaching step. Evaluation therefore follows the full sequence from objective to evidence and response.

- Plausible distractors can make a misconception visible in response patterns.
- A fresh question after targeted teaching tests improvement more directly than click counts.

02 Rule and symbols

Rule

Learning need → tool choice → activity → evaluation

03 Method

1. Name the learner objective and the misconception or difficulty to investigate.
2. Choose a suitable accessible tool and design questions that reveal reasoning.
3. Inspect response patterns and adapt teaching accordingly.

4. Use a new question or response sample to judge whether understanding improved.

For an educational-technology proposal, state the learning purpose, diagnostic task, teacher response and evidence of value.

04 Worked example

A short digital quiz can reveal misconceptions immediately, but the teacher must use the results to adapt instruction for it to support learning.

01 The brief digital quiz can make response patterns visible quickly.

02 Plausible incorrect options help reveal which misconception may be present.

03 The teacher must interpret those results and respond, for example with targeted explanation or discussion.

04 A fresh question then checks learning; the number of clicks or time on screen alone does not establish educational value.

Watch out

More screen time is not itself evidence of better learning.

Exam habit

State the learning goal before the tool name and finish with evidence of learning; programming content is outside this course pack.

05 Your turn

Propose a simple technology use to diagnose misconceptions, and state how you would evaluate its value.

If you get stuck, uncover one hint at a time:

1. Choose a simple tool that can reveal reasoning or misconceptions, rather than starting with novelty.
2. Explain how the response pattern will guide a teaching action.

3. State how a new question or response sample will test improvement, while considering access and the defined learning objective.

06 Check your understanding

What is missing if a teacher runs a digital quiz but never uses its results?

A digital quiz reveals a common misconception. Which action best turns the result into a learning benefit?

1. Judge success only by how many students clicked
2. Reteach the issue and check understanding with a new question
3. Keep the response chart but make no teaching change

Answers and explanations: page 416.

Lesson review and answers

Teaching as a profession

Return to lesson: page 392.

Your turn: No. Professional status also involves specialized preparation, knowledge, responsibility and organized standards of practice.

Concept check: 1. Correct. Preparation and ethical responsibility are distinct criteria with clear teaching links.

Written method: Specialized preparation: teachers study subject knowledge and pedagogy. Ethical duty: teachers assess fairly and protect learner dignity. Two marks for each criterion-example pair.

Professionalization and the teacher role

Return to lesson: page 395.

Your turn: It names an activity but does not explain how evidence is interpreted or used to support learning.

Concept check: 2. Correct. The teacher supports the learner in selecting and evaluating a strategy.

Written method: Assessor: identifies learning gaps using evidence and feedback. Learning guide: supports students in choosing strategies and solving problems. Two marks per role with a concrete action.

Teacher formation: before and during service

Return to lesson: page 398.

Your turn: No. A workshop during employment may be induction support or continuing development, depending on career stage and purpose.

Concept check: 3. Correct. Career stage and purpose define induction in this scenario.

Written method: Pre-service occurs before entry and develops foundational subject and pedagogical knowledge. Induction occurs during early practice and provides mentoring and support in the real workplace. One mark each for stage and purpose in both cases.

Preparation, practice and professional standards

Return to lesson: page 401.

Your turn: It identifies an issue but does not show whether the teacher acted on it or improved practice.

Concept check: 1. Correct. An action was taken, but its effect has not been assessed.

Written method: Review evidence; identify one specific weakness; plan and apply a targeted change; evaluate the next lesson using evidence. One mark per coherent step.

Plan continuing professional development

Return to lesson: page 404.

Your turn: Participation. Improved teaching or learning needs additional evidence from practice and outcomes.

Concept check: 2. Correct. This connects learning, classroom use and relevant evidence of impact.

Written method: Need: weak explanations. Goal: improve evidence-based responses. Activity: learn and apply structured questioning. Evidence: compare student response samples before and after a defined period. One mark per linked component.

Portfolios and reflective evaluation

Return to lesson: page 407.

Your turn: No. Relevance, clear organization and reflection matter more than volume alone.

Concept check: 3. Correct. A plan documents intentions and design; outcomes require additional evidence.

Written method: A lesson plan can demonstrate planning and alignment. An anonymized student-work sample with commentary can demonstrate learning and the effect of feedback. Two marks per item and explanation.

Ethics, collaboration and learning communities

Return to lesson: page 410.

Your turn: No. It can challenge practice, but uses evidence, respect and actionable suggestions.

Concept check: 1. Correct. This reports an observable event that can inform an improvement discussion.

Written method: Example: Several students could not identify the task goal; stating the success criteria and checking one example before independent work may help. Credit observable evidence, effect, actionable suggestion and respectful language.

Educational technology and evaluation of impact

Return to lesson: page 413.

Your turn: The evidence-to-action step: diagnosis has not been used to adapt instruction or check improvement.

Concept check: 2. Correct. This links diagnosis to action and evidence of improved understanding.

Written method: Use a brief concept quiz with plausible distractors. Review response patterns, reteach the misconception and check with a new question. Evaluate whether explanations or transfer improve, not only whether students clicked. One mark each for tool, diagnosis, response and evaluation.

Professional Development Seminar / Test A

20 marks / Suggested time: 20 minutes. MCQs: 1 mark each. Written problems: 4 marks each. Attempt without notes.

Part I / Choose one answer

1. Which is a professional criterion emphasized in the lecture?

   A. No responsibility to society

   B. Learning stops at graduation

   C. No preparation

   D. Specialized knowledge and ethical standards

2. Using student work to improve instruction best illustrates the teacher as:

   A. A storage clerk only

   B. An assessor and learning guide

   C. A person who never changes plans

   D. A passive spectator

3. Support during a teacher first years of practice is mainly:

   A. Only pre-service admission

   B. Induction

   C. Retirement

   D. Waste management

4. The central value of supervised field experience is:

   A. Eliminating evaluation

   B. Replacing all theory

   C. Connecting theory to practice with feedback

   D. Avoiding learners

Part II / Show your reasoning

1. Give two reasons teaching is considered a profession, each with an example.

2. Describe two teacher roles beyond explaining subject content.

3. Compare pre-service preparation with induction in two clear points.

4. Give a four-step improvement cycle for a trainee after an observed lesson.

Professional Development Seminar / Test B

20 marks / Suggested time: 20 minutes. MCQs: 1 mark each. Written problems: 4 marks each. Attempt without notes.

Part I / Choose one answer

1. Which is the strongest development goal?

   A. Improve one identified practice and evaluate its effect

   B. Collect the most certificates

   C. Become better somehow

   D. Attend anything

2. What makes a portfolio stronger?

   A. No evaluation

   B. More unrelated documents

   C. Evidence linked to goals and reflection

   D. Only decorative pages

3. Constructive professional feedback should be:

   A. Only praise regardless of evidence

   B. Unrelated to teaching

   C. Personal and humiliating

   D. Based on observation and useful next steps

4. The best reason to choose an educational tool is:

   A. It supports a defined learning objective

   B. It has the most animations

   C. Everyone must use the same tool

   D. It is newest

Part II / Show your reasoning

1. Write a short development plan for a teacher whose students struggle to explain answers.

2. Name two portfolio items and explain what each can demonstrate.

3. Rewrite this feedback constructively: Your lesson was bad.

4. Propose a simple technology use to diagnose misconceptions, and state how you would evaluate its value.

Professional Development Seminar / Test A / Answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1

MCQ: D. Professional practice combines expertise, preparation and ethical obligations.

Written solution: Specialized preparation: teachers study subject knowledge and pedagogy. Ethical duty: teachers assess fairly and protect learner dignity. Two marks for each criterion-example pair.

Return to lesson: page 392.

Question 2

MCQ: B. Assessment information guides the next teaching decision.

Written solution: Assessor: identifies learning gaps using evidence and feedback. Learning guide: supports students in choosing strategies and solving problems. Two marks per role with a concrete action.

Return to lesson: page 395.

Question 3

MCQ: B. Induction bridges initial preparation and independent professional practice.

Written solution: Pre-service occurs before entry and develops foundational subject and pedagogical knowledge. Induction occurs during early practice and provides mentoring and support in the real workplace. One mark each for stage and purpose in both cases.

Return to lesson: page 398.

Question 4

MCQ: C. Practice tests and develops professional knowledge in authentic settings.

Written solution: Review evidence; identify one specific weakness; plan and apply a targeted change; evaluate the next lesson using evidence. One mark per coherent step.

Return to lesson: page 401.

Professional Development Seminar / Test B / Answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1

MCQ: A. A focused goal connects action to measurable change.

Written solution: Need: weak explanations. Goal: improve evidence-based responses. Activity: learn and apply structured questioning. Evidence: compare student response samples before and after a defined period. One mark per linked component.

Return to lesson: page 404.

Question 2

MCQ: C. The reader can see what the evidence proves and how practice developed.

Written solution: A lesson plan can demonstrate planning and alignment. An anonymized student-work sample with commentary can demonstrate learning and the effect of feedback. Two marks per item and explanation.

Return to lesson: page 407.

Question 3

MCQ: D. Evidence-based feedback supports improvement while respecting the colleague.

Written solution: Example: Several students could not identify the task goal; stating the success criteria and checking one example before independent work may help. Credit observable evidence, effect, actionable suggestion and respectful language.

Return to lesson: page 410.

Question 4

MCQ: A. Pedagogical purpose should guide selection.

Written solution: Use a brief concept quiz with plausible distractors. Review response patterns, reteach the misconception and check with a new question. Evaluate whether explanations or transfer improve, not only whether students clicked. One mark each for tool, diagnosis, response and evaluation.

Return to lesson: page 413.

Formula sheet

Teaching as a profession p.392

Rule

$$\text{Criterion} + \text{explanation} + \text{teaching example}$$

Professionalization and the teacher role p.395

Rule

$$\text{Role} \rightarrow \text{observable professional action}$$

Teacher formation: before and during service p.398

Rule

$$\text{Pre-service} \rightarrow \text{induction} \rightarrow \text{continuing development}$$

Preparation, practice and professional standards p.401

Rule

$$\text{Knowledge} \rightarrow \text{supervised application} \rightarrow \text{feedback} \rightarrow \text{improvement}$$

Plan continuing professional development p.404

Rule

$$\text{Need} \rightarrow \text{goal} \rightarrow \text{activity} \rightarrow \text{evidence of impact}$$

Portfolios and reflective evaluation p.407

Rule

$$\text{Evidence} + \text{reflection} + \text{link to a goal}$$

Ethics, collaboration and learning communities p.410

Rule

$$\text{Observation} + \text{effect} + \text{constructive next step}$$

Educational technology and evaluation of impact p.413

Rule

$$\text{Learning need} \rightarrow \text{tool choice} \rightarrow \text{activity} \rightarrow \text{evaluation}$$

Symbol glossary

Symbols are listed with their lesson context. A symbol may represent different quantities in different rules.

Source and coverage notes

Elective lecture themes: teaching as a profession, teacher formation, professional learning, portfolios, standards and educational technology. This follows the teacher-development focus of the supplied file; it is not a generic career-skills course.

Open original course folder

The source audit inventories 159 non-programming files; 95 yielded substantial extracted text. Scanned diagrams and handwritten or image-only material were not fully transcribed. Check lecturer-specific derivations and conventions. The practice tests are original study material, not predictions of an examination.

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